BIPOLAR TRANSISTORS

CHAPTER 2

2.1 Introduction

The transistor is our most important example of an "active" component, a device that can amplify, producing an output signal with more power in it than the input signal. The additional power comes from an external source of power (the power supply, to be exact). Note that *voltage* amplification isn't what matters, since, for example, a step-up transformer, a "passive" component just like a resistor or capacitor, has voltage gain but no power gain.[1] Devices with power gain are distinguishable by their ability to make oscillators, by feeding some output signal back into the input.

It is interesting to note that the property of power amplification seemed very important to the inventors of the transistor. Almost the first thing they did to convince themselves that they had really invented something was to power a loudspeaker from a transistor, observing that the output signal sounded louder than the input signal.

The transistor is the essential ingredient of every electronic circuit, from the simplest amplifier or oscillator to the most elaborate digital computer. Integrated circuits (ICs), which have largely replaced circuits constructed from discrete transistors, are themselves merely arrays of transistors and other components built from a single chip of semiconductor material.

A good understanding of transistors is very important, even if most of your circuits are made from ICs, because you need to understand the input and output properties of the IC in order to connect it to the rest of your circuit and to the outside world. In addition, the transistor is the single most powerful resource for interfacing, whether between ICs and other circuitry or between one subcircuit and another. Finally, there are frequent (some might say too frequent) situations in which the right IC just doesn't exist, and you have to rely on discrete transistor circuitry to do the job. As you will see, transistors have an excitement all their own. Learning how they work can be great fun.

[1] It is even possible to achieve modest voltage gain in a circuit comprising only resistors and capacitors. To explore this idea, surprising even to seasoned engineers, look at Appendix J on SPICE.

There are two major species of transistors: in this chapter we will learn about bipolar junction transistors (BJTs), which historically came first with their Nobel Prize-winning invention in 1947 at Bell Laboratories. The next chapter deals with "field-effect" transistors (FETs), the now-dominant species in digital electronics. To give the coarsest comparison, BJTs excel in accuracy and low noise, whereas FETs excel in low power, high impedance, and high-current switching; there is, of course, much more to this complex subject.

Our treatment of bipolar transistors is going to be quite different from that of many other books. It is common practice to use the *h*-parameter model and equivalent circuit. In our opinion that is unnecessarily complicated and unintuitive. Not only does circuit behavior tend to be revealed to you as something that drops out of elaborate equations, rather than deriving from a clear understanding in your own mind as to how the circuit functions; you also have the tendency to lose sight of which parameters of transistor behavior you can count on and, more important, which ones can vary over large ranges.

In this chapter we will instead build up a very simple introductory transistor model and immediately work out some circuits with it. Its limitations will soon become apparent; then we will expand the model to include the respected Ebers–Moll conventions. With the Ebers–Moll equations and a simple three-terminal model, you will have a good understanding of transistors; you won't need to do a lot of calculations, and your designs will be first rate. In particular, they will be largely independent of the poorly controlled transistor parameters such as current gain.

Some important engineering notation should be mentioned. Voltage at a transistor terminal (relative to ground) is indicated by a single subscript (C, B, or E): V_C is the collector voltage, for instance. Voltage between two terminals is indicated by a double subscript: V_{BE} is the base-to-emitter voltage drop, for instance. If the same letter is repeated, that means a power-supply voltage: V_{CC} is the (positive) power-supply voltage associated with the collector,

and V_{EE} is the (negative) supply voltage associated with the emitter.[2]

Why transistor circuits are difficult

For those learning electronics for the first time, this chapter will be difficult. Here's why: all the circuits in the last chapter dealt with *two-terminal devices*, whether linear (resistors, capacitors, inductors) or nonlinear (diodes). So there was only one voltage (the voltage between the terminals) and only one current (the current flowing through the device) to think about. Transistors, by contrast, are *three-terminal devices*, which means there are two voltages and two currents to juggle.[3]

2.1.1 First transistor model: current amplifier

Let's begin. A bipolar transistor is a three-terminal device (Figure 2.1), in which a small current applied to the base controls a much larger current flowing between the collector and emitter. It is available in two flavors (*npn* and *pnp*), with properties that meet the following rules for *npn* transistors (for *pnp* simply reverse all polarities):

1. **Polarity** The collector must be more positive than the emitter.
2. **Junctions** The base–emitter and base–collector circuits behave like diodes (Figure 2.2) in which a small current applied to the base controls a much larger current flowing between the collector and emitter. Normally the base–emitter diode is conducting, whereas the base–collector diode is reverse-biased, i.e., the applied voltage is in the opposite direction to easy current flow.
3. **Maximum ratings** Any given transistor has maximum values of I_C, I_B, and V_{CE} that cannot be exceeded without costing the exceeder the price of a new transistor (for typical values, see the listing in Table 2.1 on page 74, Table 2.2 on page 106, and Table 8.1 on pages 501–502 There are also other limits, such as power dissipation ($I_C V_{CE}$), temperature, and V_{BE}, that you must keep in mind.
4. **Current amplifier** When rules 1–3 are obeyed, I_C is roughly proportional to I_B and can be written as

$$I_C = h_{FE} I_B = \beta I_B, \tag{2.1}$$

where β, the current gain (sometimes called[4] h_{FE}), is typically about 100. Both I_B and I_C flow to the emitter. Note: the collector current is not due to forward conduction of the base–collector diode; that diode is reverse-biased. Just think of it as "transistor action."

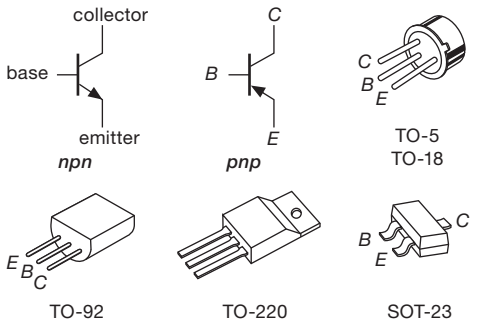


Figure 2.1. Transistor symbols and small transistor package drawings (not to scale). A selection of common transistor packages are shown in Figure 2.3.

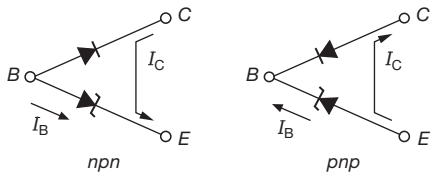


Figure 2.2. An ohmmeter's view of a transistor's terminals.

Rule 4 gives the transistor its usefulness: a small current flowing into the base controls a much larger current flowing into the collector.

An important warning: the current gain β is not a "good" transistor parameter; for instance, its value can vary from 50 to 250 for different specimens of a given transistor type. It also depends on the collector current, collector-to-emitter voltage, and temperature. *A circuit that depends on a particular value for beta is a bad circuit.*

Note particularly the effect of rule 2. This means you can't go sticking an arbitrary voltage across the base–emitter terminals, because an enormous current will flow if the base is more positive than the emitter by more than about 0.6 to 0.8 V (forward diode drop). This rule also implies that an operating transistor has $V_B \approx V_E + 0.6\,V$ ($V_B = V_E + V_{BE}$). Again, polarities are normally given for *npn* transistors; reverse them for *pnp*.

Let us emphasize again that you should not try to think of the collector current as diode conduction. It isn't,

[2] In practice, circuit designers use V_{CC} to designate the positive supply and V_{EE} the negative supply, even though logically they should be interchanged for *pnp* transistors (where all polarities are reversed).

[3] You might think that there would be three voltages and three currents; but it's slightly less complicated than that, because there are only two independent voltages and two independent currents, thanks to Kirchhoff's voltage and current laws.

[4] As the "*h*-parameter" transistor model has fallen out of popularity, you tend often to see β (instead of h_{FE}) as the symbol for current gain.

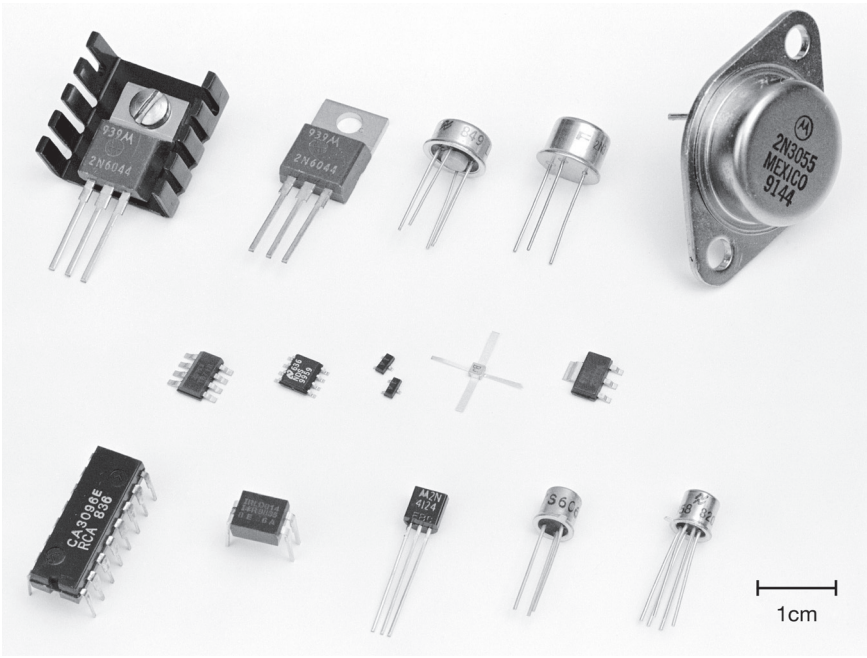


Figure 2.3. Most of the common packages are shown here, for which we give the traditional designations. Top row (power), left to right: TO-220 (with and without heatsink), TO-39, TO-5, TO-3. Middle row (surface mount): SM-8 (dual), SO-8 (dual), SOT-23, ceramic SOE, SOT-223. Bottom row: DIP-16 (quad), DIP-4, TO-92, TO-18, TO-18 (dual).

because the collector–base diode normally has voltages applied across it in the reverse direction. Furthermore, collector current varies very little with collector voltage (it behaves like a not-too-great current source), unlike forward diode conduction, in which the current rises very rapidly with applied voltage.

Table 2.1 on the following page includes a selection of commonly used bipolar transistors, with the corresponding curves of current gain[5] in Figure 2.4, and a selection of transistors intended for power applications is listed in Table 2.2 on page 106. A more complete listing can be found in Table 8.1 on pages 501–502 and Figure 8.39 in Chapter 8.

2.2 Some basic transistor circuits

2.2.1 Transistor switch

Look at the circuit in Figure 2.5. This application, in which a small control current enables a much larger current to flow in another circuit, is called a transistor switch. From the preceding rules it is easy to understand. When the mechanical switch is open, there is no base current. So, from rule 4, there is no collector current. The lamp is off.

When the switch is closed, the base rises to 0.6 V (base–emitter diode is in forward conduction). The drop across the base resistor is 9.4 V, so the base current is 9.4 mA. Blind application of Rule 4 gives $I_C = 940$ mA (for a typical beta of 100). That is wrong. Why? Because rule 4 holds only if Rule 1 is obeyed: at a collector current of 100 mA the lamp has 10 V across it. To get a higher current you would have to pull the collector below ground. A transistor can't do this, and the result is what's called *saturation* – the collector goes as close to ground as it can (typical saturation voltages are about 0.05–0.2 V, see Chapter *2x*.) and stays there. In this case, the lamp goes on, with its rated 10 V across it.

Overdriving the base (we used 9.4 mA when 1.0 mA would have barely sufficed) makes the circuit conservative; in this particular case it is a good idea, since a lamp draws more current when cold (the resistance of a lamp when cold is 5 to 10 times lower than its resistance at operating current). Also, transistor beta drops at low collector-to-base voltages, so some extra base current is necessary to bring

[5] In addition to listing typical betas (h_{FE}) and maximum allowed collector-to-emitter voltages (V_{CEO}), Table 2.1 includes the cutoff frequency (f_T, at which the beta has decreased to 1) and the feedback capacitance (C_{cb}). These are important when dealing with fast signals or high frequencies; we'll see them in §2.4.5 and Chapter *2x*.

Table 2.1 Representative Bipolar Transistors

Part # *npn* TO-92	Part # *npn* SOT-23	Part # *pnp* TO-92	Part # *pnp* SOT-23	V_{CEO} (V)	I_C (max) (mA)	h_{FE} @ mA (typ)		gain curve[d]	C_{cb}[a] (pF)	f_T[a] (MHz)	Comments
2N3904	**MMBT3904**	**2N3906**	**MMBT3906**	40	150	200	10	6	2.5	300	jellybean
2N4401	**MMBT4401**	**2N4403**	**MMBT4403**	40	500	150	150	7	7	300	'2222 and '2907 dies
BC337	**BC817**	**BC327**	**BC807**	45	750	350	40	5	10	150	jellybean
2N5089	**MMBT5089**	**2N5087**	**MMBT5087**	25	50	500	1	3	1.8	350	high beta
BC547C	**BC847C**	**BC557C**	**BC857C**	45	100	500	10	4	5	150	jellybean[b]
MPSA14	**MMBTA14**	**MPSA64**	**MMBTA64**	30	300	10000	50	-	7	125	Darlington
ZTX618	**FMMT618**	**ZTX718**	**FMMT718**	20	2500	320	3A	3a	-	120	high I_C, small pkg
PN2369	**MMBT2369**	**2N5771**	**MMBT5771**	15	150	100	10	10	3	500	fast switch, gold doped
2N5551	**MMBT5551**	**2N5401**	**MMBT5401**	150	100	100	10	5a	2.5	100	SOT-223 available
MPSA42	**MMBTA42**	**MPSA92**	**MMBTA92**	300	30	75	10	9	1.5	50	HV small signal
MPS5179	**BFS17**	**MPSH81**	**MMBTH81**	15	25	90	20	8	0.9	900	RF amplifier
—	**BFR93**[c]	—	**BFT93**[c]	12	50	50	15	10	0.5	4000	RF amp
TIP142	—	**TIP147**	—	100	10A	>1000	5A	-	high	low	TO-220, Darlington

Notes: (a) see Chapter *2x* for graphs of C_{cb} and f_T. (b) lower beta versions have an -A or -B suffix; low-noise versions are BC850 (*npn*) and BC860 (*pnp*). (c) also BFR25A and BFT25A. (d) see Figure 2.4.

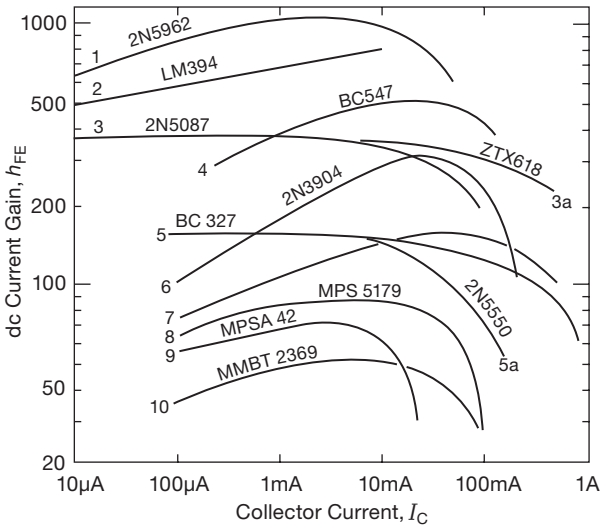


Figure 2.4. Curves of typical transistor current gain, β, for a selection of transistors from Table 2.1. These curves are taken from manufacturers' literature. You can expect production spreads of +100%, −50% from the "typical" values graphed. See also Figure 8.39 for measured beta plots for 44 types of "low-noise" transistors.

a transistor into full saturation. Incidentally, in a real circuit you would probably put a resistor from base to ground (perhaps 10k in this case) to make sure the base is at ground with the switch open. It wouldn't affect the ON operation, because it would sink only 0.06 mA from the base circuit.

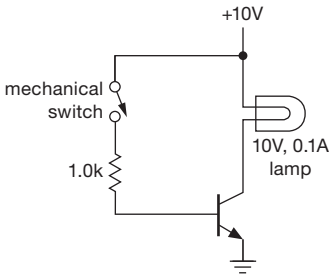


Figure 2.5. Transistor switch example.

There are certain cautions to be observed when designing transistor switches:

1. Choose the base resistor conservatively to get plenty of excess base current, especially when driving lamps, because of the reduced beta at low V_{CE}. This is also a good idea for high-speed switching, because of capacitive effects and reduced beta at very high frequencies (many megahertz).[6]
2. If the load swings below ground for some reason (e.g., it is driven from ac, or it is inductive), use a diode in series with the collector (or a diode in the reverse direction to ground) to prevent collector–base conduction on negative swings.
3. For inductive loads, protect the transistor with a diode

[6] A small "speed-up" capacitor – typically just a few picofarads – is often connected across the base resistor to improve high-speed performance.

across the load, as shown in Figure 2.6.[7] Without the diode the inductor will swing the collector to a large positive voltage when the switch is opened, most likely exceeding the collector–emitter breakdown voltage, as the inductor tries to maintain its "on" current from V_{CC} to the collector (see the discussion of inductors in §1.6.7).

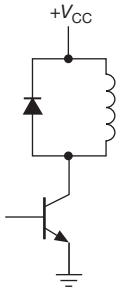


Figure 2.6. Always use a suppression diode when switching an inductive load.

You might ask why we are bothering with a transistor, and all its complexity, when we could just use that mechanical switch alone to control the lamp or other load. There are several good reasons: (a) a transistor switch can be driven *electrically* from some other circuit, for example a computer output bit; (b) transistor switches enable you to switch very rapidly, typically in a small fraction of a microsecond; (c) you can switch many different circuits with a single control signal; (d) mechanical switches suffer from wear, and their contacts "bounce" when the switch is activated, often making and breaking the circuit a few dozen times in the first few milliseconds after activation; and (e) with transistor switches you can take advantage of remote *cold switching*, in which only dc control voltages snake around through cables to reach front-panel switches, rather than the electronically inferior approach of having the signals themselves traveling through cables and switches (if you run lots of signals through cables, you're likely to get capacitive pickup as well as some signal degradation).

A. "Transistor man"

The cartoon in Figure 2.7 may help you understand some limits of transistor behavior. The little man's perpetual task in life is to try to keep $I_C = \beta I_B$; however, he is only allowed to turn the knob on the variable resistor. Thus he can go from a short circuit (saturation) to an open circuit (transistor in the OFF state), or anything in between, but he isn't allowed to use batteries, current sources, etc.

[7] Or, for faster turn-off, with a resistor, an *RC* network, or zener clamp; see §1.6.7.

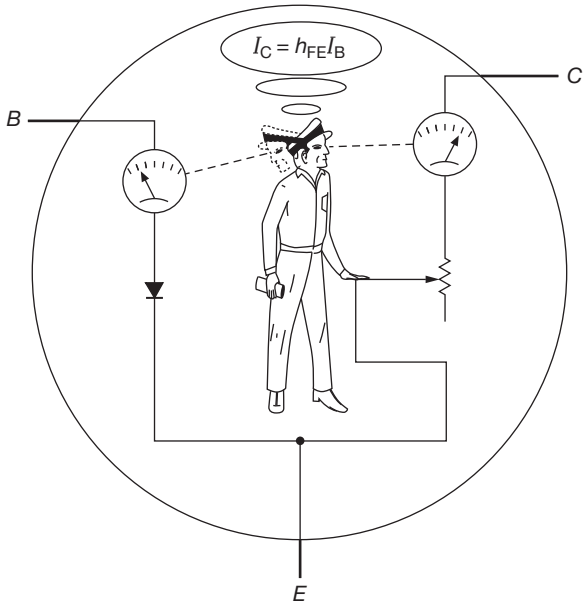


Figure 2.7. "Transistor man" observes the base current, and adjusts the output rheostat in an attempt to maintain the output current β times larger; h_{FE} and β are used interchangeably.

One warning is in order here: don't think that the collector of a transistor looks like a resistor. It doesn't. Rather, it looks approximately like a poor-quality constant-current sink (the value of current depending on the signal applied to the base), primarily because of this little man's efforts.

Another thing to keep in mind is that, at any given time, a transistor may be (a) cut off (no collector current), (b) in the active region (some collector current, and collector voltage more than a few tenths of a volt above the emitter), or (c) in saturation (collector within a few tenths of a volt of the emitter). See the discussion of transistor saturation in Chapter *2x* for more details.

2.2.2 Switching circuit examples

The transistor switch is an example of a *nonlinear* circuit: the output is not proportional to the input;[8] instead it goes to one of two possible states (cut off, or saturated). Such two-state circuits are extremely common[9] and form the basis of digital electronics. But to the authors the subject of

[8] A mathematician would define linearity by saying that the response to the sum of two inputs is the sum of the individual responses; this necessarily implies proportionality.

[9] If you took a census, asking the transistors of the world what they are doing, at least 95% would tell you they are switches.

linear circuits (such as amplifiers, current sources, and integrators) offers the most interesting challenges and the potential for great circuit creativity. We will move on to linear circuits in a moment, but this is a good time to enjoy a few circuit examples with transistors acting as switches – we like to give a feeling for the richness of electronics by showing real-world examples as soon as possible.

A. LED driver

Light-emitting diode indicators – LEDs – have replaced the incandescent lamps of yesteryear for all electronic indicator and readout applications; they're cheap, they come in lots of colors, and they last just about forever. Electrically they are similar to the ordinary silicon signal diodes we met in Chapter 1, but with a larger forward voltage drop (generally in the range of 1.5–3.5 V, rather than approximately[10] 0.6 V); that is, as you slowly increase the voltage across an LED's terminals, you find that they start conducting current at, say, 1.5 V, and the current increases rapidly as you apply somewhat more voltage (Figure 2.8). They light up, too! Typical "high-efficiency" indicator LEDs look pretty good at a few milliamps, and they'll knock your eye out at 10–20 mA.

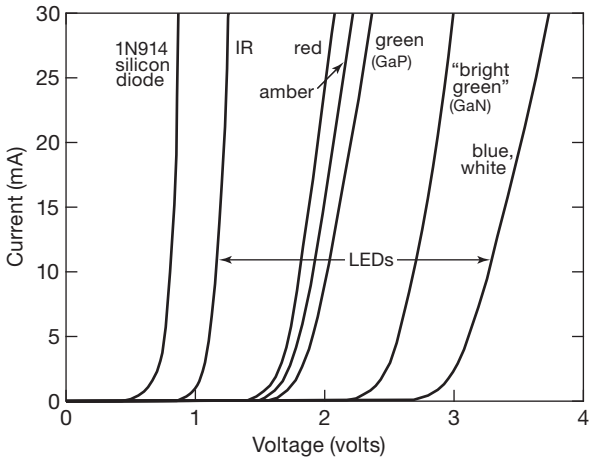


Figure 2.8. Like silicon diodes, LEDs have rapidly increasing current versus applied voltage, but with larger forward voltage drops.

We'll show a variety of techniques for driving LEDs in Chapter 12; but we can drive them already, with what we know. The first thing to realize is that we can't just switch a voltage across them, as in Figure 2.5, because of their steep I versus V behavior; for example, applying 5 V across an LED is guaranteed to blow it out. We need instead to treat it gently, coaxing it to draw the right current.

Let's assume that we want the LED to light in response to a digital signal line when it goes to a HIGH value of +3.3 V (from its normal resting voltage near ground). Let's assume also that the digital line can provide up to 1 mA of current, if needed. The procedure goes like this: first, choose an LED operating current that will provide adequate brightness, say 5 mA (you might want to try a few samples, to make sure you like the color, brightness, and viewing angle). Then use an *npn* transistor as a switch (Figure 2.9), choosing the collector resistor to provide the chosen LED current, realizing that the voltage drop across the resistor is the supply voltage minus the LED forward drop at its operating current. Finally, choose the base resistor to ensure saturation, assuming a conservatively low transistor beta ($\beta \geq 25$ is pretty safe for a typical small-signal transistor like the popular 2N3904).

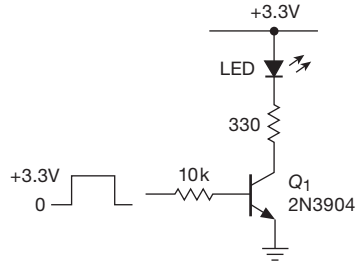


Figure 2.9. Driving an LED from a "logic-level" input signal, using an *npn* saturated switch and series current-limiting resistor.

Note that the transistor is acting as a saturated switch, with the collector resistor setting the operating current. As we'll see shortly, you can devise circuits that provide an accurate *current* output, largely independent of what the load does. Such a "current source" can also be used to drive LEDs. But our circuit is simple, and effective. There are other variations: we'll see in the next chapter that a MOSFET-type[11] transistor is often a better choice. And in Chapters 10–12 we'll see ways to drive LEDs and other optoelectronic devices directly from digital integrated circuits, without external discrete transistors.

Exercise 2.1. What is the LED current, approximately, in the circuit of Figure 2.9? What minimum beta is required for Q_1?

B. Variations on a theme

For these switch examples, one side of the load is connected to a positive supply voltage, and the other side is

[10] The larger drop is due to the use of different semiconductor materials such as GaAsP, GaAlAs, and GaN, with their larger bandgaps.

[11] metal-oxide semiconductor field-effect transistor.

switched to ground by the *npn* transistor switch. What if you want instead to ground one side of the load and switch the "high side" to a positive voltage?

It's easy enough – but you've got to use the other polarity of transistor (*pnp*), with its emitter at the positive rail, and its collector tied to the load's high side, as in Figure 2.10A. The transistor is cut off when the base is held at the emitter voltage (here +15 V), and switched into saturation by bringing the base toward the collector (i.e., toward ground). When the input is brought to ground, there's about 4 mA of base current through the 3.3 kΩ base resistor, sufficient for switching loads up to about 200 mA ($\beta > 50$).

An awkwardness of this circuit is the need to hold the input at +15 V to turn off the switch; it would be much better to use a lower control voltage, for example, +3 V and ground, commonly available in digital logic that we'll be seeing in Chapters 10–15. Figure 2.10B shows how to do that: *npn* switch Q_2 accepts the "logic-level" input of 0 V or +3 V, pulling its collector load to ground accordingly. When Q_2 is cut off, R_3 holds Q_3 off; when Q_2 is saturated (by a +3 V input), R_2 sinks base current from Q_3 to bring it into saturation.

The "divider" formed by R_2R_3 may be confusing: R_3's job is to keep Q_3 off when Q_2 is off; and when Q_2 pulls its collector low, most of its collector current comes from Q_3's base (because only ∼0.6 mA of the 4.4 mA collector current comes from R_3 – make sure you understand why). That is, R_3 does not have much effect on Q_3's saturation. Another way to say it is that the divider would sit at about +11.6 V (rather than +14.4 V), were it not for Q_3's base–emitter diode, which consequently gets most of Q_2's collector current. In any case, the value of R_3 is not critical and could be made larger; the tradeoff is slower turn-off of Q_3, owing to capacitive effects.[12]

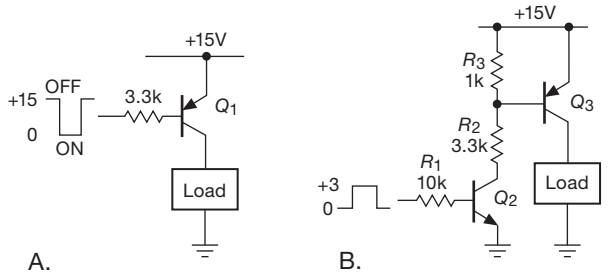


Figure 2.10. Switching the high side of a load returned to ground.

C. Pulse generator – I

By including a simple *RC*, you can make a circuit that gives a pulse output from a step input; the time constant $\tau = RC$ determines the pulse width. Figure 2.11 shows one way. Q_2 is normally held in saturation by R_3, so its output is close to ground; note that R_3 is chosen small enough to ensure Q_2's saturation. With the circuit's input at ground, Q_1 is cut off, with its collector at +5 V. The capacitor C_1 is therefore charged, with +5 V on its left terminal and approximately +0.6 V on its right terminal; i.e., it has about 4.4 V across it. The circuit is waiting for something to happen.

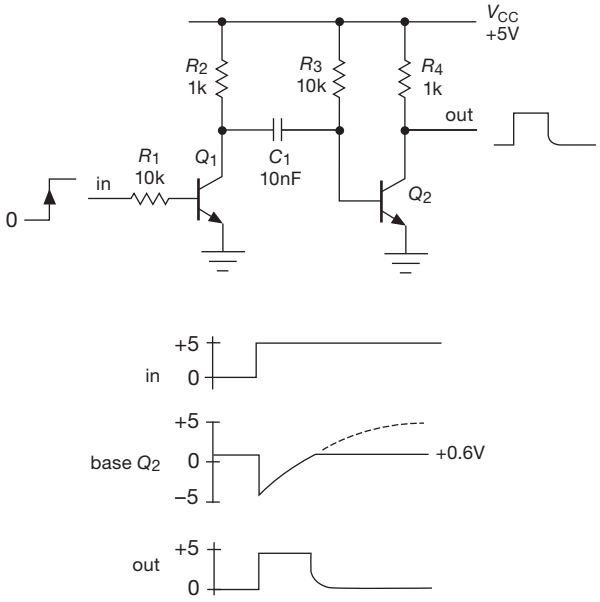


Figure 2.11. Generating a short pulse from a step input waveform.

A +5 V positive input step brings Q_1 into saturation (note the values of R_1 and R_2), forcing its collector to ground; because of the voltage across C_1, this brings the base of Q_2 momentarily negative, to about −4.4 V.[13] Q_2 is then cutoff, no current flows through R_4, and so its output jumps to +5 V; this is the beginning of the output pulse. Now for the *RC*: C_1 can't hold Q_2's base below ground forever, because current is flowing down through R_3, trying to pull it up. So the right-hand side of the capacitor charges toward +5 V, with a time constant $\tau = R_3C_1$, here equal to 100 μs. The output pulse width is set by this time constant

[12] But don't make it too small: Q_3 would not switch at all if R_3 were reduced to 100 Ω (why?). We were surprised to see this basic error in an instrument, the rest of which displayed circuit design of the highest sophistication.

[13] A caution here: this circuit should not be run from a supply voltage greater than +7 V, because the negative pulse can drive Q_2's base into reverse breakdown. This is a common oversight, even among experienced circuit designers.

and is proportional to τ. To figure out the pulse width accurately you have to look in detail at the circuit operation. In this case it's easy enough to see that the output transistor Q_2 will turn on again, terminating the output pulse, when the rising voltage on the base of transistor Q_2 reaches the $\approx$0.6 V V_{BE} drop required for turn-on. Try this problem to test your understanding.

Exercise 2.2. Show that the output pulse width for the circuit of Figure 2.11 is approximately $T_{pulse}=0.76R_3C_1=76\mu$s. A good starting point is to notice that C_1 is charging exponentially from -4.4 V toward +5 V, with the time constant as above.

D. Pulse generator – II

Let's play with this circuit a bit. It works fine as described, but note that it requires that the input remain high throughout the duration of the output pulse, at least. It would be nice to eliminate that restriction, and the circuit in Figure 2.12 shows how. To the original circuit we've added a third transistor switch Q_3, whose job is to hold the collector of Q_1 at ground once the output pulse begins, regardless of what the input signal does. Now any positive input pulse – whether longer or shorter than the desired output pulse width – produces the same output pulse width; look at the waveforms in the figure. Note that we've chosen R_5 relatively large to minimize output loading while still ensuring full saturation of Q_3.

Exercise 2.3. Elaborate on this last statement: what is the output voltage during the pulse, slightly reduced owing to the loading effect of R_5? What is the minimum required beta of Q_3 to guarantee its saturation during the output pulse?

E. Pulse generator – III

For our final act, let's fix a deficiency of these circuits, namely a tendency for the output pulse to turn off somewhat slowly. That happens because Q_2's base voltage, with its leisurely 100 μs *RC* time constant, rises smoothly (and relatively slowly) through the turn-on voltage threshold of $\approx$0.6 V. Note, by the way, that this problem does not occur at the turn-*on* of the output pulse, because at that transition Q_2's base voltage drops abruptly down to approximately -4.4 V, owing to the sharp input step waveform, which is further sharpened by the switching action of Q_1.

The cure here is to add at the output a clever circuit known as a *Schmitt trigger*, shown in its transistor implementation[14] in Figure 2.13A. It works like this: imagine a time within the positive output pulse of the previous circuits, so the input to this new Schmitt circuit is high (near

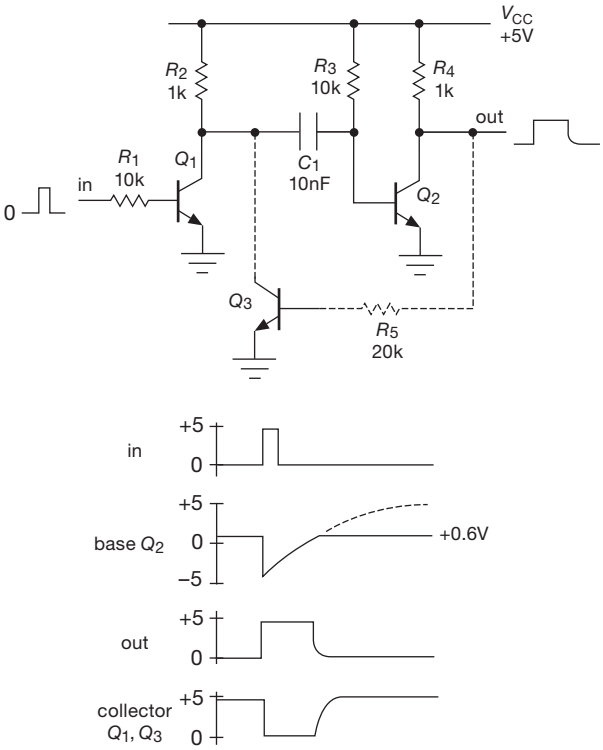


Figure 2.12. Generating a short pulse from a step or pulse input.

+5 V). That holds Q_4 in saturation, and so Q_5 is cut off, with the output at +5 V. The emitter current of Q_4 is about 5 mA, so the emitter voltage is approximately +100 mV; the base is a V_{BE} higher, approximately +700 mV.

Now imagine the trailing edge of the input pulse waveform, whose voltage smoothly drops toward ground. As it drops below 700 mV, Q_4 begins to turn off, so its collector voltage rises. If this were a simple transistor switch (i.e., if Q_5 were absent) the collector would rise to +5 V; here, however, the collector resistor R_7 instead supplies current to Q_5, putting it in saturation. So Q_5's collector drops nearly to ground.

At this simple level of analysis the circuit appears to be pretty useless, because its output is the same as its input! Let's look a little closer, though: as the input voltage drops through the 700 mV threshold and Q_5 turns on, the total emitter current rises to $\approx$10 mA (5 mA from Q_5's collector current, and another $\approx$5 mA from its base current, both of which flow out the emitter). The drop across the emitter resistor is now 200 mV, which means that the input threshold has increased to about +800 mV. So the input voltage, which had just dropped below 700 mV, now finds itself well below the new threshold, causing the

[14] We'll see other ways of making a Schmitt trigger, using op-amps or comparators, in Chapter 4.

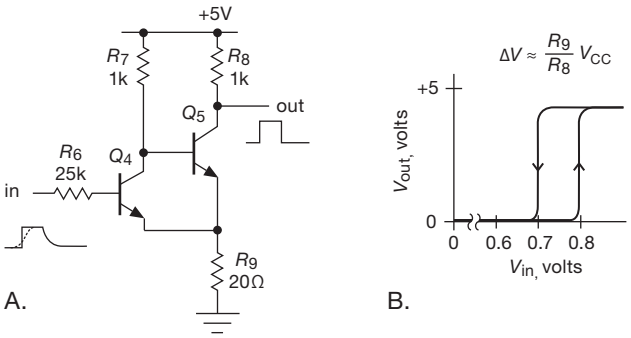


Figure 2.13. A "Schmitt trigger" produces an output with abrupt transitions, regardless of the speed of the input waveform.

output to switch abruptly. This "regenerative" action is how the Schmitt trigger turns a slowly moving waveform into an abrupt transition.

A similar action occurs as the input rises through this higher threshold; see Figure 2.13B, which illustrates how the output voltage changes as the input voltage passes through the two thresholds, an effect known as *hysteresis*. The Schmitt trigger produces rapid output transitions as the input passes through either threshold. We'll see Schmitt triggers again in Chapters 4 and 10.

There are many enjoyable applications of transistor switches, including "signal" applications like this (combined with more complex digital logic circuits), as well as "power switching" circuits in which transistors operating at high currents, high voltages, or both, are used to control hefty loads, perform power conversion, and so on. Transistor switches can also be used as substitutes for mechanical switches when we are dealing with continuous ("linear" or "analog") waveforms. We'll see examples of these in the next chapter, when we deal with FETs, which are ideally suited to such switching tasks, and again in Chapter 12, where we deal with the control of signals and external loads from logic-level signals.

We now move on to consider the first of several *linear* transistor circuits.

2.2.3 Emitter follower

Figure 2.14 shows an example of an *emitter follower*. It is called that because the output terminal is the emitter, which follows the input (the base), less one diode drop:

$$V_E \approx V_B - 0.6 \text{ volts}.$$

The output is a replica of the input, but 0.6 to 0.7 V less positive. For this circuit, V_{in} must stay at +0.6 V or more, or else the output will sit at ground. By returning the emitter resistor to a negative supply voltage, you can permit negative voltage swings as well. Note that there is no collector resistor in an emitter follower.

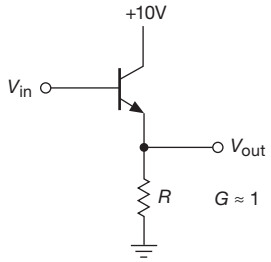


Figure 2.14. Emitter follower.

At first glance this circuit may appear quite thoroughly useless, until you realize that the input impedance is much larger than the output impedance, as will be demonstrated shortly. This means that the circuit requires less power from the signal source to drive a given load than would be the case if the signal source were to drive the load directly. Or a signal of some internal impedance (in the Thévenin sense) can now drive a load of comparable or even lower impedance without loss of amplitude (from the usual voltage-divider effect). In other words, an emitter follower has current gain, even though it has no voltage gain. It has *power* gain. Voltage gain isn't everything!

A. Impedances of sources and loads

This last point is very important and is worth some more discussion before we calculate in detail the beneficial effects of emitter followers. In electronic circuits, you're always hooking the output of something to the input of something else, as suggested in Figure 2.15. The signal source might be the output of an amplifier stage (with Thévenin equivalent series impedance $\mathbf{Z}_{\text{out}}$), driving the next stage or perhaps a load (of some input impedance $\mathbf{Z}_{\text{in}}$). In general, the loading effect of the following stage causes a reduction of signal, as we discussed earlier in §1.2.5A. For this reason it is usually best to keep $Z_{\text{out}} \ll Z_{\text{in}}$ (a factor of 10 is a comfortable rule of thumb).

In some situations it is OK to forgo this general goal of making the source stiff compared with the load. In particular, if the load is always connected (e.g., within a circuit) and if it presents a known and constant $\mathbf{Z}_{\text{in}}$, it is not too serious if it "loads" the source. However, it is always nicer if signal levels don't change when a load is connected. Also, if $\mathbf{Z}_{\text{in}}$ varies with signal level, then having a stiff source

($Z_{out} \ll Z_{in}$) ensures linearity, where otherwise the level-dependent voltage divider would cause distortion.[15]

Finally, as we remarked in §1.2.5A, there are two situations in which $Z_{out} \ll Z_{in}$ is actually the wrong thing to do: in radiofrequency circuits we usually *match* impedances ($\mathbf{Z}_{out} = \mathbf{Z}_{in}$), for reasons we'll describe in Appendix H. A second exception applies if the signal being coupled is a *current* rather than a voltage. In that case the situation is reversed, and we strive to make $Z_{in} \ll Z_{out}$ ($Z_{out} = \infty$, for a current source).

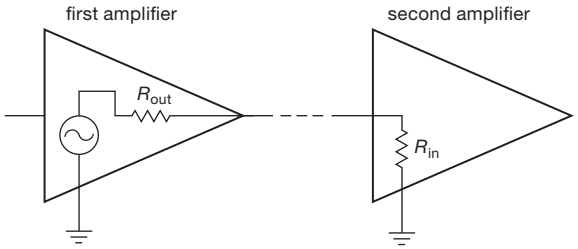


Figure 2.15. Illustrating circuit "loading" as a voltage divider.

B. Input and output impedances of emitter followers

As we've stated, the emitter follower is useful for changing impedances of signals or loads. To put it starkly, that's really the whole point of an emitter follower.

Let's calculate the input and output impedances of the emitter follower. In the preceding circuit we consider R to be the load (in practice it sometimes *is* the load; otherwise the load is in parallel with R, but with R dominating the parallel resistance anyway). Make a voltage change ΔV_B at the base; the corresponding change at the emitter is $\Delta V_E = \Delta V_B$. Then the change in emitter current is

$$\Delta I_E = \Delta V_B / R,$$

so

$$\Delta I_B = \frac{1}{\beta + 1} \Delta I_E = \frac{\Delta V_B}{R(\beta + 1)}$$

(using $I_E = I_C + I_B$). The input resistance is $\Delta V_B / \Delta I_B$. Therefore

$$r_{in} = (\beta + 1)R. \tag{2.2}$$

The transistor small-signal (or "incremental") current gain (β, or h_{fe}) is typically about 100, so a low-impedance load looks like a much higher impedance at the base; it is easier to drive.

In the preceding calculation we used the *changes* in the voltages and currents, rather than the steady (dc) values of those voltages (or currents), to arrive at our input resistance r_{in}. Such a "small-signal" analysis is used when the variations represent a possible signal, as in an audio amplifier, riding on a steady dc "bias" (see §2.2.7). Although we indicated changes in voltage and current explicitly (with "ΔV," etc.), the usual practice is to use lowercase symbols for small-signal variations (thus $\Delta V \leftrightarrow v$); with this convention the above equation for ΔI_E, for example, would read $i_E = v_B / R$.

The distinction between dc current gain (h_{FE}) and small-signal current gain (h_{fe}) isn't always made clear, and the term beta is used for both. That's alright, since $h_{fe} \approx h_{FE}$ (except at very high frequencies), and you never assume you know them accurately, anyway.

Although we used resistances in the preceding derivation, we could generalize to complex impedances by allowing ΔV_B, ΔI_B, etc., to become complex numbers. We would find that the same transformation rule applies for impedances:

$$\mathbf{Z}_{in} = (\beta + 1)\mathbf{Z}_{load}. \tag{2.3}$$

We could do a similar calculation to find that the output impedance $\mathbf{Z}_{out}$ of an emitter follower (the impedance looking into the emitter) driven from a source of internal impedance $\mathbf{Z}_{source}$ is given by

$$\mathbf{Z}_{out} = \frac{\mathbf{Z}_{source}}{\beta + 1}. \tag{2.4}$$

Strictly speaking, the output impedance of the circuit should also include the parallel resistance of R, but in practice $\mathbf{Z}_{out}$ (the impedance looking into the emitter) dominates.

Exercise 2.4. Show that the preceding relationship is correct. *Hint*: hold the source voltage fixed and find the change in output current for a given forced change in output voltage. Remember that the source voltage is connected to the base through a series resistor.

Because of these nice properties, emitter followers find application in many situations, e.g., making low-impedance signal sources within a circuit (or at outputs), making stiff voltage references from higher-impedance references (formed from voltage dividers, say), and generally isolating signal sources from the loading effects of subsequent stages.

[15] We use the boldface symbol $\mathbf{Z}$ when the complex nature of impedance is important. In common usage the term "impedance" can refer loosely to the *magnitude* of impedance, or even to a purely real impedance (e.g., transmission-line impedance); for such instances we use the ordinary math-italic symbol Z.

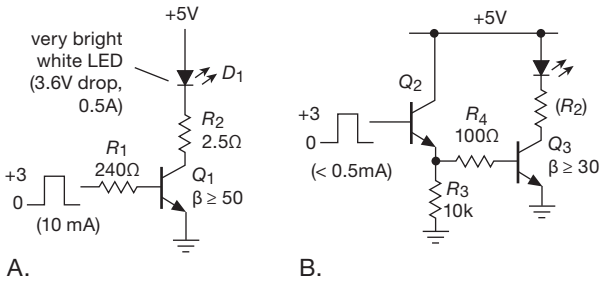


Figure 2.16. Putting an emitter follower in front of a switch makes it easy for a low-current control signal to switch a high-current load.

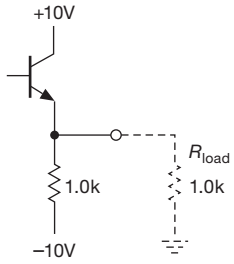


Figure 2.17. An *npn* emitter follower can source plenty of current through the transistor, but can sink limited current only through its emitter resistor.

Exercise 2.5. Use a follower with the base driven from a voltage divider to provide a stiff source of +5 volts from an available regulated +15 V supply. Load current (max) = 25 mA. Choose your resistor values so that the output voltage doesn't drop more than 5% under full load.

C. Follower drives switch

Figure 2.16 shows a nice example of an emitter follower rescuing an awkward circuit. We're trying to switch a really bright white LED (the kind you use for "area lighting"), which drops about 3.6 V at its desired 500 mA of forward current. And we've got a 0–3 V digital logic signal available to control the switch. The first circuit uses a single *npn* saturated switch, with a base resistor sized to produce 10 mA of base current, and a 2.5 Ω current-limiting resistor in series with the LED.

This circuit is OK, sort of. But it draws an uncomfortably large current from the control input; and it requires Q_1 to have plenty of current gain at the full load current of 0.5 A. In the second circuit (Figure 2.16B) an emitter follower has come to the rescue, greatly reducing the input current (because of its current gain), and at the same time relaxing the minimum beta requirement of the switch (Q_3). To be fair, we should point out that a low-threshold MOSFET provides an even simpler solution here; we'll tell you how, in Chapters 3 and 12.

D. Important points about followers

Current flow in one direction only. Notice (§2.1.1, rule 4) that in an emitter follower the *npn* transistor can only *source* (as opposed to *sink*) current. For instance, in the loaded circuit shown in Figure 2.17 the output can swing to within a transistor saturation voltage drop of V_{CC} (about +9.9 V), but it cannot go more negative than −5 volts. That is because on the extreme negative swing, the transistor can do no better than to turn off completely, which it does at −4.4 volts input (−5 V output, set by the divider formed by the load and emitter resistors). Further negative swing at the input results in back-biasing of the base–emitter junction, but no further change in output. The output, for a 10 volt amplitude sinewave input, looks as shown in Figure 2.18.

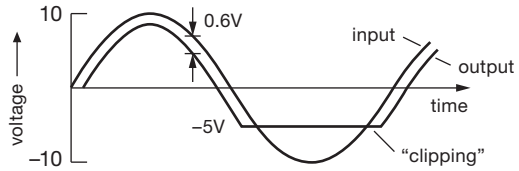


Figure 2.18. Illustrating the asymmetrical current drive capability of the *npn* emitter follower.

Another way to view the problem is to say that the emitter follower has a low value of *small-signal* output impedance, whereas its large-signal output impedance is much higher (as large as R_E). The output impedance changes over from its small-signal value to its large-signal value at the point where the transistor goes out of the active region (in this case at an output voltage of −5 V). To put this point another way, a low value of small-signal output impedance doesn't necessarily mean that the circuit can generate large signal swings into a low resistance load. A low small-signal output impedance doesn't imply a large output current capability.

Possible solutions to this problem involve either decreasing the value of the emitter resistor (with greater power dissipation in resistor and transistor), using a *pnp* transistor (if all signals are negative only), or using a "push–pull" configuration, in which two complementary transistors (one *npn*, one *pnp*) are used (§2.4.1). This sort of problem can also come up when the load that an emitter follower is driving contains voltage or current sources of its own, and thus can force a current in the "wrong" direction. This happens most often with

regulated power supplies (the output is usually an emitter follower) driving a circuit that has other power supplies.

Base-emitter breakdown. Always remember that the base–emitter reverse breakdown voltage for silicon transistors is small, quite often as little as 6 volts. Input swings large enough to take the transistor out of conduction can easily result in breakdown (causing permanent degradation of current gain β) unless a protective diode is added (Figure 2.19).

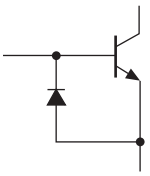


Figure 2.19. A diode prevents base–emitter reverse voltage breakdown.

Gain is slightly less than unity. The voltage gain of an emitter follower is actually slightly less than 1.0, because the base–emitter voltage drop is not really constant, but depends slightly on collector current. You will see how to handle that later in the chapter, when we have the Ebers–Moll equation.

2.2.4 Emitter followers as voltage regulators

The simplest regulated supply of voltage is simply a zener (Figure 2.20). Some current must flow through the zener, so you choose

$$\frac{V_{\text{in}}(\text{min}) - V_{\text{out}}}{R} > I_{\text{out}}(\text{max}).$$

Because V_{in} isn't regulated, you use the lowest value of V_{in} that might occur. Designing for satisfactory operation under the worst combination (here minimum V_{in} and maximum I_{out}) is known as "worst-case" design. In practice, you would also worry about component tolerances, line-voltage limits, etc., designing to accommodate the worst possible combination that would ever occur.

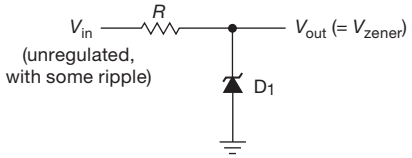


Figure 2.20. Simple zener voltage regulator.

The zener must be able to dissipate

$$P_{\text{zener}} = \left(\frac{V_{\text{in}} - V_{\text{out}}}{R} - I_{\text{out}}\right) V_{\text{zener}}.$$

Again, for worst-case design, you would use $V_{\text{in}}(\text{max})$ and $I_{\text{out}}(\text{min})$.

Exercise 2.6. Design a $+10\dot{\text{V}}$ regulated supply for load currents from 0 to 100 mA; the input voltage is $+20$ to $+25$ V. Allow at least 10 mA zener current under all (worst-case) conditions. What power rating must the zener have?

This simple zener-regulated supply is sometimes used for noncritical circuits or circuits using little supply current. However, it has limited usefulness, for several reasons:

- V_{out} isn't adjustable or settable to a precise value.
- Zener diodes give only moderate ripple rejection and regulation against changes of input or load, owing to their finite dynamic impedance.
- For widely varying load currents a high-power zener is often necessary to handle the dissipation at low load current.[16]

By using an emitter follower to isolate the zener, you get the improved circuit shown in Figure 2.21. Now the situation is much better. Zener current can be made relatively independent of load current, since the transistor base current is small, and far lower zener power dissipation is possible (reduced by as much as a factor of β). The collector resistor R_{C} can be added to protect the transistor from momentary output short circuits by limiting the current, even though it is not essential to the emitter follower function. Choose R_{C} so that the voltage drop across it is less than the drop across R for the highest normal load current (i.e., so that the transistor does not saturate at maximum load).

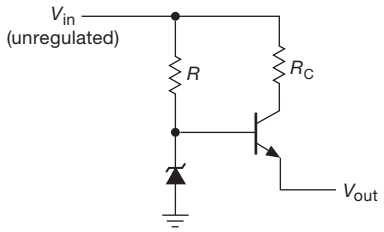


Figure 2.21. Zener regulator with follower, for increased output current. R_{C} protects the transistor by limiting maximum output current.

[16] This is a property shared by all *shunt regulators*, of which the zener is the simplest example.

Exercise 2.7. Design a +10 V supply with the same specifications as in Exercise 2.6. Use a zener and emitter follower. Calculate worst-case dissipation in transistor and zener. What is the percentage change in zener current from the no-load condition to full load? Compare with your previous circuit.

A nice variation of this circuit aims to eliminate the effect of ripple current (through R) on the zener voltage by supplying the zener current from a current source, which is the subject of §2.2.6. An alternative method uses a lowpass filter in the zener bias circuit (Figure 2.22). R is chosen such that the series pair provides sufficient zener current. Then C is chosen large enough so that $RC \gg 1/f_{\text{ripple}}$.[17]

Later you will see better voltage regulators, ones in which you can vary the output easily and continuously by using feedback. They are also better voltage sources, with output impedances measured in milliohms, temperature coefficients of a few parts per million per degree centigrade, and other desirable features.

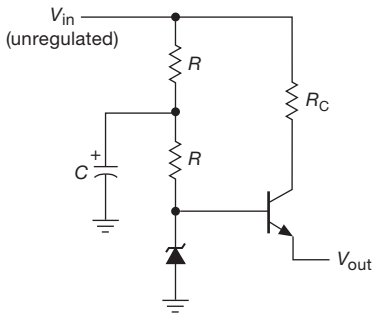


Figure 2.22. Reducing ripple in the zener regulator.

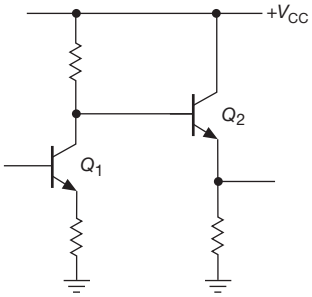


Figure 2.23. Biasing an emitter follower from a previous stage.

2.2.5 Emitter follower biasing

When an emitter follower is driven from a preceding stage in a circuit, it is usually OK to connect its base directly to the previous stage's output, as shown in Figure 2.23.

[17] In a variation of this circuit, the upper resistor is replaced with a diode.

Because the signal on Q_1's collector is always within the range of the power supplies, Q_2's base will be between V_{CC} and ground, and therefore Q_2 is in the active region (neither cut off nor saturated), with its base–emitter diode in conduction and its collector at least a few tenths of a volt more positive than its emitter. Sometimes, though, the input to a follower may not be so conveniently situated with respect to the supply voltages. A typical example is a capacitively coupled (or ac-coupled) signal from some external source (e.g., an audio signal input to a stereo amplifier). In that case the signal's average voltage is zero, and direct coupling to an emitter follower will give an output like that in Figure 2.24.

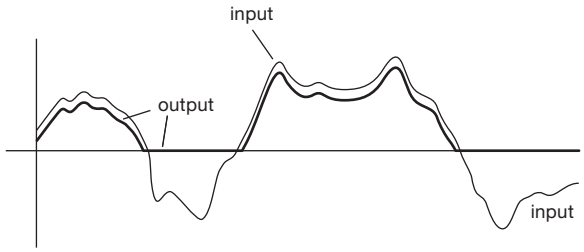


Figure 2.24. A transistor amplifier powered from a single positive supply cannot generate negative voltage swings at the transistor output terminal.

It is necessary to *bias* the follower (in fact, any transistor amplifier) so that collector current flows during the entire signal swing. In this case a voltage divider is the simplest way (Figure 2.25). R_1 and R_2 are chosen to put the base halfway between ground and V_{CC} when there is no input signal, i.e., R_1 and R_2 are approximately equal. The process of selecting the operating voltages in a circuit, in the absence of applied signals, is known as setting the *quiescent point*. In this case, as in most cases, the quiescent point is chosen to allow maximum symmetrical signal swing of the output waveform without *clipping* (flattening of the top or bottom of the waveform). What values should R_1 and R_2 have? Applying our general principle (§1.2.5A, §2.2.3A), we make the impedance of the dc bias source (the impedance looking into the voltage divider) small compared with the load it drives (the dc impedance looking into the base of the follower). In this case,

$$R_1 \| R_2 \ll \beta R_E.$$

This is approximately equivalent to saying that the current flowing in the voltage divider should be large compared with the current drawn by the base.

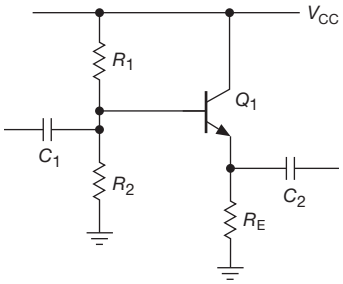


Figure 2.25. An ac-coupled emitter follower. Note base bias voltage divider.

A. Emitter follower design example

As an actual design example, let's make an emitter follower for audio signals (20 Hz to 20 kHz). V_{CC} is $+15$ V, and quiescent current is to be 1 mA.

Step 1. *Choose* V_E. For the largest possible symmetrical swing without clipping, $V_E = 0.5V_{CC}$, or $+7.5$ volts.

Step 2. *Choose* R_E. For a quiescent current of 1 mA, $R_E = 7.5$k.

Step 3. *Choose* R_1 *and* R_2. V_B is $V_E + 0.6$ V, or 8.1 V. This determines the ratio of R_1 to R_2 as 1:1.17. The preceding loading criterion requires that the parallel resistance of R_1 and R_2 be about 75k or less (one-tenth of $7.5\text{k} \times \beta$). Suitable standard values are $R_1 = 130$k, $R_2 = 150$k.

Step 4. *Choose* C_1. The capacitor C_1 forms a highpass filter with the impedance it sees as a load, namely the impedance looking into the base in parallel with the impedance looking into the base voltage divider. If we assume that the load this circuit will drive is large compared with the emitter resistor, then the impedance looking into the base is βR_E, about 750k. The divider looks like 70k. So the capacitor sees a load of about 63k, and it should have a value of at least 0.15 μF so that the 3 dB point will be below the lowest frequency of interest, 20 Hz.

Step 5. *Choose* C_2. The capacitor C_2 forms a highpass filter in combination with the load impedance, which is unknown. However, it is safe to assume that the load impedance won't be smaller than R_E, which gives a value for C_2 of at least 1.0 μF to put the 3 dB point below 20 Hz. Because there are now two cascaded highpass filter sections, the capacitor values should be increased somewhat to prevent excessive attenuation (reduction of signal amplitude, in this case 6 dB) at the lowest frequency of interest. $C_1 = 0.47$ μF and $C_2 = 3.3$ μF might be good choices.[18]

[18] These values may seem curiously "unround." But they are chosen from the widely available EIA "E6" decade values (see Appendix C); and in fact "round-number" values of 0.5 μF and 3.0 μF are harder to find.

From our simple transistor model, the output impedance at the emitter is just $\mathbf{Z}_{out} = R_E \| [(\mathbf{Z}_{in} \| R_1 \| R_2)/\beta]$, where $\mathbf{Z}_{in}$ is the (Thévenin) output resistance of the signal that drives this circuit. So, taking $\beta \approx 100$, a signal source with 10 kΩ output resistance would result in an output impedance (at the emitter) of about 87 Ω. As we'll see later in the chapter (§2.3), there's an effect (the intrinsic emitter impedance, r_e) that adds an additional resistance of $0.025/I_E$ effectively in series with the emitter; so the output impedance here (with 10 kΩ source) would be about 110 Ω.

B. Followers with split supplies

Because signals often are "near ground," it is convenient to use symmetrical positive and negative supplies. This simplifies biasing and eliminates coupling capacitors (Figure 2.26).

Warning: you must always provide a dc path for base bias current, even if it goes only to ground. In this circuit it is assumed that the signal source has a dc path to ground. If not (e.g., if the signal is capacitively coupled), you must provide a resistor to ground (Figure 2.27). R_B could be about one-tenth of βR_E, as before.

Exercise 2.8. Design an emitter follower with ± 15 V supplies to operate over the audio range (20 Hz to 20 kHz). Use 5 mA quiescent current and capacitive input coupling.

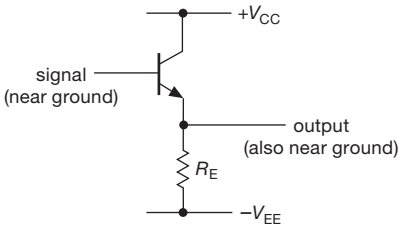


Figure 2.26. A dc-coupled emitter follower with split supply.

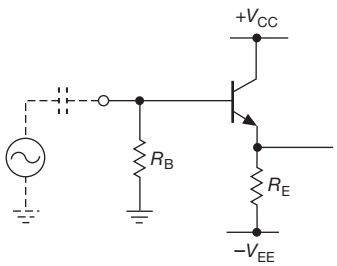


Figure 2.27. Always provide a dc bias path.

C. Bad biasing

You sometimes see sadness-inducing circuits like the disaster shown in Figure 2.28. The designer chose R_B by assuming a particular value for beta (100), estimating the base current, and then hoping for a 7 V drop across R_B. This is a bad design; beta is not a good parameter and will vary considerably. By using voltage biasing with a stiff voltage divider, as in the detailed example presented earlier, the quiescent point is insensitive to variations in transistor beta. For instance, in the previous design example the emitter voltage will increase by only 0.35 V (5%) for a transistor with $\beta = 200$ instead of the nominal $\beta = 100$. And, as with this emitter follower example, it is just as easy to fall into this trap and design bad transistor circuits in the other transistor configurations (notably the common-emitter amplifier, which we will treat later in this chapter).

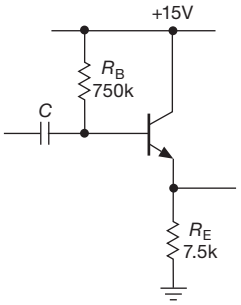


Figure 2.28. Don't do this!

D. Cancelling the offset – I

Wouldn't it be nice if an emitter follower did not cause an offset of the output signal by the $V_{BE} \approx 0.6$ V base–emitter drop? Figure 2.29 shows how to cancel the dc offset, by cascading a *pnp* follower (which has a positive V_{BE} offset) with an *npn* follower (which has a comparable negative V_{BE} offset). Here we've configured the circuit with ± 10 V symmetrical split supplies; and we've used equal-value emitter resistors so that the two transistors have a comparable quiescent current for an input signal near 0 V.

This is a nice trick, useful to know about and often helpful. But the cancellation isn't perfect, for reasons we'll see later in the chapter (V_{BE} depends somewhat on collector current, and on transistor size, §2.3), and again in Chapter 5. But, as we'll see in Chapter 4, it *is* in fact rather easy to make a follower, using *operational amplifiers*, with nearly perfect zero offset (10 μV or less); and as a bonus you get input impedances in the gigaohms (or more), input currents in the nanoamps (or less), and output impedances

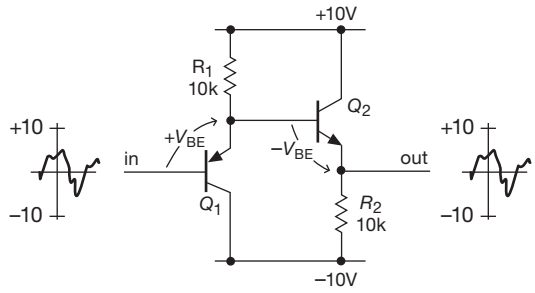


Figure 2.29. Cascading a *pnp* and an *npn* follower produces approximate cancellation of the V_{BE} offsets.

measured in fractions of an ohm. Take a look ahead at Chapter 4.

2.2.6 Current source

Current sources, although often neglected, are as important and as useful as voltage sources. They often provide an excellent way to bias transistors, and they are unequaled as "active loads" for super-gain amplifier stages and as emitter sources for differential amplifiers. Integrators, sawtooth generators, and ramp generators need current sources. They provide wide-voltage-range pullups within amplifier and regulator circuits. And, finally, there are applications in the outside world that require constant current sources, e.g., electrophoresis or electrochemistry.

A. Resistor plus voltage source

The simplest approximation to a current source is shown in Figure 2.30. As long as $R_{load} \ll R$ (in other words, $V_{load} \ll V$), the current is nearly constant and is approximately

$$I \approx V/R.$$

The load doesn't have to be resistive. A capacitor will charge at a constant rate, as long as $V_{cap} \ll V$; this is just the first part of the exponential charging curve of an *RC*.

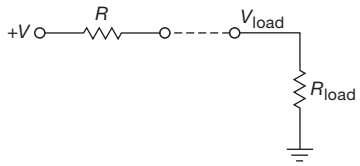


Figure 2.30. Current-source approximation.

There are several drawbacks to a simple resistor current source. To make a good approximation to a current source, you must use large voltages, with lots of power dissipation in the resistor. In addition, the current isn't easily

programmable, i.e., controllable over a large range by means of a voltage somewhere else in the circuit.

Exercise 2.9. If you want a current source constant to 1% over a load voltage range of 0 to +10 volts, how large a voltage source must you use in series with a single resistor?

Exercise 2.10. Suppose you want a 10 mA current in the preceding problem. How much power is dissipated in the series resistor? How much gets to the load?

B. Transistor current source

Happily, it is possible to make a very good current source with a transistor (Figure 2.31). It works like this: applying V_B to the base, with $V_B > 0.6$ V, ensures that the emitter is always conducting:

$$V_E = V_B - 0.6 \text{ volts}.$$

So

$$I_E = V_E/R_E = (V_B - 0.6 \text{ volts})/R_E.$$

But, since $I_E \approx I_C$ for large beta,

$$I_C \approx (V_B - 0.6 \text{ volts})/R_E, \tag{2.5}$$

independent of V_C, as long as the transistor is not saturated ($V_C \gtrsim V_E + 0.2$ volts).

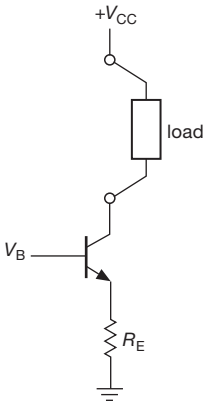


Figure 2.31. Transistor current source: basic concept.

C. Current-source biasing

The base voltage can be provided in a number of ways. A voltage divider is OK, as long as it is stiff enough. As before, the criterion is that its impedance should be much less than the dc impedance looking into the base (βR_E). Or you can use a zener diode (or a two-terminal IC reference like the LM385), biased from V_{CC}, or even a few forward-biased diodes[19] in series from base to the corresponding emitter supply. Figure 2.32 shows some examples. In the last example (Figure 2.32C), a *pnp* transistor *sources* current to a load returned to ground. The other examples (using *npn* transistors) should properly be called current *sinks*, but the usual practice is to refer to them all loosely as "current sources."[20] In the first circuit, the voltage-divider impedance of ∼1.3k is stiff compared with the impedance looking into the base of about 100k (for $\beta = 100$), so any changes in beta with collector voltage will not much affect the output current by causing the base voltage to change. In the other two circuits the biasing resistors are chosen to provide several milliamps to bring the diodes into conduction.

D. Compliance

A current source can provide constant current to the load only over some finite range of load voltage. To do otherwise would be equivalent to providing infinite power. The output voltage range over which a current source behaves well is called its output *compliance*. For the preceding transistor current sources, the compliance is set by the requirement that the transistors stay in the active region. Thus in the first circuit the voltage at the collector can go down until the transistor is almost in saturation, perhaps +1.1 V at the collector. The second circuit, with its higher emitter voltage, can sink current down to a collector voltage of about +5.1 V.

In all cases the collector voltage can range from a value near saturation all the way up to the supply voltage. For example, the last circuit can source current to the load for any voltage between zero and about +8.6 V across the load. In fact, the load might even contain batteries or power supplies of its own, which could carry the collector beyond the supply voltage (Figure 2.32A,B) or below ground (Figure 2.32C). That's OK, but you must watch out for transistor breakdown (V_{CE} must not exceed BV_{CEO}, the specified collector–emitter breakdown voltage) and also for excessive power dissipation (set by $I_C V_{CE}$). As you will see in §§3.5.1B, 3.6.4C, and 9.4.2, there is an additional safe-operating-area constraint on power transistors.

Exercise 2.11. You have +5 and +15 V regulated supplies available in a circuit. Design a 5 mA *npn* current sink using the +5 V to bias the base. What is the output compliance?

[19] A red LED, with its forward voltage drop of ≈1.6 V, is a convenient substitute for a string of three diodes.

[20] "Sink" and "source" simply refer to the direction of current flow: if a circuit *supplies* (positive) current to a point, it is a *source*, and vice versa.

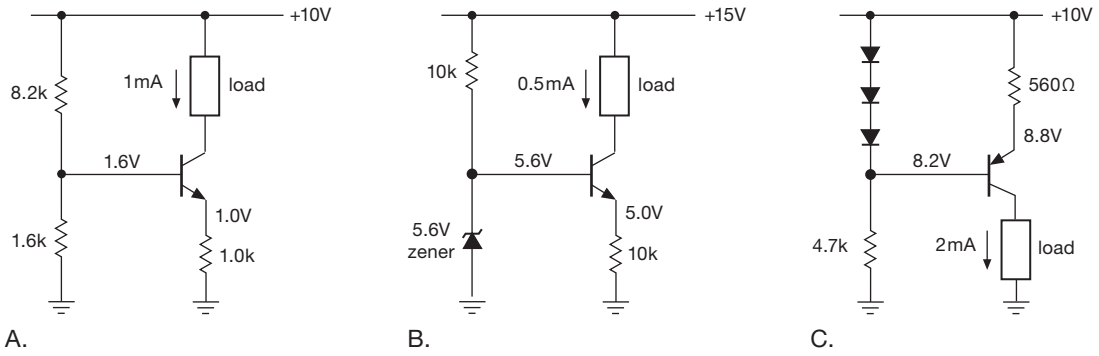


Figure 2.32. Transistor current-source circuits, illustrating three methods of base biasing; *npn* transistors *sink* current, whereas *pnp* transistors *source* current. The circuit in C illustrates a load returned to ground. See also Figure 3.26.

A current source doesn't have to have a fixed voltage at the base. By varying V_B you get a voltage-programmable current source. The input signal swing v_{in} (recall that lowercase symbols mean *variations*) must stay small enough so that the emitter voltage never drops to zero, if the output current is to reflect input-voltage variations smoothly. The result will be a current source with variations in output current proportional to the variations in input voltage, $i_{out} = v_{in}/R_E$. This is the basis of the amplifier we'll see next (§2.2.7).

E. Cancelling the offset – II

It's a minor drawback of these current source circuits that you have to apply a base voltage that is offset by $V_{BE} \approx 0.6\,V$ from the voltage that you want to appear across the emitter resistor; and it is of course the latter that sets the output current. It's the same offset issue as with an emitter follower; and you can use the same trick (§2.2.5D) to bring about approximate cancellation of the offset in situations in which that is a problem.

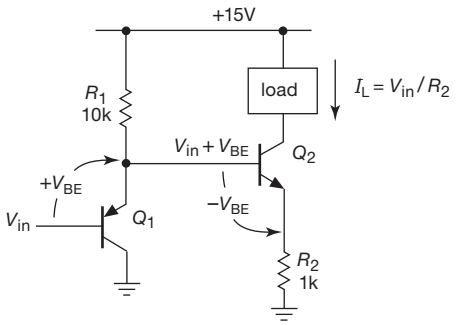


Figure 2.33. Compensating the V_{BE} drop in a current source.

Look at Figure 2.33. It has our standard current-source output stage Q_2, with the current set by the voltage across the emitter resistor: $I_L = V_E/R_2$. So Q_2's base needs to be a V_{BE} higher (the offset), but that's just what the *pnp* input follower does anyway. So, voilà, the voltage at Q_2's emitter winds up being approximately equal to V_{in} that you apply; and so the output current is simply $I_L = V_{in}/R_2$, with no ifs, ands, buts, or V_{BE} offsets. Cute!

We hasten to point out, though, that this is not a particularly accurate cancellation, because the two transistors will in general have different collector currents, and therefore somewhat different base–emitter drops (§2.3). But it's a first-order hack, and a lot better than nothing. And, once again, the magic of operational amplifiers (Chapter 4) will provide a way to make current sources in which the output current is accurately programmed by an input voltage, without that pesky V_{BE} offset.

F. Deficiencies of current sources

These transistor current-source circuits perform well, particularly when compared with a simple resistor biased from a fixed voltage (Figure 2.30). When you look closely, though, you find that they do depart from the ideal at some level of scrutiny – that is, the load current does show some (relatively small) variation with voltage. Another way to say the same thing is that the current source has a finite ($R_{Th} < \infty$) Thévenin equivalent resistance.

We discuss the causes of these deficiencies, and some very clever circuit fixes, later in the chapter, and also in Chapter *2x*.

2.2.7 Common-emitter amplifier

Consider a current source with a resistor as load (Figure 2.34). The collector voltage is

$$V_C = V_{CC} - I_C R_C$$

We could capacitively couple a signal to the base to cause the collector voltage to vary. Consider the example in

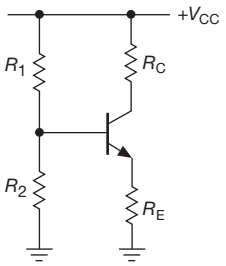


Figure 2.34. Current source driving a resistor as load: an *amplifier*!

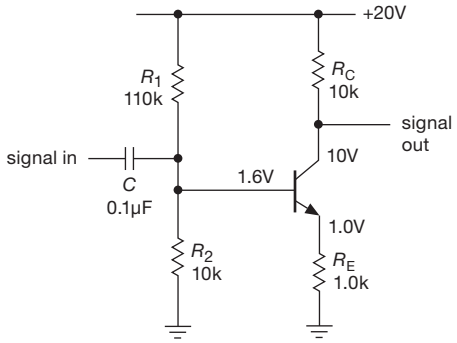


Figure 2.35. An ac common-emitter amplifier with emitter degeneration. Note that the output terminal is the collector rather than the emitter.

Figure 2.35. Blocking capacitor C is chosen so that all frequencies of interest are passed by the highpass filter it forms in combination with the parallel resistance of the base biasing resistors[21]; that is,

$$C \geq \frac{1}{2\pi f(R_1 \| R_2)}.$$

The quiescent collector current is 1.0 mA because of the applied base bias and the 1.0k emitter resistor. That current puts the collector at +10 volts (+20 V, minus 1.0 mA through 10k). Now imagine an applied wiggle in base voltage v_B. The emitter follows with $v_E = v_B$, which causes a wiggle in emitter current

$$i_E = v_E/R_E = v_B/R_E$$

and nearly the same change in collector current (β is large). So the initial wiggle in base voltage finally causes a collector voltage wiggle

$$v_C = -i_C R_C = -v_B(R_C/R_E)$$

Aha! It's a *voltage amplifier*, with a voltage amplification (or "gain") given by

$$\text{gain} = v_{out}/v_{in} = -R_C/R_E \qquad (2.6)$$

In this case the gain is −10,000/1000, or −10. The minus sign means that a positive wiggle at the input gets turned into a negative wiggle (10 times as large) at the output. This is called a *common-emitter amplifier* with emitter degeneration.

A. Input and output impedances of the common-emitter amplifier

We can easily determine the input and output impedances of the amplifier. The input signal sees, in parallel, 110k, 10k, and the impedance looking into the base. The latter is about 100k (β times R_E), so the input impedance (dominated by the 10k) is about 8k. The input coupling capacitor thus forms a highpass filter, with the 3 dB point at 200 Hz. The signal driving the amplifier sees 0.1 μF in series with 8k, which to signals of normal frequencies (well above the 3 dB point) just looks like 8k.

The output impedance is 10k in parallel with the impedance looking into the collector. What is that? Well, remember that if you snip off the collector resistor, you're simply looking into a current source. The collector impedance is very large (measured in megohms), and so the output impedance is just the value of the collector resistor, 10k. It is worth remembering that the impedance looking into a transistor's collector is high, whereas the impedance looking into the emitter is low (as in the emitter follower). Although the output impedance of a common-emitter amplifier will be dominated by the collector load resistor, the output impedance of an emitter follower will not be dominated by the emitter load resistor, but rather by the impedance looking into the emitter.

2.2.8 Unity-gain phase splitter

Sometimes it is useful to generate a signal and its inverse, i.e., two signals 180° out of phase. That's easy to do – just use an emitter-degenerated amplifier with a gain of −1 (Figure 2.36). The quiescent collector voltage is set to $0.75V_{CC}$, rather than the usual $0.5V_{CC}$, in order to achieve the same result – maximum symmetrical output swing without clipping at either output. The collector can swing from $0.5V_{CC}$ to V_{CC}, whereas the emitter can swing from ground to $0.5V_{CC}$.

Note that the phase-splitter outputs must be loaded with equal (or very high) impedances at the two outputs to maintain gain symmetry.

[21] The impedance looking into the base itself will usually be much larger because of the way the base resistors are chosen, and it can generally be ignored.

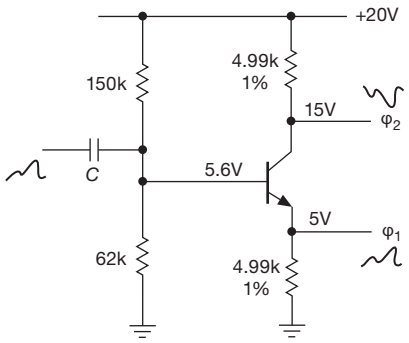


Figure 2.36. Unity-gain phase splitter.

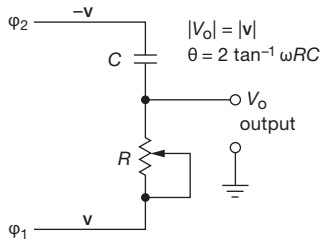


Figure 2.37. Constant-amplitude phase shifter.

A. Phase shifter

A nice use of the phase splitter is shown in Figure 2.37. This circuit gives (for a sinewave input) an output sinewave of adjustable phase (from zero to 180°) and with constant amplitude. It can be best understood with a phasor diagram of voltages (§1.7.12); with the input signal represented by a unit vector along the real axis, the signals look as shown in Figure 2.38.

Signal vectors $\mathbf{v}_R$ and $\mathbf{v}_C$ must be at right angles, and they must add to form a vector of constant length along the real axis. There is a theorem from geometry that says that the locus of such points is a circle. So the resultant vector (the output voltage) always has unit length, i.e., the same amplitude as the input, and its phase can vary from nearly zero to nearly 180° relative to the input wave as *R* is varied from nearly zero to a value much larger than X_C at the operating frequency. However, note that the phase shift depends on the frequency of the input signal for a given setting of the potentiometer *R*. It is worth noting that a simple *RC* highpass (or lowpass) network could also be used as an adjustable phase shifter. However, its output amplitude would vary over an enormous range as the phase shift was adjusted.

An additional concern here is the ability of the phase-splitter circuit to drive the *RC* phase shifter as a load. Ideally, the load should present an impedance that is large compared with the collector and emitter resistors. As a result, this circuit is of limited utility where a wide range of phase shifts is required. You will see improved phase-splitter techniques in Chapter 4, where we use op-amps as impedance buffers, and in Chapter 7, where a cascade of several phase-shifter sections generates a set of "quadrature" signals that extends the phase-shifting range to a full 0° to 360°.

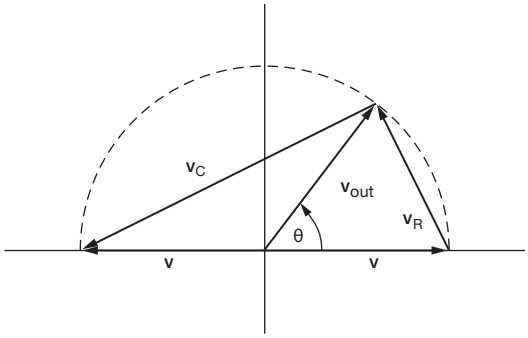


Figure 2.38. Phasor diagram for phase shifter, for which $\theta = 2\arctan(\omega RC)$.

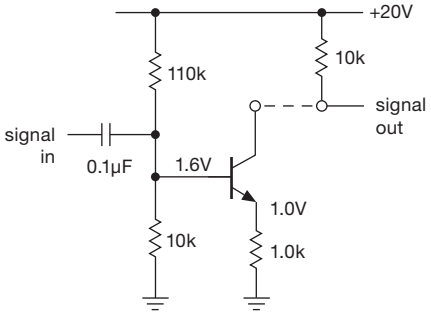


Figure 2.39. The common-emitter amplifier is a transconductance stage driving a (resistive) load.

2.2.9 Transconductance

In the preceding section we figured out the operation of the emitter-degenerated amplifier by (a) imagining an applied base voltage swing and seeing that the emitter voltage had the same swing, then (b) calculating the emitter current swing; then, ignoring the small base current contribution, we got the collector current swing and thus (c) the collector voltage swing. The voltage gain was then simply the ratio of collector (output) voltage swing to base (input) voltage swing.

There's another way to think about this kind of amplifier. Imagine breaking it apart, as in Figure 2.39. The first

part is a voltage-controlled current source, with quiescent current of 1.0 mA and gain of −1 mA/V. Gain means the ratio of output to input; in this case the gain has units of current/voltage, or 1/resistance. The inverse of resistance is called *conductance*.[22] An amplifier whose gain has units of conductance is called a *transconductance* amplifier; the ratio of changes $\Delta I_{\rm out}/\Delta V_{\rm in}$ (usually written with small-signal changes indicated in lowercase: $i_{\rm out}/v_{\rm in}$) is called the transconductance, g_m:

$$g_m = \frac{\Delta I_{\rm out}}{\Delta V_{\rm in}} = \frac{i_{\rm out}}{v_{\rm in}}. \tag{2.7}$$

Think of the first part of the circuit as a transconductance amplifier, i.e., a voltage-to-current amplifier with transconductance g_m (gain) of −1 mA/V (1000 μS, or 1 mS, which is just $1/R_{\rm E}$). The second part of the circuit is the load resistor, an "amplifier" that converts current to voltage. This resistor could be called a *transresistance* converter, and its gain (r_m) has units of voltage/current, or resistance. In this case its quiescent voltage is $V_{\rm CC}$, and its gain (transresistance) is 10 V/mA (10kΩ), which is just $R_{\rm C}$. Connecting the two parts together gives you a voltage amplifier. You get the overall gain by multiplying the two gains. In this case the voltage gain $G_V = g_m R_{\rm C} = -R_{\rm C}/R_{\rm E}$, or −10, a unitless number equal to the ratio (output voltage change)/(input voltage change).

This is a useful way to think about an amplifier, because you can analyze performance of the sections independently. For example, you can analyze the transconductance part of the amplifier by evaluating g_m for different circuit configurations or even different devices, such as field-effect transistors FETs. Then you can analyze the transresistance (or load) part by considering gain versus voltage swing tradeoffs. If you are interested in the overall voltage gain, it is given by $G_{\rm V} = g_m r_m$, where r_m is the transresistance of the load. Ultimately the substitution of an active load (current source), with its extremely high transresistance, can yield single-stage voltage gains of 10,000 or more. The *cascode* configuration, which we will discuss later, is another example easily understood with this approach.

In Chapter 4, which deals with operational amplifiers, you will see further examples of amplifiers with voltages or currents as inputs or outputs: voltage amplifiers (voltage to voltage), current amplifiers (current to current), and transresistance amplifiers (current to voltage).

[22] The inverse of reactance is *susceptance* (and the inverse of impedance is *admittance*), and has a special unit, the *siemens* ("S," not to be confused with lowercase "s," which means seconds), which used to be called the *mho* (ohm spelled backward, symbol "℧").

A. Turning up the gain: limitations of the simple model

The voltage gain of the emitter-degenerated amplifier is $-R_{\rm C}/R_{\rm E}$, according to our model. What happens as $R_{\rm E}$ is reduced toward zero? The equation predicts that the gain will rise without limit. But if we made actual measurements of the preceding circuit, keeping the quiescent current constant at 1 mA, we would find that the gain would level off at about 400 when $R_{\rm E}$ is zero, i.e., with the emitter grounded. We would also find that the amplifier would become significantly nonlinear (the output would not be a faithful replica of the input), the input impedance would become small and nonlinear, and the biasing would become critical and unstable with temperature. Clearly our transistor model is incomplete and needs to be modified to handle this circuit situation, as well as others we will talk about presently. Our fixed-up model, which we will call the transconductance model, will be accurate enough for the remainder of the book.

B. Recap: the "four topologies"

Before jumping into the complexity just ahead, let's remind ourselves of the four transistor circuits we've seen, namely the switch, emitter follower, current source, and common-emitter amplifier. We've drawn these very schematically in Figure 2.40, omitting details like biasing, and even the polarity of transistor (i.e., *npn* or *pnp*). For completeness we've included also a fifth circuit, the *common-base amplifier*, which we'll meet soon enough (§2.4.5B).

2.3 Ebers–Moll model applied to basic transistor circuits

We've enjoyed seeing some nice feats that can be accomplished with the simplest BJT model – switch, follower, current source, amplifier – but we've run up against some serious limitations (hey, would you believe, *infinite* gain?!). Now it's time to go a level deeper, to address these limitations. The material that follows will suffice for our purposes. And – good news – for many BJT applications the simple model you've already seen is completely adequate.

2.3.1 Improved transistor model: transconductance amplifier

The important change is in rule 4 (§2.1.1), where we said earlier that $I_{\rm C} = \beta I_{\rm B}$. We thought of the transistor as a current amplifier whose input circuit behaved like a diode. That's roughly correct, and for some applications it's good enough. But to understand differential amplifiers,

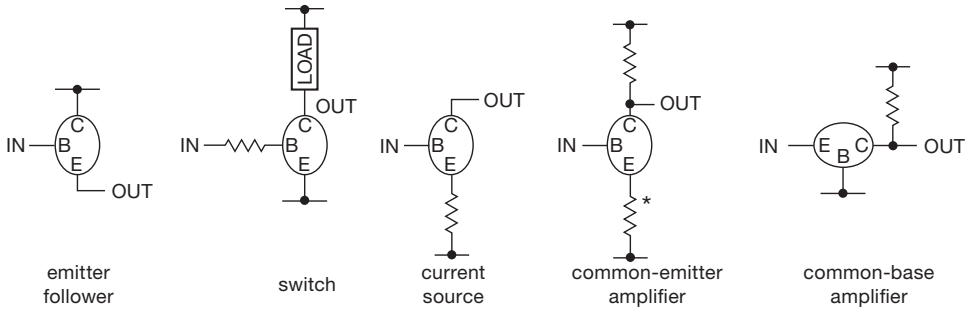


Figure 2.40. Five basic transistor circuits. Fixed voltages (power supplies or ground) are indicated by connections to horizontal line segments. For the switch, the load may be a resistor, to produce a full-swing voltage output; for the common-emitter amplifier, the emitter resistor may be bypassed or omitted altogether.

logarithmic converters, temperature compensation, and other important applications, you must think of the transistor as a *transconductance* device – collector current is determined by base-to-emitter *voltage*.

Here's the modified rule 4.

4. Transconductance amplifier When rules 1–3 (§2.1.1) are obeyed, I_C is related to V_{BE} by[23]

$$I_C = I_S(T)\left(e^{V_{BE}/V_T} - 1\right), \tag{2.8}$$

or, equivalently,

$$V_{BE} = \frac{kT}{q}\log_e\left(\frac{I_C}{I_S(T)} + 1\right), \tag{2.9}$$

where

$$V_T = kT/q = 25.3\,\text{mV} \tag{2.10}$$

at room temperature (68°F, 20°C), q is the electron charge (1.60×10^{-19} coulombs), k is Boltzmann's constant (1.38×10^{-23} joules/K, sometimes written k_B), T is the absolute temperature in degrees Kelvin (K=°C+273.16), and $I_S(T)$ is the *saturation current* of the particular transistor (which depends strongly on temperature, T, as we'll see shortly). Then the base current, which also depends on V_{BE}, can be approximated by

$$I_B = I_C/\beta,$$

where the "constant" β is typically in the range 20 to 1000, but depends on transistor type, I_C, V_{CE}, and temperature. $I_S(T)$ approximates the reverse leakage current (roughly 10^{-15} A for a small-signal transistor like the 2N3904). In the active region $I_C \gg I_S$, and therefore the -1 term can be neglected in comparison with the exponential:

$$I_C \approx I_S(T)e^{V_{BE}/V_T}. \tag{2.11}$$

The equation for I_C is known as the Ebers–Moll equation.[24] It also describes approximately the current versus voltage for a diode, if V_T is multiplied by a correction factor m between 1 and 2. For transistors it is important to realize that the collector current is accurately determined by the base–emitter voltage, rather than by the base current (the base current is then roughly determined by β), and that this exponential law is accurate over an enormous range of currents, typically from nanoamps to milliamps. Figure 2.41 makes the point graphically.[25] If you measure the base current at various collector currents, you will get a graph of β versus I_C like that in Figure 2.42. Transistor beta versus collector current is discussed further in Chapter 2*x*.

Although the Ebers–Moll equation tells us that the base–emitter voltage "programs" the collector current, this property is not easy to use in practice (biasing a transistor by applying a base voltage) because of the large temperature coefficient of base–emitter voltage. You will see later how the Ebers–Moll equation provides insight and solutions to this problem.

2.3.2 Consequences of the Ebers–Moll model: rules of thumb for transistor design

From the Ebers–Moll equation (2.8) we get these simple (but handy) "ratio rules" for collector current: $I_{C2}/I_{C1} = \exp(\Delta V_{BE}/V_T)$ and $\Delta V_{BE} = V_T \log_e(I_{C2}/I_{C1})$. We

[23] We indicate the important temperature dependence of I_S by explicitly showing it in functional form – "$I_S(T)$".

[24] J. J. Ebers & J. L. Moll, "Large-signal behavior of junction transistors," *Proc. IRE* **42**, 1761 (1954).

[25] This is sometimes called a Gummel plot.

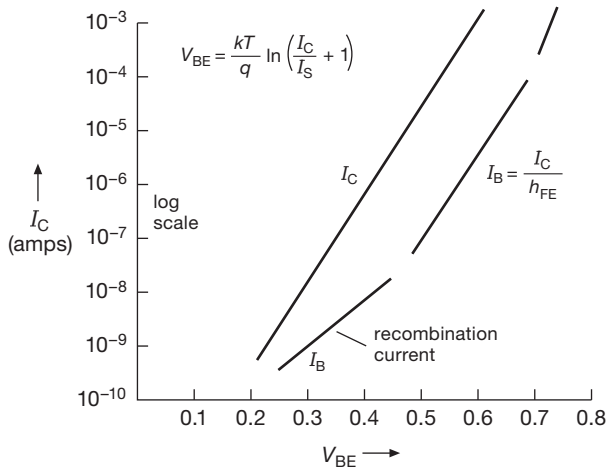


Figure 2.41. Transistor base and collector currents as functions of base-to-emitter voltage V_{BE}.

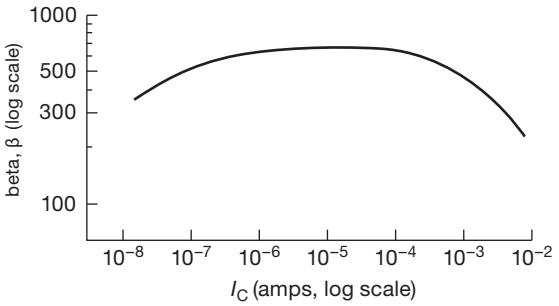


Figure 2.42. Typical transistor current gain (β) versus collector current.

also get the following important quantities we will be using often in circuit design.

A. The steepness of the diode curve.

How much do we need to increase V_{BE} to increase I_C by a factor of 10? From the Ebers–Moll equation, that's just $V_T \log_e 10$, or 58.2 mV at room temperature. We like to remember this as *base–emitter voltage increases approximately 60 mV per decade of collector current*. (Two other formulations: collector current doubles for each 18 mV increase in base–emitter voltage; collector current increases 4% per millivolt increase in base–emitter voltage.) Equivalently, $I_C = I_{C0} e^{\Delta V/25}$, where ΔV is in millivolts.[26]

B. The small-signal impedance looking into the emitter, r_e, for the base held at a fixed voltage.

Taking the derivative of V_{BE} with respect to I_C, you get

$$r_e = V_T/I_C = 25/I_C \text{ ohms}, \tag{2.12}$$

where I_C is in milliamps.[27] The numerical value $25/I_C$ is for room temperature. This *intrinsic* emitter resistance, r_e, acts as if it is in series with the emitter in all transistor circuits. It limits the gain of a grounded-emitter amplifier, causes an emitter follower to have a voltage gain of slightly less than unity, and prevents the output impedance of an emitter follower from reaching zero. Note that the transconductance[28] of a grounded emitter amplifier is just

$$g_m = I_C/V_T = 1/r_e \ \ (= 40 I_C \text{ at room temp}). \tag{2.13}$$

C. The temperature dependence of V_{BE}.

A glance at the Ebers–Moll equation suggests that V_{BE} (at constant I_C) has a positive temperature coefficient because of the multiplying factor of T in V_T. However, the strong temperature dependence of $I_S(T)$ more than compensates for that term, such that V_{BE} (at constant I_C) *decreases* about 2.1 mV/°C. It is roughly proportional to $1/T_{abs}$, where T_{abs} is the absolute temperature. Sometimes it's useful to cast this instead in terms of the temperature dependence of I_C (at constant V_{BE}): I_C *increases* about 9%/°C; it doubles for an 8°C rise.

There is one additional quantity we will need on occasion, although it is not derivable from the Ebers–Moll equation. It is known as the Early effect,[29] and it sets important limits on current-source and amplifier performance.

D. Early effect.

V_{BE} (at constant I_C) varies slightly with changing V_{CE}. This effect is caused by the variation of effective base width as V_{CE} changes, and it is given, approximately, by

$$\Delta V_{BE} = -\eta \Delta V_{CE}, \tag{2.14}$$

where $\eta \approx 10^{-4}$–10^{-5}. (As an example, the *npn* 2N5088 has $\eta = 1.3 \times 10^{-4}$, thus a 1.3 mV change of V_{BE} to maintain constant collector current when V_{CE} changes by 10 V.)

[26] The "25" in this and the following discussion is more precisely 25.3 mV, the value of $k_B T/q$ at room temperature. It's proportional to absolute temperature – engineers like to say "PTAT," pronounced *pee'-tat*. This has interesting (and useful) consequences, for example the opportunity to make a "silicon thermometer." We'll see more of this in Chapter 2*x*, and again in Chapter 9.

[27] We like to remember the fact that $r_e = 25\,\Omega$ at a collector current of 1 mA. Then we just scale inversely for other currents: thus $r_e = 2.5\Omega$ at $I_C = 10$ mA, etc.

[28] At the next level of sophistication we'll see that, since the quantity r_e is proportional to absolute temperature, a grounded emitter amplifier whose collector current is PTAT has transconductance (and gain) independent of temperature. More in Chapter 2*x*.

[29] J. M. Early, "Effects of space-charge layer widening in junction transistors," *Proc. IRE* **40**, 1401 (1952). James Early died in 2004.

This is often described instead as a linear increase of collector current with increasing collector voltage when V_{BE} is held constant; you see it expressed as

$$I_C = I_{C0}\left(1 + \frac{V_{CE}}{V_A}\right), \tag{2.15}$$

where V_A (typically 50–500 V) is known as the Early voltage.[30] This is shown graphically in Figure 2.59 in §2.3.7A. A low Early voltage indicates a low collector output resistance; *pnp* transistors tend to have low V_A, see measured values in Table 8.1. We treat the Early effect in more detail in Chapter *2x*.[31]

These are the essential quantities we need. With them we will be able to handle most problems of transistor circuit design, and we will have little need to refer to the Ebers–Moll equation itself.[32]

2.3.3 The emitter follower revisited

Before looking again at the common-emitter amplifier with the benefit of our new transistor model, let's take a quick look at the humble emitter follower. The Ebers–Moll model predicts that an emitter follower should have nonzero output impedance, even when driven by a voltage source, because of finite r_e (item B in the above list). The same effect also produces a voltage gain slightly less than unity, because r_e forms a voltage divider with the load resistor.

These effects are easy to calculate. With fixed base voltage, the impedance looking back into the emitter is just $R_{out} = dV_{BE}/dI_E$; but $I_E \approx I_C$, so $R_{out} \approx r_e$, the intrinsic emitter resistance [recall $r_e = 25/I_C(\text{mA})$]. For example, in Figure 2.43A, the load sees a driving impedance of $r_e = 25\,\Omega$, because $I_C = 1$ mA. (This is paralleled by the emitter resistor R_E, if used; but in practice R_E will always be much larger than r_e.) Figure 2.43B shows a more typical situation, with finite source resistance R_S (for simplicity we've omitted the obligatory biasing components – base divider and blocking capacitor – which are shown in Figure 2.43C). In this case the emitter follower's output impedance is just r_e in series with $R_s/(\beta+1)$ (again paralleled by an unimportant R_E, if present). For example, if $R_s = 1$k and $I_C = 1$ mA, $R_{out} = 35\,\Omega$ (assuming $\beta = 100$). It is easy to show that the intrinsic emitter r_e also figures into an emitter follower's *input* impedance, just as if it were in series with the load (actually, parallel combination of load resistor and emitter resistor). In other words, for the emitter follower circuit the effect of the Ebers–Moll model is simply to add a series emitter resistance r_e to our earlier results.[33]

The voltage gain of an emitter follower is slightly less than unity, owing to the voltage divider produced by r_e and the load. It is simple to calculate, because the output is at the junction of r_e and R_{load}: $G_V = v_{out}/v_{in} = R_L/(r_e + R_L)$. Thus, for example, a follower running at 1 mA quiescent current, with 1k load, has a voltage gain of 0.976. Engineers sometimes like to write the gain in terms of the transconductance, to put it in a form that holds for FETs also (see §3.2.3A); in that case (using $g_m = 1/r_e$) you get $G_V = R_L g_m/(1 + R_L g_m)$.

2.3.4 The common-emitter amplifier revisited

Previously we got wrong answers for the voltage gain of the common-emitter amplifier with emitter resistor (sometimes called emitter degeneration) when we set the emitter resistor equal to zero; recall that our wrong answer was $G_V = -R_C/R_E = \infty$!

The problem is that the transistor has $25/I_C(\text{mA})$ ohms of built-in (intrinsic) emitter resistance r_e that must be added to the actual external emitter resistor. This resistance is significant only when small emitter resistors (or none at all) are used.[34] So, for instance, the amplifier we considered previously will have a voltage gain of $-10\text{k}/r_e$, or -400, when the external emitter resistor is zero. The input impedance is not zero, as we would have predicted earlier (βR_E); it is approximately βr_e, or in this case (1 mA quiescent current) about 2.5k.[35]

[30] The connection between Early voltage and η is $\eta = V_T/(V_A + V_{CE}) \approx V_T/V_A$; see Chapter *2x*.

[31] Previewing some of the results there, the Early effect (a) determines a transistor's collector output resistance $r_o = V_A/I_C$; (b) sets a limit on single-stage voltage gain; and (c) limits the output resistance of a current source. Other things being equal, *pnp* transistors tend to have low Early voltages, as do transistors with high beta; high-voltage transistors usually have high Early voltages, along with low beta. These trends can be seen in the measured Early voltages listed in Table 8.1.

[32] The computer circuit-analysis program SPICE includes accurate transistor simulation with the Ebers–Moll formulas and Gummel–Poon charge models. It's a lot of fun to "wire up" circuits on your computer screen and set them running with SPICE. For more detail see the application of SPICE to BJT amplifier distortion in Chapter *2x*.

[33] There's more, if you look deeper: at high frequencies (above f_T/β) the effective current gain drops inversely with frequency; so you get a linearly rising output impedance from an emitter follower that is driven with low R_s. That is, it looks like an inductance, and a capacitive load can cause ringing or even oscillation; these effects are treated in Chapter *2x*.

[34] Or, equivalently, when the emitter resistor is bypassed with a capacitor whose impedance at signal frequencies is comparable with, or less than, r_e.

[35] These estimates of gain and input impedance are reasonably good, as long as we stay away from operation at very high frequencies or

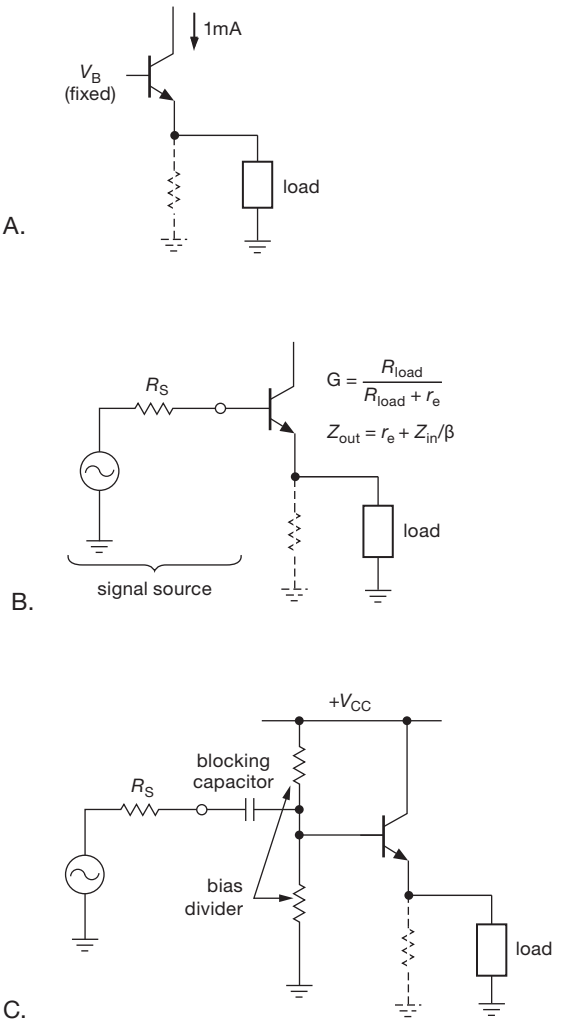


Figure 2.43. Output impedance of emitter followers (see text).

The terms "grounded emitter" and "common emitter" are sometimes used interchangeably, and they can be confusing. We will use the phrase "grounded-emitter amplifier" to mean a common-emitter amplifier with $R_E = 0$ (or equivalent bypassing). A common-emitter amplifier stage may have an emitter resistor; what matters is that the emitter circuit is common to the input circuit and the output circuit.

from circuits in which the collector load resistor is replaced with a current source "active load" ($R_C \to \infty$). The ultimate voltage gain of a grounded-emitter amplifier, in the latter situation, is limited by the Early effect; this is discussed in more detail Chapter 2*x*.

A. Shortcomings of the single-stage grounded emitter amplifier

The extra voltage gain you get by using $R_E = 0$ comes at the expense of other properties of the amplifier. In fact, the grounded-emitter amplifier, in spite of its popularity in textbooks, should be avoided except in circuits with overall negative feedback. In order to see why, consider Figure 2.44.

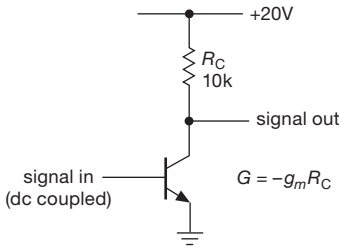


Figure 2.44. Common-emitter amplifier without emitter degeneration.

1. Nonlinearity. The voltage gain is $G = -g_m R_C = -R_C/r_e = -R_C I_C(\text{mA})/25$, so for a quiescent current of 1 mA, the gain is -400. But I_C varies as the output signal varies. For this example, the gain will vary from -800 ($V_{out} = 0$, $I_C = 2$ mA) down to zero ($V_{out} = V_{CC}$, $I_C = 0$). For a triangle-wave input, the output will look as in Figure 2.45. The amplifier has lots of distortion, or poor linearity. The grounded-emitter amplifier without feedback is useful only for small-signal swings about the quiescent point. By contrast, the emitter-degenerated amplifier has a gain almost entirely independent of collector current, as long as $R_E \gg r_e$, and can be used for undistorted amplification even with large-signal swings.

It's easy to estimate the distortion, both with and without an external emitter resistor. With a *grounded* emitter, the incremental (small-signal) gain is $G_V = -R_C/r_e = -I_C R_C/V_T = -V_{drop}/V_T$, where V_{drop} is the instantaneous voltage drop across the collector resistor. Because the gain is proportional to the drop across the collector resistor, the nonlinearity (fractional change of gain with swing) equals the ratio of instantaneous swing to average quiescent drop across the collector resistor: $\Delta G/G \approx \Delta V_{out}/V_{drop}$, where V_{drop} is the average, or quiescent, voltage drop across the collector resistor R_C. Because this represents the extreme variation of gain (i.e., at the peaks of the swing), the overall waveform "distortion" (usually stated as the amplitude of the residual waveform after subtraction of the strictly linear component) will be smaller by roughly a factor of 3. Note that the distortion depends on only the ratio of swing to quiescent drop, and not directly on the operating current, etc.

As an example, in a grounded emitter amplifier powered from +10 V, biased to half the supply (i.e., $V_{\rm drop} = 5$ V), we measured a distortion of 0.7% at 0.1 V output sinewave amplitude and 6.6% at 1 V amplitude; these values are in good agreement with the predicted values. Compare this with the situation with an added external emitter resistor $R_{\rm E}$, in which the voltage gain becomes $G_{\rm V} = -R_{\rm C}/(r_{\rm e} + R_{\rm E}) = -I_{\rm C}R_{\rm C}/(V_{\rm T} + I_{\rm C}R_{\rm E})$. Only the first term in the denominator contributes distortion, so the distortion is reduced by the ratio of $r_{\rm e}$ to the total effective emitter resistance: the nonlinearity becomes $\Delta G/G \approx (\Delta V_{\rm out}/V_{\rm drop})[r_{\rm e}/(r_{\rm e} + R_{\rm E})] = (\Delta V_{\rm out}/V_{\rm drop})[V_{\rm T}/(V_{\rm T} + I_{\rm E}R_{\rm E})]$; the second term is the factor by which the distortion is reduced. When we added an emitter resistor, chosen to drop 0.25 V at the quiescent current – which by this estimate should reduce the nonlinearity by a factor of 10 – the measured distortion of the previous amplifier dropped to 0.08% and 0.74% for 0.1 V and 1 V output amplitudes, respectively. Once again, these measurement agree well with our prediction.

Exercise 2.12. Calculate the predicted distortion for these two amplifiers at the two output levels that were measured.

As we remarked, the nonlinearity of a common-emitter amplifier, when driven by a triangle wave, takes the form of the asymmetric "barn roof" distortion sketched in Figure 2.45.[36] For comparison we took a real-life 'scope (oscilloscope) trace of a grounded emitter amplifier (Figure 2.46); we used a 2N3904 with a 5k collector resistor to a +10 V supply, biased (carefully!) to half the supply. With a ruler we estimated the incremental gain at output voltages of +5 V (halfway to V_+) and at +7.5 V, as shown, where the collector current is 1 mA and 0.5 mA, respectively. The gain values are in pretty good agreement with the predictions ($G = R_{\rm C}/r_{\rm e} = I_{\rm C}({\rm mA})R_{\rm C}/25\Omega$) of $G = -200$ and $G = -100$, respectively. By comparison, Figure 2.47 shows what happened when we added a 225 Ω emitter resistor: the gain is reduced by a factor of 10 at the quiescent point ($G = R_{\rm C}/(R_{\rm E} + r_{\rm e}) \approx R_{\rm C}/250\,\Omega$), but with much improved linearity (because changes in $r_{\rm e}$ contribute little to the overall resistance in the denominator, which is now dominated by the fixed 225 Ω external emitter resistor).

For sinusoidal input, the output contains all harmonics of the fundamental wave. Later in the chapter we'll see how to make differential amplifiers with a pair of transistors; for these the residual distortion is symmetric, and contains only the odd harmonics. And in Chapter *2x* we'll see some very clever methods for cancelling distortion in differential

[36] Because the gain (i.e., the slope of $V_{\rm out}$ versus $V_{\rm in}$) is proportional to the distance from the $V_{\rm CC}$ line, the shape of the curve is in fact an exponential.

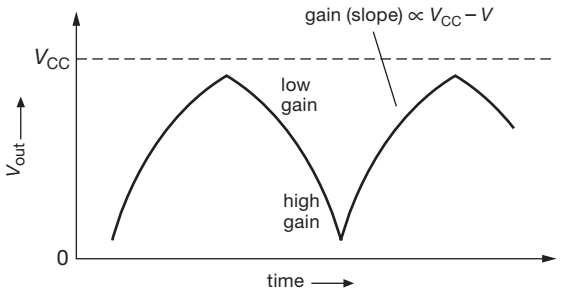


Figure 2.45. Nonlinear output waveform from grounded-emitter amplifier.

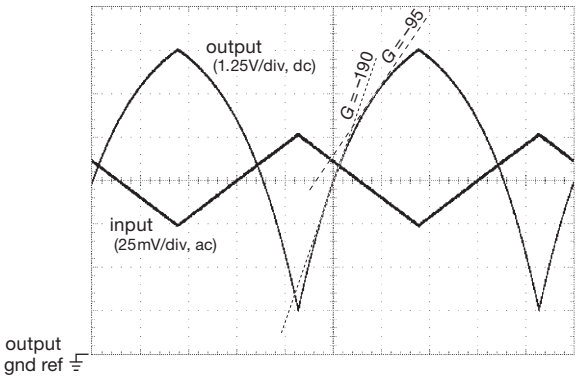


Figure 2.46. *Real life!* The grounded-emitter amplifier of Figure 2.44, with $R_{\rm C} = 5$k, $V_+ = +10$ V, and a 1 kHz triangle wave input. Top and bottom of the screen are +10 V and ground for the dc-coupled output trace (note sensitive scale for the ac-coupled input signal). Gain estimates (tangent lines) are at $V_{\rm out}$ values of $0.5V_+$ and $0.75V_+$. Horizontal: 0.2 ms/div.

amplifiers, along with the use of SPICE simulation software for rapid analysis and circuit iteration. Finally, to set things in perspective, we should add that any amplifier's residual distortion can be reduced dramatically by use of *negative feedback*. We'll introduce feedback later in this chapter (§2.5), after you've gained familiarity with common transistor circuits. Feedback will finally take center stage when we get to *operational amplifiers* in Chapter 4.

2. Input impedance. The input impedance is roughly $\mathbf{Z}_{\rm in} = \beta r_{\rm e} = 25\beta/I_{\rm C}({\rm mA})$ ohms. Once again, $I_{\rm C}$ varies over the signal swing, giving a varying input impedance. Unless the signal source driving the base has low impedance, you will wind up with nonlinearity because of the nonlinear (variable) voltage divider formed from the signal source and the amplifier's input impedance. By contrast, the input impedance of an emitter-degenerated amplifier is nearly constant, and high.

3. Biasing. The grounded emitter amplifier is difficult to

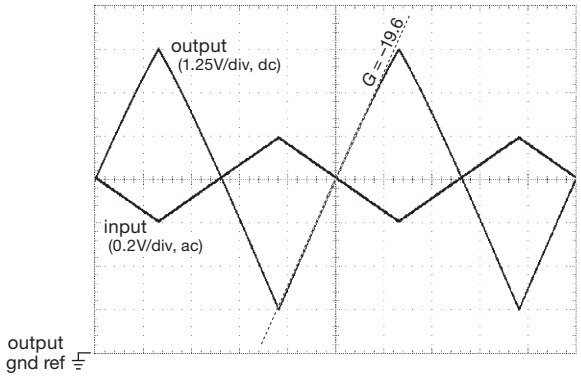


Figure 2.47. Adding a 225 Ω emitter resistor improves the linearity dramatically at the expense of gain (which drops by a factor of 10 at the quiescent point). Horizontal: 0.2 ms/div.

bias. It might be tempting just to apply a voltage (from a voltage divider) that gives the right quiescent current according to the Ebers–Moll equation. That won't work, because of the temperature dependence of V_{BE} (at fixed I_C), which varies about 2.1 mV/°C [it actually decreases with increasing T because of the variation of $I_S(T)$ with temperature; as a result, V_{BE} is roughly proportional to $1/T$, the absolute temperature]. This means that the collector current (for fixed V_{BE}) will increase by a factor of 10 for a 30°C rise in temperature (which corresponds to a 60 mV change in V_{BE}), or about 9%/°C. Such unstable biasing is useless, because even rather small changes in temperature will cause the amplifier to saturate. For example, a grounded emitter stage biased with the collector at half the supply voltage will go into saturation if the temperature rises by 8°C.

Exercise 2.13. Verify that an 8°C rise in ambient temperature will cause a base-voltage-biased grounded emitter stage to saturate, assuming that it was initially biased for $V_C = 0.5V_{CC}$.

Some solutions to the biasing problem are discussed in the following sections. By contrast, the emitter-degenerated amplifier achieves stable biasing by applying a voltage to the base, most of which appears across the emitter resistor, thus determining the quiescent current.

B. Emitter resistor as feedback

Adding an external series resistor to the intrinsic emitter resistance r_e (emitter degeneration) improves many properties of the common-emitter amplifier, but at the expense of gain. You will see the same thing happening in Chapters 4 and 5, when we discuss *negative feedback*, an important technique for improving amplifier characteristics by feeding back some of the output signal to reduce the effective input signal. The similarity here is no coincidence – the emitter-degenerated amplifier itself uses a form of negative feedback. Think of the transistor as a transconductance device, determining collector current (and therefore output voltage) according to the voltage applied between the base and emitter; but the input to the amplifier is the voltage from base to *ground*. So the voltage from base to emitter is the input voltage, *minus a sample of the output* (namely $I_E R_E$). That's negative feedback, and that's why emitter degeneration improves most properties of the amplifier (here improved linearity and stability and increased input impedance.[37]) Later in the chapter, in §2.5, we'll make these statements quantitative when we first look at feedback. And there are great things to look forward to, with the full flowering of feedback in Chapters 4 and 5!

2.3.5 Biasing the common-emitter amplifier

If you must have the highest possible gain (or if the amplifier stage is inside a feedback loop), it is possible to arrange successful biasing of a common-emitter amplifier. There are three solutions that can be applied, separately or in combination: bypassed emitter resistor, matched biasing transistor, and dc feedback.

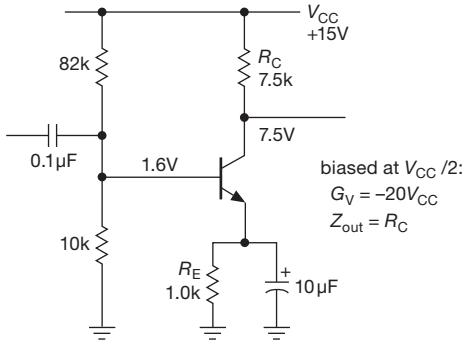


Figure 2.48. A bypassed emitter resistor can be used to improve the bias stability of a grounded-emitter amplifier.

A. Bypassed emitter resistor

You can use a bypassed emitter resistor, biasing as for the degenerated amplifier, as shown in Figure 2.48. In this case R_E has been chosen about $0.1R_C$, for ease of biasing; if R_E is too small, the emitter voltage will be much smaller than the base–emitter drop, leading to temperature instability of the quiescent point as V_{BE} varies with temperature. The emitter bypass capacitor is chosen to make its impedance small compared with r_e (not R_E – why?) at the

[37] And, as we'll learn, the output impedance would be reduced – a desirable feature in a voltage amplifier – if the feedback were taken directly from the collector.

lowest frequency of interest. In this case its impedance is 25 Ω at 650 Hz. At signal frequencies the input coupling capacitor sees an impedance of 10k in parallel with the base impedance, in this case $\beta \times 25\,\Omega$, or roughly 2.5k. At dc, the impedance looking into the base is much larger (β times the emitter resistor, or about 100k).

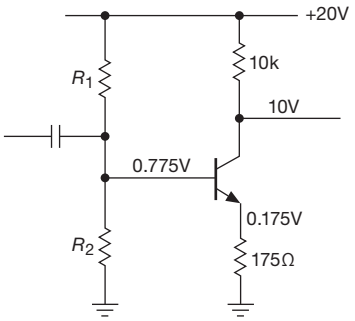


Figure 2.49. Gain-of-50 stage presents bias stability problem.

A variation on this circuit consists of using two emitter resistors in series, one of them bypassed. For instance, suppose you want an amplifier with a voltage gain of 50, quiescent current of 1 mA, and V_{CC} of +20 volts, for signals from 20 Hz to 20 kHz. If you try to use the emitter-degenerated circuit, you will have the circuit shown in Figure 2.49. The collector resistor is chosen to put the quiescent collector voltage at $0.5V_{CC}$. Then the emitter resistor is chosen for the required gain, including the effects of the r_e of $25/I_C(\text{mA})$. The problem is that the emitter voltage of only 0.175 V will vary significantly as the ∼0.6 V of base–emitter drop varies with temperature (−2.1 mV/°C, approximately), since the base is held at constant voltage by R_1 and R_2; for instance, you can verify that an increase of 20°C will cause the collector current to increase by nearly 25%.

Exercise 2.14. Show that this statement is correct.

The solution here is to add some bypassed emitter resistance for stable biasing, with no change in gain at signal frequencies (Figure 2.50). As before, the collector resistor is chosen to put the collector at 10 volts ($0.5V_{CC}$). Then the unbypassed emitter resistor is chosen to give a gain of 50, including the intrinsic emitter resistance $r_e = 25/I_C(\text{mA})$. Enough bypassed emitter resistance is added to make stable biasing possible (one-tenth of the collector resistance is a good guideline). The base voltage is chosen to give 1 mA of emitter current, with impedance about one-tenth the dc impedance looking into the base (in this case about 100k). The emitter bypass capacitor is chosen to have low impedance compared with 180 + 25 Ω at the lowest signal frequencies. Finally, the input coupling capacitor is chosen to have low impedance compared with the *signal-frequency* input impedance of the amplifier, which is equal to the voltage-divider impedance in parallel with $\beta \times (180 + 25)\,\Omega$ (the 820 Ω is bypassed and looks like a short at signal frequencies).

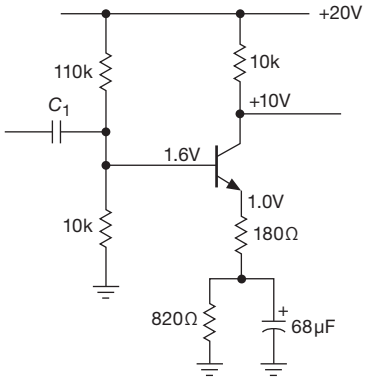


Figure 2.50. A common-emitter amplifier combining bias stability, linearity, and large voltage gain.

An alternative circuit splits the signal and dc paths (Figure 2.51). This lets you vary the gain (by changing the 180 Ω resistor) without bias change.

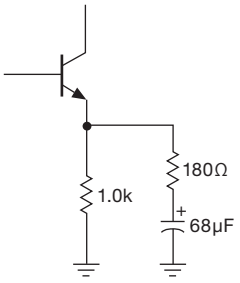


Figure 2.51. Equivalent emitter circuit for Figure 2.50.

B. Matched biasing transistor

You can use a matched transistor to generate the correct base voltage for the required collector current; this ensures automatic temperature compensation (Figure 2.52).[38] Q_1's collector is drawing 1 mA, since it is guaranteed to be near ground (about one V_{BE} drop above ground, to be exact); if Q_1 and Q_2 are a matched pair (available as a single device, with the two transistors on one piece of silicon), then Q_2 will also be biased to draw 1 mA, putting its collector at

[38] R. Widlar, "Some circuit design techniques for linear integrated circuits," *IEEE Trans. Circuit Theory* **CT-12**, 586 (1965). See also US Patent 3,364,434.

+10 volts and allowing a full ±10 V symmetrical swing on its collector. Changes in temperature are of no importance as long as both transistors are at the same temperature. This is a good reason for using a "monolithic" dual transistor.

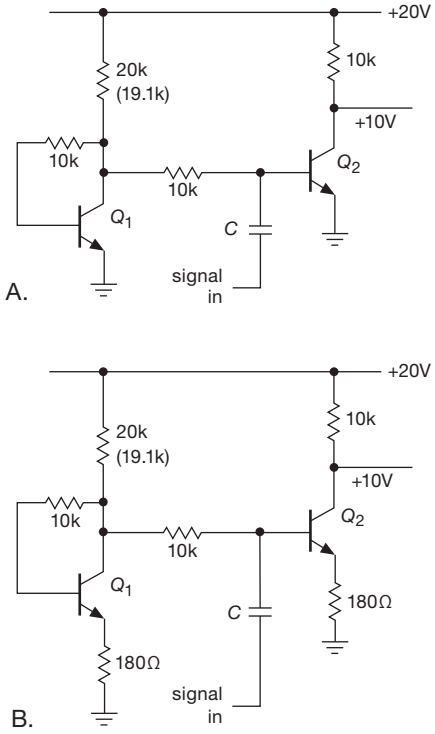


Figure 2.52. Biasing scheme with compensated V_{BE} drop, for both grounded emitter (A) and degenerated emitter (B) stages. With the values shown, V_C would be approximately 10.5 V; reducing the 20k resistor to 19.1k (a standard value) would take into account the effects of V_{BE} and finite β and put V_C at 10 V.

C. Feedback at dc

You can use dc feedback to stabilize the quiescent point. Figure 2.53A shows one method. By taking the bias voltage from the collector, rather than from V_{CC}, you get some measure of bias stability. The base sits one diode drop above ground – and because its bias comes from a 10:1 divider, the collector must be at 11 diode drops above ground, or about 7 volts. Any tendency for the transistor to saturate (e.g., if it happens to have unusually high beta) is stabilized, since the dropping collector voltage will reduce the base bias. This scheme is acceptable if great stability is not required. The quiescent point is liable to drift a volt or so as the ambient (surrounding) temperature changes, because the base–emitter voltage has a significant temperature coefficient (Ebers–Moll, again). Better stability is possible if several stages of amplification are included within the feedback loop. You will see examples later in connection with feedback.

A better understanding of feedback is really necessary to understand this circuit. For instance, feedback acts to reduce the input and output impedances. The input signal sees R_1's resistance effectively reduced by the voltage gain of the stage. In this case it is equivalent to a resistor of about 200 Ω to ground (not pleasant at all!). Later in this chapter (and again in Chapter 4) we treat feedback in enough detail so that you will be able to figure the voltage gain and terminal impedance of this circuit.

Figures 2.53B–D illustrate some variations on the basic dc-feedback bias scheme: circuit B adds some emitter degeneration to improve linearity and predictability of gain; circuit C adds to that an input follower to increase the input impedance (with appropriately increased R_1R_2 divider values and changed ratio to accommodate the additional V_{BE} drop); and circuit D combines the methods of Figure 2.51 with circuit B to achieve greater bias stability.

Note that the base bias resistor values in these circuits could be increased to raise the input impedance, but you should then take into account the non-negligible base current. Suitable values might be $R_1 = 220$k and $R_2 = 33$k. An alternative approach might be to bypass the feedback resistance in order to eliminate feedback (and therefore lowered input impedance) at signal frequencies (Figure 2.54).[39]

D. Comments on biasing and gain

One important point about grounded emitter amplifier stages: you might think that the voltage gain can be raised by increasing the quiescent current, since the intrinsic emitter resistance r_e drops with rising current. Although r_e does decrease with increasing collector current, the smaller collector resistor you need to obtain the same quiescent collector voltage just cancels the advantage. In fact, you can show that the small-signal voltage gain of a grounded-emitter amplifier biased to $0.5V_{CC}$ is given by $G = 20V_{CC}$ (in volts), independent of quiescent current.

Exercise 2.15. Show that the preceding statement is true.

If you need more voltage gain in one stage, one approach is to use a current source as an *active load*. Because its impedance is very high, single-stage voltage gains of

[39] But *caution*: the cascaded *RC* sections (33k into 10 μF, 33k into the input capacitor) can cause peaking or instability, unless care is taken (for example by avoiding similar *RC* products).

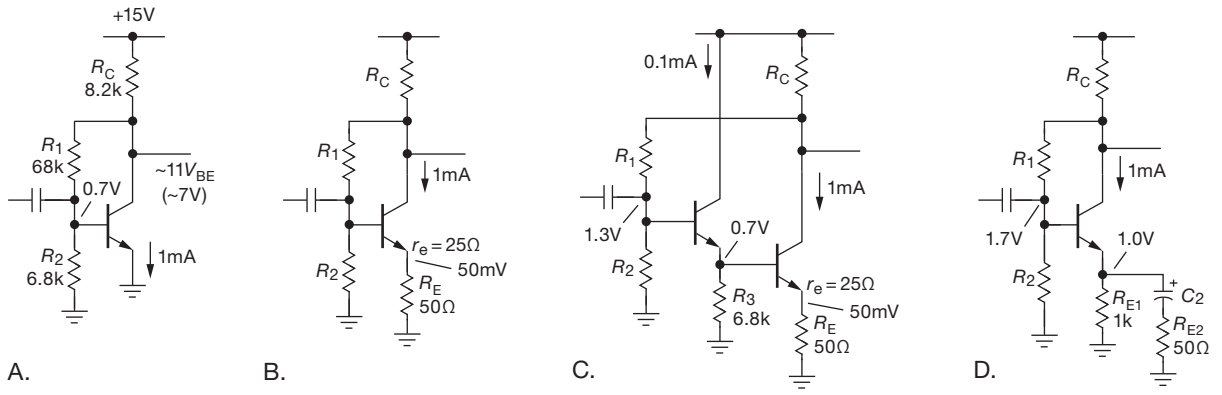


Figure 2.53. Bias stability is improved by feedback.

1000 or more are possible.[40] Such an arrangement cannot be used with the biasing schemes we have discussed, but must be part of an overall dc feedback loop, a subject we will discuss in Chapter 4. You should be sure such an amplifier looks into a high-impedance load; otherwise the gain obtained by high collector load impedance will be lost. Something like an emitter follower, an FET, or an op-amp presents a good load.

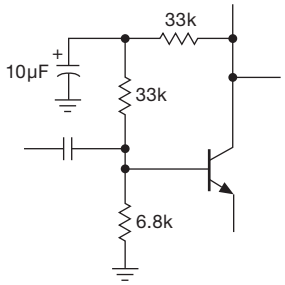


Figure 2.54. Eliminating impedance-lowering feedback at signal frequencies.

In RF amplifiers intended for use only over a narrow frequency range, it is common to use a parallel *LC* circuit as a collector load. In that case a very high voltage gain is possible since the *LC* circuit has high impedance (like a current source) at the signal frequency, with low impedance at dc. Because the *LC* is "tuned," out-of-band interfering signals (and distortion) are effectively rejected. Additional bonuses are the possibility of peak-to-peak (pp) output swings of $2V_{CC}$, and the use of transformer coupling from the inductor.

[40] Ultimately limited by the transistor's finite collector output resistance (a consequence of the Early effect); see the Early effect discussion in Chapter *2x*.

Exercise 2.16. Design a tuned common-emitter amplifier stage to operate at 100 kHz. Use a bypassed emitter resistor, and set the quiescent current at 1.0 mA. Assume $V_{CC} = +15$ V and $L = 1.0$ mH, and put a 6.2k resistor across the *LC* to set $Q = 10$ (to get a 10% bandpass; see §1.7.14). Use capacitive input coupling.

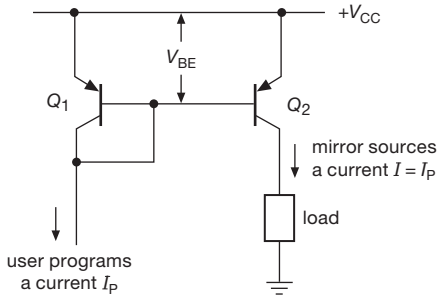


Figure 2.55. Classic bipolar-transistor matched-pair current mirror. Note the common convention of referring to the positive supply as V_{CC}, even when *pnp* transistors are used.

2.3.6 An aside: the perfect transistor

Looking at BJT transistor properties like the non-zero (and temperature-dependent) V_{BE}, the finite (and current-dependent) emitter impedance r_e and transconductance g_m, the collector current that varies with collector voltage (Early effect) etc., one is tempted to ask which transistor is better? Is there a "best" transistor, or perhaps even a *perfect* transistor? If you go through our transistor tables, e.g., Tables 2.1 and 2.2, and especially Table 8.1 for small-signal transistors, you'll see there is no best transistor candidate. That's because all physical bipolar transistors are subject to the same device physics, and their parameters tend to scale with die size and current, etc.

However, it turns out that there *is* a candidate for a "perfect transistor," if you don't limit yourself to a single *npn* or *pnp* structure; see Figure 2.56. This device has nearly ideal properties: V_{BE}=0 V (!), along with very high g_m (thus low r_e), and very high beta. And just to top it off, current can flow in either direction – it's ambidextrous, or "bipolarity" (saying it's bipolarity is better than saying it's a bipolar bipolar transistor). Like a regular BJT, it's a transconductance device: when driven with a positive V_{BE} input signal, it sources an output current that is g_m times greater, and vice versa (with a negative V_{BE} it sinks a current). Unlike a BJT, though, it's noninverting. All signals are referenced to ground. Very nice.

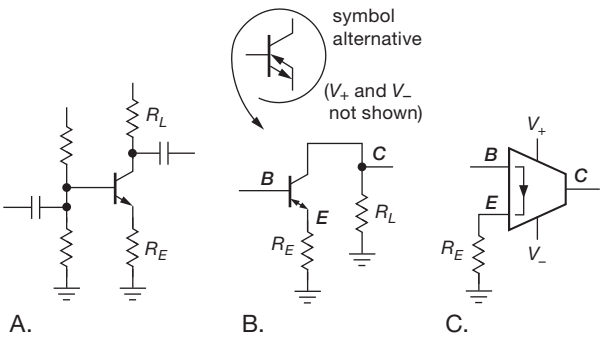


Figure 2.56. A. An ordinary common-emitter BJT amplifier stage, with an emitter-degeneration resistor R_E and a load resistor R_L. B. In a common-emitter amplifier built with the "perfect" transistor, all signals are referenced to ground, to which the load R_L is also returned (the power supplies are not shown). C. The OTA symbol for the perfect transistor, implemented as an Operational Transconductance Amplifier device. The truncated apex symbol means the device has a current output.

How does the perfect transistor work? Figure 2.57 shows a four-transistor circuit known as a *diamond transistor stage*. This circuit is a variation of the cascaded *pnp-npn* emitter follower in Figure 2.29: a complementary *npn-pnp* input follower is wired in parallel, and biased with current sources; the emitter outputs ($2V_{BE}$ apart) drive a matched push–pull output follower, which therefore runs at the same quiescent current. The common node is the effective emitter, *E*. Finally, a pair of current mirrors brings the two individual collector currents to a common output, the effective collector, *C*, where the output current is zero if the input voltage (between terminals *B* and *E*) is zero. As with an ordinary BJT, any current into (or out of) the emitter has to appear at the collector. The part does require two power supply connections. We'll have more to say about this interesting component in Chapters *2x* and *4x*.

Texas Instruments calls their perfect transistor (its part

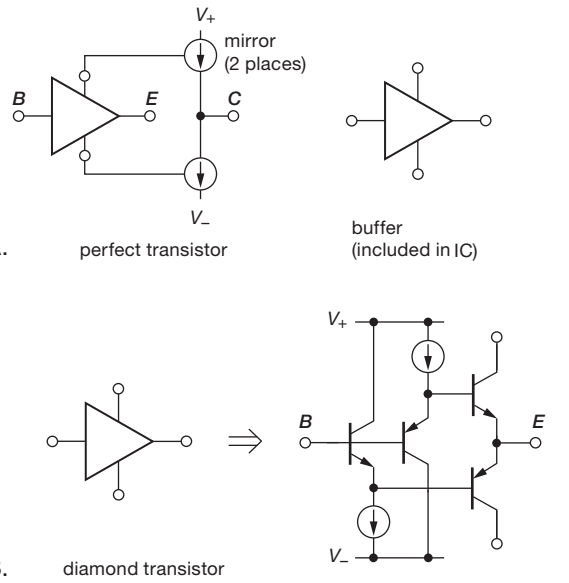


Figure 2.57. A. The OPA860 perfect transistor includes a diamond transistor (the triangle) and a pair of current mirrors. A second diamond transistor acts as an output buffer. B. The diamond transistor consists of a complementary pair of matched offset-cancelled emitter followers.

number is OPA860[41]) an Operational Transconductance Amplifier (OTA). Other names they use are "Voltage-Controlled Current source," "Transconductor," "Macro Transistor," and "positive second-generation current conveyor" (CCII+). We fear it has a branding identity crisis, so, with characteristic understatement, we're calling it a "perfect transistor."

How close to perfection do these parts come? The OPA860 and OPA861 perfect transistors have these specs: V_{os}=3 mV typ (12 mV max), g_m=95 mS, r_e=10.5 Ω, Z_{out}=54kΩ‖2 pF, Z_{in}=455kΩ‖2 pF, $I_{out(max)}$=±15 mA. Its maximum gain is 5100. Hardly perfect, but, hey, not half bad. You can create many nice circuits with these puppies (e.g., active filter, wideband current summing circuit, or integrator for nanosecond-scale pulses); see the OPA860 datasheet for details.

[41] TI's OP861 version omits the output buffer and is available in a small SOT-23 package. That's one of our favorite surface-mount package styles, available for many of the other transistors mentioned in our tables. Knowledgeable readers will recognize the circuitry from inside a current-feedback, or CFB opamp. Some of these devices (for example the AD844) make the internal node available.

2.3.7 Current mirrors

The technique of matched base–emitter biasing can be used to make what is called a *current mirror*, an interesting current-source circuit that simply reverses the sign of a "programming" current. (Figure 2.55). You program the mirror by sinking a current from Q_1's collector. That causes a V_{BE} for Q_1 appropriate to that current at the circuit temperature and for that transistor type. Q_2, matched to Q_1,[42] is thereby programmed to source the same current to the load. The small base currents are unimportant.[43]

One nice feature of this circuit is voltage compliance of the output transistor current source to within a few tenths of a volt of V_{CC}, as there is no emitter resistor drop to contend with. Also, in many applications it is handy to be able to program a current with a current. An easy way to generate the control current I_P is with a resistor (Figure 2.58). Because the bases are a diode drop below V_{CC}, the 14.4k resistor produces a control current, and therefore an output current, of 1 mA. Current mirrors can be used in transistor circuits whenever a current source is needed. They're very popular in integrated circuits, where (a) matched transistors abound and (b) the designer tries to make circuits that will work over a large range of supply voltages. There are even resistorless IC op-amps in which the operating current of the whole amplifier is set by one external resistor, with all the quiescent currents of the individual amplifier stages inside being determined by current mirrors.

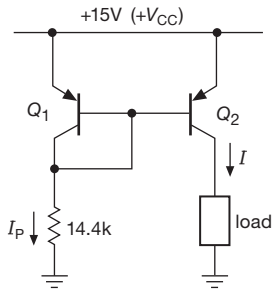


Figure 2.58. Programming current-mirror current.

A. Current-mirror limitations due to the Early effect

One problem with the simple current mirror is that the output current varies a bit with changes in output voltage, i.e., the output impedance is not infinite. This is because of the slight variation of V_{BE} with collector voltage at a given current in Q_2 (which is due to the Early effect); said a different way, the curve of collector current versus collector–emitter voltage at a fixed base-emitter voltage is not flat (Figure 2.59). In practice, the current might vary 25% or so over the output compliance range – much poorer performance than that of the current source with an emitter resistor discussed earlier.

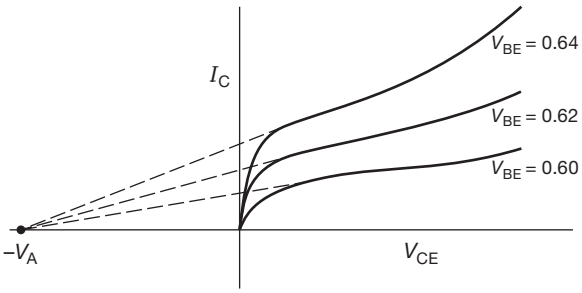


Figure 2.59. Early effect: collector current varies with V_{CE}. (Interestingly, you get a very similar curve, with comparable V_A, if you apply instead a family of constant base *currents*.)

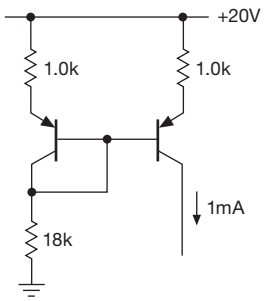


Figure 2.60. Improved current mirror with emitter resistors.

One solution, if a better current source is needed (it often isn't), is the circuit shown in Figure 2.60. The emitter resistors are chosen to have at least a few tenths of a volt drop; this makes the circuit a far better current source, since the small variations of V_{BE} with V_{CE} are now negligible in determining the output current. Again, matched transistors should be used. Note that this circuit loses its effectiveness if operation over a wide range of programming current is intended (figure out why).[44]

[42] A monolithic dual transistor is ideal; Table 8.1b on page 502 lists most available matched transistors. Some, such as the DMMT3904 and 3906, are matched to 1 mV and are quite affordable, $0.36 in small quantities.

[43] This circuit is often called the Widlar current mirror; see the reference on page 97 and US Patent 3,320,439.

[44] Current sources and current mirrors are discussed in more detail in Chapter 2*x*.

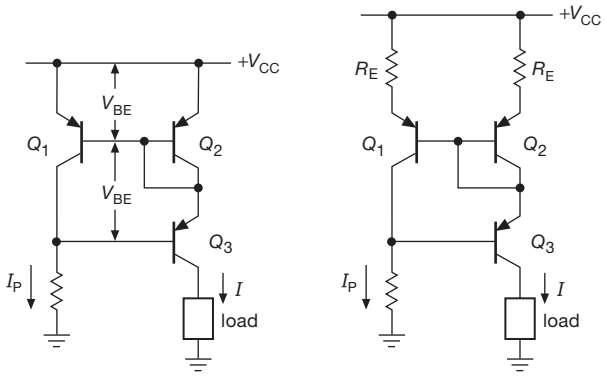


Figure 2.61. Wilson current mirror. Good stability with load variations is achieved through cascode transistor Q_3, which reduces voltage variations across Q_1. Adding a pair of emitter resistors R_E, as shown, reduces output current error caused by V_{BE} mismatch, when chosen such that $I_P R_E$ is of order 100 mV or more.

B. Wilson mirror

Another current mirror with improved consistency of current is shown in the clever circuit of Figure 2.61. Q_1 and Q_2 are in the usual mirror configuration, but Q_3 now keeps Q_1's collector fixed at two diode drops below V_{CC}. That circumvents the Early effect in Q_1, whose collector is now the programming terminal, with Q_2 now sourcing the output current. The result is that both current-determining transistors (Q_1 and Q_2) have fixed collector–emitter drops; you can think of Q_3 as simply passing the output current through to a variable-voltage load (a similar trick is used in the cascode connection, which you will see later in the chapter). Transistor Q_3, by the way, does not have to be matched to Q_1 and Q_2; but if it has the same beta, then you get an exact cancellation of the (small) base current error that afflicts the simple mirror of Figure 2.55 (or the beta-enhanced mirror in Chapter *2x*).

Exercise 2.17. Show that this statement is true.

There are additional nice tricks you can do with current mirrors, such as generating multiple independent outputs, or an output that is a fixed multiple of the programming current. One trick (invented by the legendary Widlar) is to unbalance the R_E's in Figure 2.61; as a rough estimate, the output current ratio is approximately the ratio of resistor values (because the base-emitter drops are roughly equal). But to get it right you need to take into account the difference of V_{BE}'s (because the transistors are running at different currents), for which the graph in Figure 2.62 is helpful. This graph is also useful for estimating the current unbalance in a current mirror built with discrete (i.e., not matched) transistors. We treat current mirrors further in Chapter *2x* (§§*2x.3* and *2x.11*).

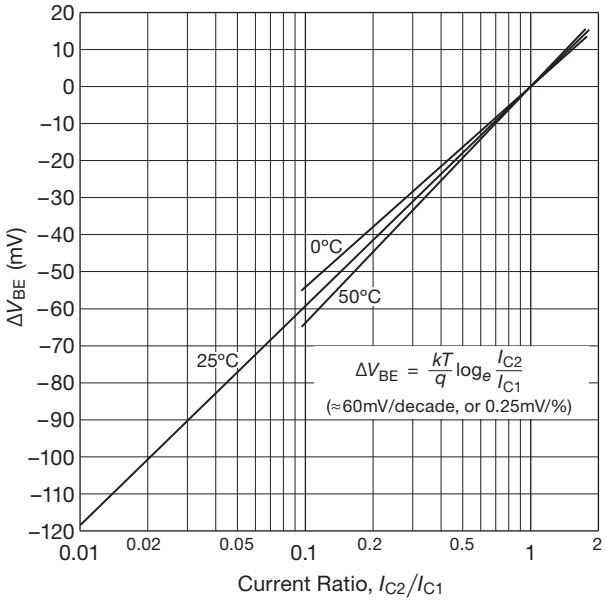


Figure 2.62. Collector current ratios for matched transistors, as determined by the difference in applied base-emitter voltages. See Table 8.1b for low-noise matched BJTs.

2.3.8 Differential amplifiers

The differential amplifier is a very common configuration used to amplify the difference voltage between two input signals. In the ideal case the output is entirely independent of the individual signal levels – only the difference matters.

Differential amplifiers are important in applications in which weak signals are contaminated by "pickup" and other miscellaneous noise. Examples include digital and RF signals transferred over twisted-pair cables, audio signals (the term "balanced" means differential, usually 600 Ω impedance, in the audio business), local-area-network signals (such as 100BASE-TX and 1000BASE-T Ethernet), electrocardiogram voltages, magnetic disk head amplifiers, and numerous other applications. A differential amplifier at the receiving end restores the original signal if the interfering "common-mode" signals (see below) are not too large. Differential amplifiers are universally used in operational amplifiers, an essential building block that is the subject of Chapter 4. They're very important in dc amplifier design (amplifiers that amplify clear down to dc, i.e., have no coupling capacitors) because their symmetrical design is inherently compensated against thermal drifts.

Some nomenclature: when both inputs change levels together, that's a *common-mode* input change. A differential change is called *normal mode*, or sometimes *differential mode*. A good differential amplifier has a high *common-mode rejection ratio* (CMRR), the ratio of response for a normal-mode signal to the response for a common-mode signal of the same amplitude. CMRR is usually specified in decibels. The common-mode input range is the voltage level over which the inputs may vary. The differential amplifier is sometimes called a "long-tailed pair."

Figure 2.63 shows the basic circuit. The output is taken off one collector with respect to ground; that is called a *single-ended output* and is the most common configuration. You can think of this amplifier as a device that amplifies a difference signal and converts it to a single-ended signal so that ordinary subcircuits (followers, current sources, etc.) can make use of the output. (If, instead, a differential output is desired, it is taken between the collectors.)

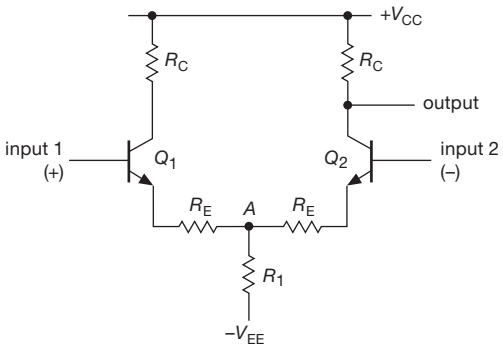


Figure 2.63. Classic transistor differential amplifier.

What is the gain? That's easy enough to calculate: imagine a symmetrical input signal wiggle, in which input 1 rises by v_{in} (a small-signal variation) and input 2 drops by the same amount. As long as both transistors stay in the active region, point *A* remains fixed. You then determine the gain as with the single transistor amplifier, remembering that the input change is actually twice the wiggle on either base: $G_{diff} = R_C/2(r_e + R_E)$. Typically R_E is small, 100 Ω or less, or it may be omitted entirely. Differential voltage gains of a few hundred are possible.

You can determine the common-mode gain by putting identical signals v_{in} on both inputs. If you think about it correctly[45] (remembering that R_1 carries both emitter currents), you'll find $G_{CM} = -R_C/(2R_1 + R_E)$. Here we've ignored the small r_e, because R_1 is typically large, at least a few thousand ohms. We really could have ignored R_E as well. The CMRR is thus roughly $R_1/(r_e + R_E)$. Let's look at a typical example (Figure 2.64) to get some familiarity with differential amplifiers.

Collector resistor R_C is chosen for a quiescent current of 100 μA. As usual, we put the collector at $0.5V_{CC}$ for large dynamic range. Q_1's collector resistor can be omitted, since no output is taken there.[46] R_1 is chosen to give total emitter current of 200 μA, split equally between the two sides when the (differential) input is zero. From the formulas just derived, this amplifier has a differential gain of 10 and a common-mode gain of 0.55. Omitting the 1.0k resistors raises the differential gain to 50, but drops the (differential) input impedance from about 250k to about 50k (you can substitute Darlington transistors[47] in the input stage to raise the impedance into the megohm range, if necessary).

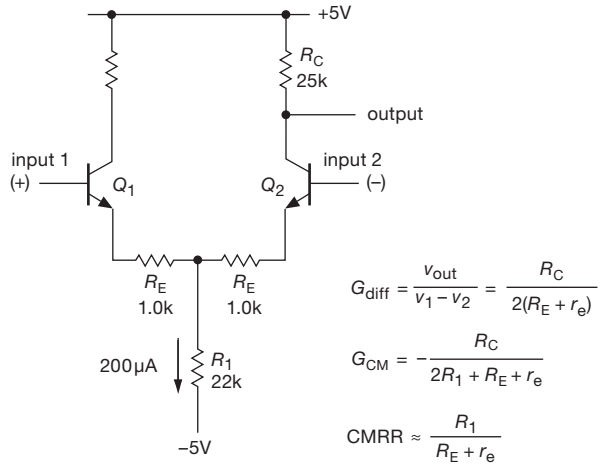


Figure 2.64. Calculating differential amplifier performance.

Remember that the maximum gain of a single-ended grounded emitter amplifier biased to $0.5V_{CC}$ is $20V_{CC}$ (in volts). In the case of a differential amplifier the maximum differential gain ($R_E = 0$) is half that figure, or (for arbitrary quiescent point) 20 times the voltage (in volts) across the collector resistor. The corresponding maximum CMRR (again with $R_E = 0$) is equal to 20 times the voltage (in volts) across R_1. As with the single-ended common-emitter amplifier, the emitter resistors R_E reduce distortion, at the

[45] *Hint*: replace R_1 with a parallel pair, each of resistance $2R_1$; then notice that you can cut the wire connecting them together at point *A* (because no current flows); take it from there.

[46] *Can* be omitted, but at the expense of accurately balanced base-emitter drops: you get better balance if you retain both collector resistors (avoiding Early effect); but you suppress Miller effect (§2.4.5) at input 1 if you omit Q_1's collector resistor.

[47] See §2.4.2.

expense of gain. See the extensive discussion of BJT amplifier distortion in Chapter *2x*.

Exercise 2.18. Verify that these expressions are correct. Then design a differential amplifier to run from ±5 V supply rails, with $G_{\text{diff}} = 25$ and $R_{\text{out}} = 10\text{k}$. As usual, put the collector's quiescent point at half of V_{CC}.

A. Biasing with a current source

The common-mode gain of the differential amplifier can be reduced enormously by the substitution of a current source for R_1. Then R_1 effectively becomes very large and the common-mode gain is nearly zero. If you prefer, just imagine a common-mode input swing; the emitter current source maintains a constant total emitter current, shared equally by the two collector circuits, by symmetry. The output is therefore unchanged. Figure 2.65 shows an example. The CMRR of this circuit, using an LM394 monolithic transistor pair for Q_1 and Q_2, will be around 100,000:1 (100 dB) at dc. The common-mode input range for this circuit goes from −3.5 V to +3 V; it is limited at the low end by the compliance of the emitter current source and at the high end by the collector's quiescent voltage.[48]

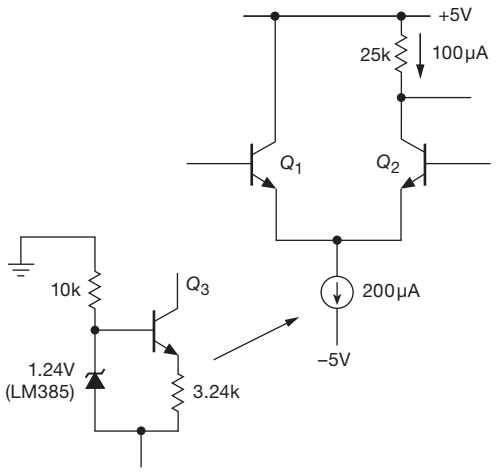


Figure 2.65. Improving CMRR of the differential amplifier with a current source.

Be sure to remember that this amplifier, like all transistor amplifiers, must have a dc bias path to the bases. If the input is capacitively coupled, for instance, you must have base resistors to ground. An additional caution for differential amplifiers, particularly those without inter-emitter resistors: bipolar transistors can tolerate only 6 volts of base–emitter reverse bias before breakdown; thus, applying a differential input voltage larger than this will destroy the input stage (if there is no inter-emitter resistor). An inter-emitter resistor limits the breakdown current and prevents destruction, but the transistors may be degraded nonetheless (in beta, noise, etc.). In either case the input impedance drops drastically during reverse conduction.

An interesting aside: the emitter current sink shown in Figure 2.65 has some variation with temperature, because V_{BE} decreases with increasing temperature (to the tune of approximately −2.1 mV/°C, §2.3.2), causing the current to increase. More explicitly, if we call the 1.24 V zener-like reference "V_{ref}," then the drop across the emitter resistor equals $V_{\text{ref}} - V_{\text{BE}}$; the current is proportional, thus increasing with temperature. As it happens, this is in fact *beneficial*: it can be shown from basic transistor theory that the quantity $V_{g0} - V_{\text{BE}}$ is approximately proportional to absolute temperature (PTAT), where V_{g0} is the silicon bandgap voltage (extrapolated to absolute zero), approximately 1.23 V. So, by choosing our V_{ref} voltage equal to the bandgap voltage, we have an emitter current that increases PTAT; and this cancels the temperature dependence of differential-pair voltage gain ($g_m \propto 1/T_{\text{abs}}$, §2.3.2). We'll explore this sort of cleverness a bit more in §9.10.2. And in Chapter 9 there's extensive discussion of the differential amplifier and the closely related "instrumentation amplifier."

B. Use in single-ended dc amplifiers

A differential amplifier makes an excellent dc amplifier, even for single-ended inputs. You just ground one of the inputs and connect the signal to the other (Figure 2.66). You might think that the "unused" transistor could be eliminated. Not so! The differential configuration is inherently compensated for temperature drifts, and even when one input is at ground that transistor is still doing something: a temperature change causes both V_{BE}'s to change the same amount, with no change in balance or output. That is, changes in V_{BE} are not amplified by G_{diff} (only by G_{CM}, which can be made essentially zero). Furthermore, the cancellation of V_{BE}'s means that there are no 0.6 V drops at the input to worry about. The quality of a dc amplifier constructed this way is limited only by mismatching of input V_{BE}'s or their temperature coefficients. Commercial monolithic transistor pairs and commercial differential amplifier ICs are available with extremely good matching (e.g., the MAT12 *npn* monolithic matched pair has a typical drift of

[48] You can make good current sinks also with JFETs (see the discussion in §3.2.2C), but BJTs are better for this task in many ways. See for example Figure 3.26, where we show four configurations of BJT current sinks that improve upon the JFET alternative.

V_{BE} between the two transistors of 0.15 μV/°C). See Table 8.1b on page 502 for a listing of matched BJTs.

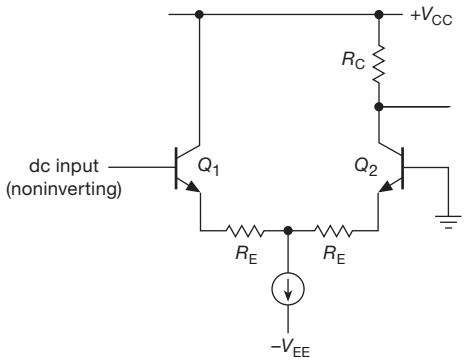


Figure 2.66. A differential amplifier can be used as a precision single-ended dc amplifier.

Either input could have been grounded in the preceding circuit example. The choice depends on whether or not the amplifier is supposed to invert the signal. (The configuration shown is preferable at high frequencies, however, because of the *Miller effect*; see §2.4.5.) The connection shown is noninverting, and so the inverting input has been grounded. This terminology carries over to op-amps, which are versatile high-gain differential amplifiers.

C. Current-mirror active load

As with the simple grounded-emitter amplifier, it is sometimes desirable to have a single-stage differential amplifier with very high gain. An elegant solution is a current-mirror active load (Figure 2.67). Q_1Q_2 is the differential pair with emitter current source. Q_3 and Q_4, a current mirror, form the collector load. The high effective collector load impedance provided by the mirror yields voltage gains of 5000 or more, assuming no load at the amplifier's output.[49] Such an amplifier is very common as the input stage in a larger circuit, and is usually used only within a feedback loop, or as a comparator (discussed in the next section). Be sure to keep the load impedance of such an amplifier very high, or the gain will drop enormously.

D. Differential amplifiers as phase splitters

The collectors of a symmetrical differential amplifier generate equal signal swings of opposite phase. By taking outputs from both collectors, you've got a phase splitter. Of course, you could also use a differential amplifier with both differential inputs and differential outputs. This differential output signal could then be used to drive an additional differential amplifier stage, with greatly improved overall common-mode rejection.

[49] The dc gain is limited primarily by the Early effect; see §2.3.2 and the discussion in Chapter *2x*.

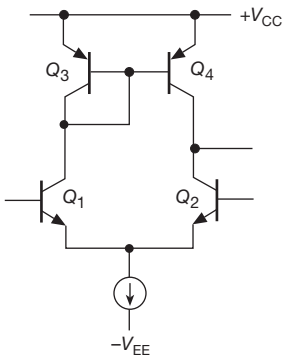


Figure 2.67. Differential amplifier with active current mirror load.

E. Differential amplifiers as comparators

Because of its high gain and stable characteristics, the differential amplifier is the main building block of the *comparator* (which we saw in §1.4.2E), a circuit that tells which of two inputs is larger. They are used for all sorts of applications: switching on lights and heaters, generating square waves from triangles, detecting when a level in a circuit exceeds some particular threshold, class-D amplifiers and pulse-code modulation, switching power supplies, etc. The basic idea is to connect a differential amplifier so that it turns a transistor switch on or off, depending on the relative levels of the input signals. The linear region of amplification is ignored, with one or the other of the two input transistors cut off at any time. A typical hookup is illustrated in §2.6.2 by a temperature-controlling circuit that uses a resistive temperature sensor (thermistor).

2.4 Some amplifier building blocks

We've now seen most of the basic – and important – transistor circuit configurations: switch, follower, current source (and mirror), and common-emitter amplifier (both single-ended and differential). For the remainder of the chapter we look at some circuit elaborations and their consequences: push-pull, Darlington and Sziklai, bootstrapping, Miller effect, and the cascode configuration. We'll finish with an introduction to the wonderful (and essential) technique of *negative feedback*. Chapter *2x* deals with follow-on transistor circuits and techniques at a greater level of sophistication.

Table 2.2 Bipolar Power Transistors[a]

NPN	PNP	Case	V_{CEO} max (V)	I_C max[b] (A)	P_{diss} max[b,h] (W)	$R_{\theta JC}$[c] (°C/W)	h_{FE} min	h_{FE} typ	at I_C (A)	f_T min (MHz)	multiple manf?
standard BJT											
BD139	BD140	TO-126	80	1.5	12.5	10	40[e]	100	0.15	50	•
2N3055	2N2955	TO-3	60	15	115	1.5	20	--	4	2.5	•
2N6292	2N6107	TO-220	70	7	40	3.1	30	--	2	4	–
TIP31C	TIP32C	TO-220	100	3	40	3.1	25	100	1	3	•
TIP33C	TIP34C	TO-218[d]	100	10	80	1.6	40	100	1	3	•
TIP35C	TIP36C	TO-218[d]	100	25	125	1.0	25	150	1.5	3	•
MJ15015	MJ15016	TO-3	120	15	180	1.0	20	35	4	0.8[g]	–
MJE15030	MJE15031	TO-220	150	8	50	2.5	40	80	3	30	•,z
MJE15032	MJE15033	TO-220	250	8	50	2.5	50	100	1	30	•
2SC5200	2SA1943	TO-264	230	17	150	0.8	55	80	1	30	•
2SC5242[k]	2SA1962[k]	TO-3P	250	s	s	s	s	s	s	s	•
MJE340	MJE350	TO-126	300	0.5	20	6	30	--	0.05	--	•
TIP47	MJE5730	TO-220	250	1	40	3.1	30	--	0.3	10	•
TIP50[u]	MJE5731A[u]	TO-220	400	s	s	s	s	--	s	s	•
MJE13007	MJE5852	TO-220	400[f]	8	80	1.6	8[g]	20[g]	2	14[t]	–
Darlington											
MJD112	MJD117	DPak	100	2	20	6.3	1000	2000	2	25	•
TIP122	TIP127	TO-220	100	5	65	1.9	1000	--	3	--	•
TIP142	TIP147	TO-218	100	10	125	1.0	1000	--	5	--	•
MJ11015	MJ11016	TO-3	120	30	200	0.9	1000	--	20	4	•
MJ11032	MJ11033	TO-3	120	50	300	0.6	1000	--	25	--	•
MJH11019	MJH11020	TO-218	200[v]	15	150	0.8	400	--	10	3	•

Notes: (a) sorted more or less by voltage, current and families; see also additional tables in *Chapter 2x*. (b) with case at 25C. (c) P_{diss}(reality) =($T_{J[your\text{-}max\text{-}value]} - T_{amb}$) / ($R_{\theta JC} + R_{\theta CS} + R_{\theta SA}$); this is a much lower number than the "spec," especially if you're careful with T_J max, say 100°C. (d) similar to TO-247. (e) higher gain grades are available. (f) much higher V_{CES} ``blocking'' capability (compared with Vceo), e.g. 700V for MJE13007. (g) higher for the PNP device. (h) P_{diss}(max) = (150°C–25°C) / $R_{\theta JC}$; this is a classic datasheet specsmanship value. (k) larger pkg version of above. (s) same as above. (t) typical. (u) higher voltage version of above. (v) there are also 150V and 250V versions. (z) if these are hard to get, try the '028 and '029 versions (120V rather than 150V).

2.4.1 Push–pull output stages

As we mentioned earlier in the chapter, an *npn* emitter follower cannot sink current and a *pnp* follower cannot source current. The result is that a single-ended follower operating between split supplies can drive a ground-returned load only if a high quiescent current is used.[50] The quiescent current must be at least as large as the maximum output current during peaks of the waveform, resulting in high quiescent power dissipation. For example, Figure 2.68 shows a follower circuit to drive an 8 Ω loudspeaker load with up to 10 watts of audio.

An explanation of the driver stage: the *pnp* follower Q_1 is included to reduce drive requirements and to cancel Q_2's V_{BE} offset (0 V input gives approximately 0 V output). Q_1 could, of course, be omitted for simplicity. The hefty current source in Q_1's emitter load is used to ensure that there is sufficient base drive to Q_2 at the top of the signal swing. A resistor as emitter load would be inferior because it would have to be a rather low value (50 Ω or less) in order to guarantee at least 50 mA of base drive to Q_2 at the peak of the swing, when load current would be maximum and the drop across the resistor would be minimum; the resultant quiescent current in Q_1 would be excessive.

The output of this example circuit can swing to nearly ±15 volts (peak) in both directions, giving the desired output power (9 V rms across 8 Ω). However, the output transistor dissipates 55 watts with no signal (hence the heatsink symbol), and the emitter resistor dissipates another 110 watts. Quiescent power dissipation many times greater than the maximum output power is characteristic of this kind of class-A circuit (transistor always in conduction);

[50] An amplifier in which current flows in the output transistor over the full waveform swing is sometimes called a "class-A" amplifier.

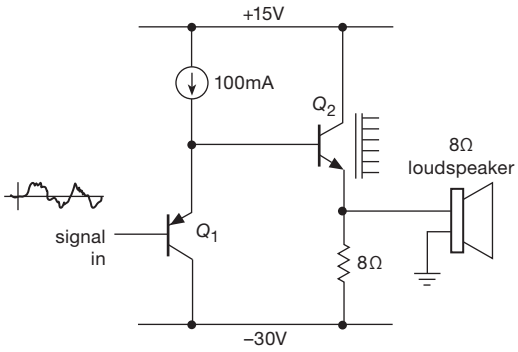


Figure 2.68. A 10 W loudspeaker amplifier, built with a single-ended emitter follower, dissipates 165 W of quiescent power!

this obviously leaves a lot to be desired in applications in which any significant amount of power is involved.

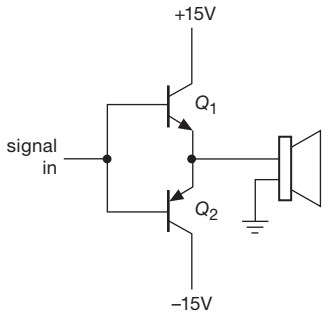


Figure 2.69. Push–pull emitter follower.

Figure 2.69 shows a *push–pull* follower doing the same job. Q_1 conducts on positive swings, Q_2 on negative swings. With zero input voltage, there is no collector current and no power dissipation. At 10 watts of output power there is less than 10 watts of dissipation in each transistor.[51]

A. Crossover distortion in push-pull stages

There is a problem with the preceding circuit as drawn. The output trails the input by a V_{BE} drop; on positive swings the output is about 0.6 V less positive than the input, and the reverse for negative swings. For an input sine wave, the output would look as shown in Figure 2.70. In the language of the audio business, this is called *crossover distortion*. The best cure (feedback offers another method, although by itself it is not entirely satisfactory; see §4.3.1E) is to bias the push–pull stage into slight conduction, as in Figure 2.71.

The bias resistors R bring the diodes into forward conduction, holding Q_1's base a diode drop above the input

[51] An amplifier like this, with half-cycle conduction in each of the output transistors, is sometimes called a "class-B" amplifier.

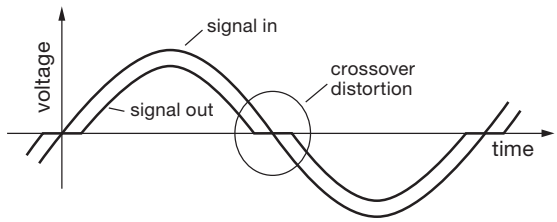


Figure 2.70. Crossover distortion in the push–pull follower.

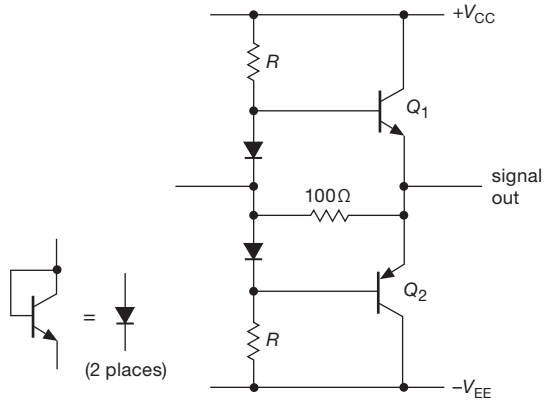


Figure 2.71. Biasing the push–pull follower to eliminate crossover distortion.

signal and Q_2's base a diode drop below the input signal. Now, as the input signal crosses through zero, conduction passes from Q_2 to Q_1; one of the output transistors is always on. The value R of the base resistors is chosen to provide enough base current for the output transistors at the peak output swing. For instance, with ± 20 V supplies and an 8 Ω load running up to 10 watts sinewave power, the peak base voltage is about 13.5 volts and the peak load current is about 1.6 amps. Assuming a transistor beta of 50 (power transistors generally have lower current gain than small-signal transistors), the 32 mA of necessary base current will require base resistors of about 220 Ω (6.5 V from V_{CC} to base at peak swing).

In this circuit we've added a resistor from input to output (this could have been done in Figure 2.69 as well). This serves to eliminate the "dead zone" as conduction passes from one transistor to the other (particularly in the first circuit), which is desirable especially when this circuit is included within a larger feedback circuit. However, it does not substitute for the better procedure of linearizing by biasing, as in Figure 2.71, to achieve transistor conduction over the full output waveform. We have more to say about this in Chapter *2x*.

B. Thermal stability in class-B push–pull amplifiers

The preceding amplifier has one unfortunate feature: it is not thermally stable. As the output transistors warm up (and they will get hot, because they are dissipating power when signal is applied), their V_{BE} drops, causing quiescent current to flow. The added heat this produces causes the situation to get worse, with the strong possibility of what is called *thermal runaway* (whether it runs away or not depends on a number of factors, including how large a "heatsink" is used, how well the diode's temperature tracks the transistor's temperature, etc.). Even without runaway, better control over the circuit is needed, usually with the sort of arrangement shown in Figure 2.72.

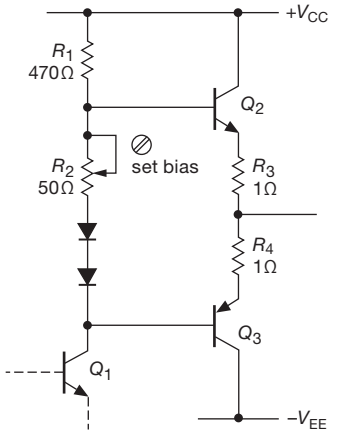


Figure 2.72. Adding (small) emitter resistors improve thermal stability in the push–pull follower.

For variety, the input is shown coming from the collector of the previous stage; R_1 now serves the dual purpose of being Q_1's collector resistor, and also providing current to bias the diodes and bias-setting resistor in the push–pull base circuit. Here R_3 and R_4, typically a few ohms or less, provide a "cushion" for the critical quiescent current biasing: the voltage between the bases of the output transistors must now be a bit greater than two diode drops, and you provide the extra with adjustable biasing resistor R_2 (often replaced with a third series diode, or, better, with the more elegant biasing circuit of Figure 2.78 on page 111). With a few tenths of a volt across R_3 and R_4, the temperature variation of V_{BE} doesn't cause the current to rise very rapidly (the larger the drop across R_3 and R_4, the less sensitive it is), and the circuit will be stable. Stability is improved by mounting the diodes[52] in physical contact with the output transistors (or their heatsinks).

You can estimate the thermal stability of such a circuit by remembering that the base–emitter drop decreases by about 2.1 mV for each 1°C rise, and that the collector current increases by a factor of 10 for every 60 mV increase in base–emitter voltage (or 4% per mV). For example, if R_2 were replaced with a diode, you would have three diode drops between the bases of Q_2 and Q_3, leaving about one diode drop across the series combination of R_3 and R_4. (The latter would then be chosen to give an appropriate quiescent current, perhaps 100 mA for an audio power amplifier.) The worst case for thermal stability occurs if the biasing diodes are not thermally coupled to the output transistors.

Let us assume the worst and calculate the increase in output-stage quiescent current corresponding to a 30°C temperature rise in output transistor temperature. That's not a lot for a power amplifier, by the way. For that temperature rise, the V_{BE} of each output transistor will decrease by about 63 mV at constant current, raising the voltage across R_3 and R_4 by about 50% (i.e., the quiescent current will rise by about 50%). The corresponding figure for the preceding amplifier circuit without emitter resistors (Figure 2.71) will be a factor of 10 rise in quiescent current (recall that I_C increases a decade per 60 mV increase in V_{BE}), i.e., 1000%. The improved thermal stability of this biasing arrangement (even without having the diodes thermally coupled to the output transistors) is evident. And you'll do significantly better when the diodes (or diode-connected transistors, or, best of all, V_{BE}-referenced biasing as shown in Figure 2.78) ride along on the heatsink.

This circuit has the additional advantage that, by adjusting the quiescent current, you have some control over the amount of residual crossover distortion. A push–pull amplifier biased in this way to obtain substantial quiescent current at the crossover point is sometimes referred to as a "class-AB" amplifier, meaning that both transistors conduct simultaneously during a significant portion of the cycle. In practice, you choose a quiescent current that is a good compromise between low distortion and excessive quiescent dissipation. Feedback, introduced later in this chapter (and exploited shamelessly, and with joy, in Chapter 4), is almost always used to reduce distortion still further.

We will see a further evolution of this circuit in §2.4.2, where we supplement it with the intriguingly named techniques of V_{BE}-referenced biasing, collector bootstrapping, and β-boosting complementary Darlington output stage.

[52] Or, better, diode-connected transistors: connect base and collector together as "anode" with emitter as "cathode."

C. "Class-D" amplifiers

An interesting solution to this whole business of power dissipation (and distortion) in class-AB linear power amplifiers is to abandon the idea of a linear stage entirely and use instead a *switching* scheme: imagine that the push–pull follower transistors Q_2 and Q_3 in Figure 2.72 are replaced with a pair of transistor *switches*, with one ON and the other OFF at any time, so that the output is switched completely to $+V_{CC}$ or to $-V_{CC}$ at any instant. Imagine also that these switches are operated at a high frequency (say at least 10 times the highest audio frequency) and that their relative timing is controlled (by techniques we'll see later, in Chapters 10–13) such that the *average* output voltage is equal to the desired analog output. Finally, we add an *LC* lowpass filter to kill the high switching signal, leaving the desired (lower-frequency) analog output intact.

This is a *Class-D*, or *switching* amplifier. It has the advantage of very high efficiency, because the switching transistors are either off (no current) or in saturation (near-zero voltage); that is, the power dissipated in the switching transistors (the product $V_{CE} \times I_C$) is always small. There's also no worry about thermal runaway. The downsides are the problems of emission of high-frequency noise, switching feedthrough to the output, and the difficulty of achieving excellent linearity.

Class-D amplifiers are nearly universal in inexpensive audio equipment, and they are increasingly finding their way into high-end audio equipment. Figure 2.73 shows measured waveforms of an inexpensive (and tiny!) class-D amplifier IC driving a 5 Ω load with a sinewave at the high end of the audio range (20 kHz). This particular IC uses a 250 kHz switching frequency, and can drive 20 watts each into a pair of stereo speakers; pretty much everything you need (except for the output *LC* filters) is on the chip, which costs about $3 in small quantities. Pretty neat.

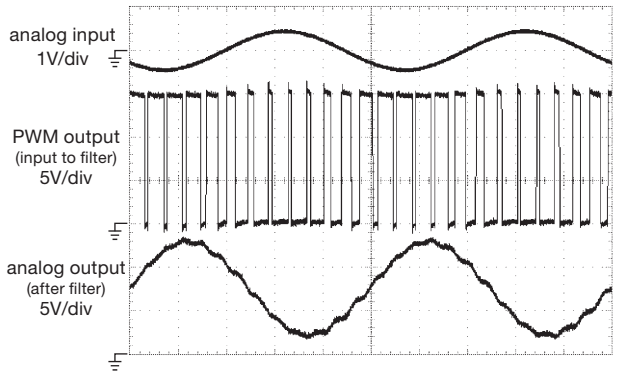


Figure 2.73. Class-D amplifier waveforms: a 20 kHz input sinewave controls the "duty cycle" (fraction of time the output is HIGH) of a push–pull switched output. These waveforms are from a TPA3123 stereo amplifier chip running from +15 V, and show the prefiltered PWM (pulse-width modulated) output, and the final smoothed output after the *LC* lowpass output filter. Horizontal: 10 μs/div.

2.4.2 Darlington connection

If you hook two transistors together as in Figure 2.74, the result – called a *Darlington connection*[53] (or *Darlington pair*) – behaves like a single transistor with beta equal to the product of the two transistor betas.[54] This can be very handy where high currents are involved (e.g., voltage regulators or power amplifier output stages), or for input stages of amplifiers where very high input impedance is necessary.

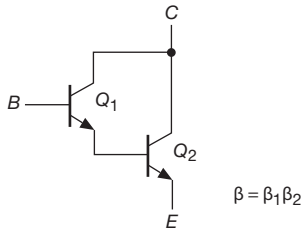


Figure 2.74. Darlington transistor configuration.

For a Darlington transistor the base-emitter drop is twice normal and the saturation voltage is at least one diode drop (since Q_1's emitter must be a diode drop above Q_2's emitter). Also, the combination tends to act like a rather slow transistor because Q_1 cannot turn off Q_2 quickly. This problem is usually taken care of by including a resistor from base to emitter of Q_2 (Figure 2.75). Resistor R also prevents leakage current through Q_1 from biasing Q_2 into conduction;[55] its value is chosen so that Q_1's leakage current (nanoamps for small-signal transistors, as much as hundreds of microamps for power transistors) produces less than a diode drop across R, and so that R doesn't sink a large proportion of Q_2's base current when it has a diode drop across it. Typically R might be a few hundred ohms in

[53] Sidney Darlington, US Patent 2,663,806: "Semiconductor Signal Translating Device." Darlington wanted the patent to cover any number of transistors in one package, but the lawyers at Bell Laboratories overruled him, thus forgoing a patent that would have covered every IC.

[54] At the operating current of each transistor, of course.

[55] And, by stabilizing Q_1's collector current, it improves the predictability of the Darlington's total V_{BE}.

a power transistor Darlington, or a few thousand ohms for a small-signal Darlington.

Darlington transistors are available as single packages, usually with the base–emitter resistor included. A typical example is the *npn* power Darlington MJH6284 (and *pnp* cousin MJH6287), with a current gain of 1000 (typical) at a collector current of 10 amps. Another popular power Darlington is the inexpensive *npn* TIP142 (and *pnp* cousin TIP147): these cost $1 in small quantities and have typical β=4000 at I_C=5 A. And for small-signal applications we like the widely available MPSA14 or MMBTA14 (in TO-92 and SOT23 packages, respectively), with a minimum beta of 10,000 at 10 mA and 20,000 at 100 mA. These 30 volt parts have no internal base–emitter resistor (so you can use them at very low currents); they cost less than $0.10 in small quantities. Figure 2.76 shows beta versus collector current for these parts; note the pleasantly high values of beta, but with substantial dependence both on temperature and on collector current.

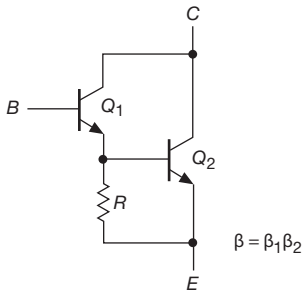


Figure 2.75. Improving turn-off speed in a Darlington pair. (The beta formula is valid as long as R does not rob significant base current from Q_2.)

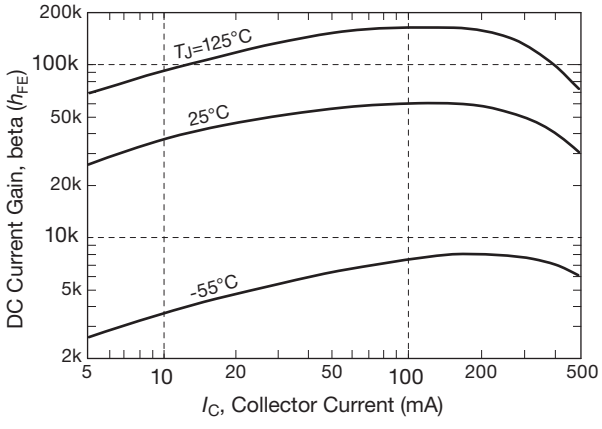


Figure 2.76. Typical beta versus collector current for the popular MPSA14 *npn* Darlington (adapted from the datasheet).

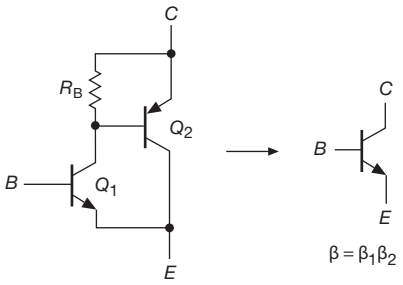


Figure 2.77. Sziklai connection ("complementary Darlington").

A. Sziklai connection

A similar beta-boosting configuration is the Sziklai connection,[56] sometimes referred to as a complementary Darlington (Figure 2.77). This combination behaves like an *npn* transistor, again with large beta. It has only a single base–emitter drop, but (like the Darlington) it also cannot saturate to less than a diode drop. A resistor from base to emitter of Q_2 is advisable, for the same reasons as with the Darlington (leakage current; speed; stability of V_{BE}). This connection is common in push–pull power output stages in which the designer may wish to use one polarity of high-current output transistor only. However, even when used as complementary polarity pairs, it is generally to be preferred over the Darlington for amplifiers and other linear applications; that is because it has the advantage of a single V_{BE} drop (versus two), and that voltage drop is stabilized by the base–emitter resistor of the output transistor. For example, if R_B is chosen such that its current (with a nominal V_{BE} drop across it) is 25% of the output transistor's base current at peak output, then the driver transistor sees a collector current that ranges over only a factor of 5; so its V_{BE} (which is the Sziklai's V_{BE}) varies only 40 mV ($V_T \ln 5$) over the full output current swing. The Sziklai configuration is discussed in more detail in Chapter 2x (see §*2x.10*); and you'll find nice examples of circuits that rely on the

[56] George C. Sziklai, "Symmetrical properties of transistors and their applications," *Proc. IRE* **41**, 717–24 (1953), and US patents 2,762,870 and 2,791,644. His new complementary configuration is buried as Figure 8, where he remarks that "The complementary symmetry of transistors finds an interesting application when it is applied to the cascading of push-pull amplifier stages." The circuit evidently was devised by Sziklai, Lohman, and Herzog, for a transistorized TV demonstration at RCA; the common wisdom was that transistors weren't good enough for the task. In early ICs, where only poor *pnp* transistors were available, an additional *npn* was added, in Sziklai fashion, to boost the current capability of the *pnp*; the combination was called a "composite lateral *pnp*."

Sziklai's unique properties in that chapter's section "BJT amplifier distortion: a SPICE exploration."

Figure 2.78 shows a nice example of a push–pull Sziklai output stage. This has an important advantage compared with the Darlington alternative, namely that the biasing of the Q_3Q_5 pair into class-AB conduction (to minimize crossover distortion) has just two base–emitter drops, rather than four; and, more importantly, Q_3 and Q_5 are running cool compared with the output transistors (Q_4 and Q_6), so they can be relied upon to have a stable base–emitter drop. This allows higher quiescent currents than with the conventional Darlington, where you have to leave a larger safety margin; bottom line, lower distortion.[57]

In this circuit Q_2 functions as an "adjustable V_{BE} multiplier" for biasing, here settable from 1 to 3.5 V_{BE}'s; it is bypassed at signal frequencies. Another circuit trick is the "bootstrapping" of Q_1's collector resistor by C_1 (see §2.4.3), raising its effective resistance at signal frequencies and increasing the amplifier's loop gain to produce lower distortion.

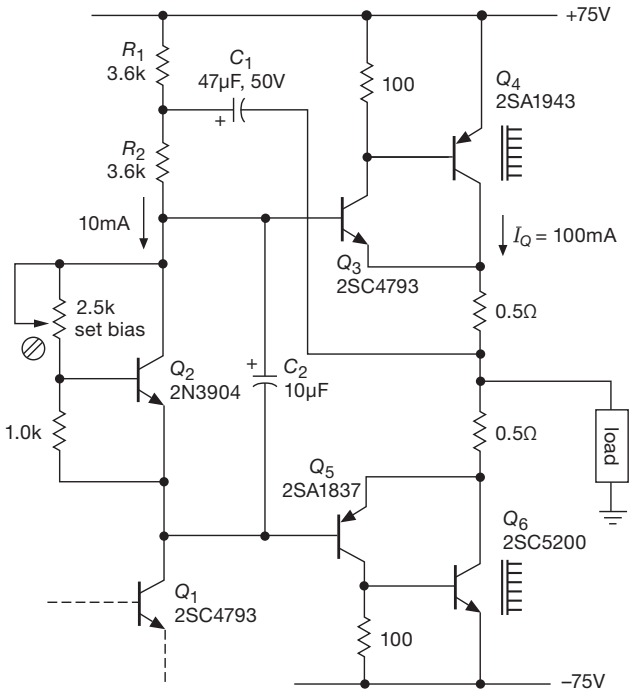


Figure 2.78. Push–pull power stage with Sziklai-pair output transistors, capable of output swings to ±70 V and output currents of ±2 A peak.

B. Superbeta transistor

The Darlington connection and its near relatives should not be confused with the so-called superbeta transistor, a device with very high current gain achieved through the manufacturing process. A typical superbeta transistor is the 2N5962, with a guaranteed minimum current gain of 450 at collector currents from 10 µA to 10 mA (see, for example, Table 8.1a on page 501. Superbeta matched pairs are available for use in low-level amplifiers that require matched characteristics, for example the differential amplifier of §2.3.8. Legendary examples are the LM394 and MAT-01 series; these provide high-gain *npn* transistor pairs whose V_{BE}'s are matched to a fraction of a millivolt (as little as 50 µV in the best versions) and whose betas are matched to about 1%. The MAT-03 is a *pnp* matched pair (see Table 8.1b on page 502). Some commercial op-amps use superbeta differential input stages to achieve input (i.e., base bias) currents as low as 50 *pico*amps this way; examples are the LT1008 and LT1012.

2.4.3 Bootstrapping

When biasing an emitter follower, for instance, you choose the base voltage-divider resistors so that the divider presents a stiff voltage source to the base, i.e., their parallel impedance is much less than the impedance looking into the base. For this reason the resulting circuit has an input impedance dominated by the voltage divider – the driving signal sees a much lower impedance than would otherwise be necessary. Figure 2.79 shows an example. The input resistance of about 9.1k is mostly due to the voltage-divider impedance of 10k. It is always desirable to keep input impedances high, and anyway it's a shame to load the input with the divider, which, after all, is only there to bias the transistor.

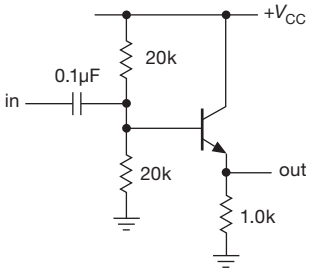


Figure 2.79. Bias network lowers input impedance.

"Bootstrapping" is the colorful name given to a technique that circumvents this problem (Figure 2.80). The

[57] To handle higher power, a common practice is to connect in parallel several identical Q_3Q_4 stages (each with its 0.5 Ω emitter resistor), and similarly for Q_5Q_6. See §2.4.4.

transistor is biased by the divider R_1R_2 through series resistor R_3. Capacitor C_2 is chosen to have low impedance at signal frequencies compared with the bias resistors. As always, bias is stable if the dc impedance seen from the base (in this case 9.7k) is much less than the dc impedance looking into the base (in this case approximately 100k). But now the signal-frequency input impedance is no longer the same as the dc impedance. Look at it this way: an input wiggle v_{in} results in an emitter wiggle $v_E \approx v_{in}$. So the change in current through bias resistor R_3 is $i = (v_{in} - v_E)/R_3 \approx 0$, i.e., Z_{in} (of the bias string) $= v_{in}/i_{in} \approx$ infinity. We've made the loading (shunt) impedance of the bias network very large *at signal frequencies*.

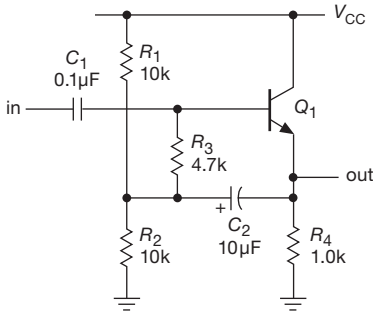


Figure 2.80. Raising the input impedance of an emitter follower at signal frequencies by bootstrapping the base bias divider.

Another way of seeing this is to notice that R_3 always has the same voltage across it at signal frequencies (since both ends of the resistor have the same voltage changes), i.e., it's a current source. But a current source has infinite impedance. In reality the effective impedance is less than infinity because the gain of a follower is slightly less than unity. That is so because the base–emitter drop depends on collector current, which changes with the signal level. You could have predicted the same result from the voltage-dividing effect of the impedance looking into the emitter [$r_e = 25/I_C$(mA) ohms] combined with the emitter resistor. If the follower has voltage gain A (slightly less than unity), the effective value of R_3 at signal frequencies is

$$R_3/(1-A).$$

The voltage gain of a follower can be written $A{=}R_L/(R_L{+}r_e)$, where R_L is the total load seen at the emitter (here $R_1 \parallel R_2 \parallel R_4$), so the effective value of bias resistor R_3 at signal frequencies can be written as $R_3 \rightarrow R_3(1+R_L/r_e)$. In practice the value of R_3 is effectively increased by a hundred or so, and the input impedance is then dominated by the transistor's base impedance. The emitter-degenerated amplifier can be bootstrapped in the same way, since the signal on the emitter follows the base. The bias divider circuit is driven by the low-impedance emitter output at signal frequencies, which is what isolates the input signal from this usual task, and makes possible the beneficial increase of input impedance.

A. Bootstrapping collector load resistors

The bootstrap principle can be used to increase the effective value of a transistor's collector load resistor, if that stage drives a follower. That can increase the voltage gain of the stage substantially – recall that $G_V = -g_m R_C$, with $g_m = 1/(R_E + r_e)$. This technique is used in Figure 2.78, where we bootstrapped Q_1's collector load resistor (R_2), forming an approximate current-source load. This serves two useful functions: (a) it raises the voltage gain of Q_1, and (b) it provides base drive current to Q_3Q_4 that does not drop off toward the top of the swing (as would a resistive load, just when you need it most).

2.4.4 Current sharing in paralleled BJTs

It's not unusual in power electronics design to find that the power transistor you've chosen is not able to handle the required power dissipation, and needs to share the job with additional transistors. This is a fine idea, but you need a way to ensure that each transistor handles an equal portion of the power dissipation. In §9.13.5 we illustrate the use of transistors *in series*. This can simplify the problem, because we know they'll all be running at the same current. But it's often more attractive to divide up the current by connecting the transistors in parallel, as in Figure 2.81A.

There are two problems with this approach. First, we know the bipolar transistor is a transconductance device, with its collector current determined in a precise way by its base-to-emitter voltage V_{BE}, as given by the Ebers–Moll equations 2.8 and 2.9. As we saw in §2.3.2, the temperature coefficient of V_{BE} (at constant collector current) is about -2.1 mV/°C; or, equivalently, I_C *increases* with temperature for a fixed V_{BE}.[58] This is unfortunate, because if the junction of one of the transistors becomes hotter than the rest, it takes more of the total current, thereby heating up even more. It's in danger of the dreaded thermal runaway.

The second problem is that transistors of the same part number are not identical. They come off the shelf with

[58] This result comes directly from $\partial I_C/\partial T = -g_m \partial V_{BE}/\partial T$, which, after substituting $g_m = I_C/V_T$ tells us that the fractional change of collector current is just $(\partial I_C/\partial T)/I_C = -(\partial V_{BE}/\partial T)/V_T$. So the collector current increases fractionally by about 2.1mV/25mV (or 8.4%) per °C – a rather large amount!

differing values of V_{BE} for a given I_C. This is true even for parts made at the same time on the same fabrication line, and from the same silicon wafer. To see how large a variation you are likely to get, we measured 100 adjacent ZTX851 transistors on a reel, with an observed spread of about 17 mV, shown in Figure 8.44. This really represents a "best case," because you cannot be certain that a batch of incoming transistors derive from a single lot, much less a single wafer. When you first build something, the V_{BE}'s of "identical" transistors may be within 20–50 mV of each other, but that matching is lost when one of them has to be replaced someday. It's always safer to assume a possible 100 mV or so spread of base-emitter voltages. Recalling that ΔV_{BE}=60 mV corresponds to a factor of ten current ratio, it's clear that you cannot get away with a direct parallel connection like Figure 2.81A.

The usual solution to this problem is the use of small resistors in the emitters, as shown in Figure 2.81B. These are called *emitter-ballasting* resistors, and their value is chosen to drop at least a few tenths of a volt at the higher end of the anticipated operating current range. That voltage drop must be adequate to swamp the V_{BE} spread of the individual transistors, and is ordinarily chosen somewhere in the range of 300–500 mV.

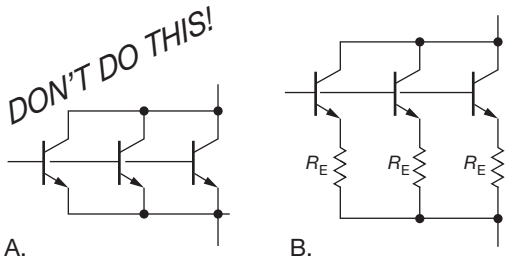


Figure 2.81. To equalize the currents of parallel transistors, use emitter ballasting resistors R_E, as in circuit B.

At high currents the resistors may suffer from an inconveniently-high power dissipation, so you may want to use the current-sharing trick shown in Figure 2.82. Here the current-sensing transistors Q_4–Q_6 adjust the base drive to the "paralleled" power transistors Q_1–Q_3 to maintain equal emitter currents (you can think of Q_4–Q_6 as a high-gain differential amplifier with three inputs). This "active ballast" technique works well with power Darlington BJTs, and it works particularly well with MOSFETs (see Figure 3.117), thanks to their negligible input (gate) current, thus making MOSFETs a good choice for circuits with lots of power dissipation.[59]

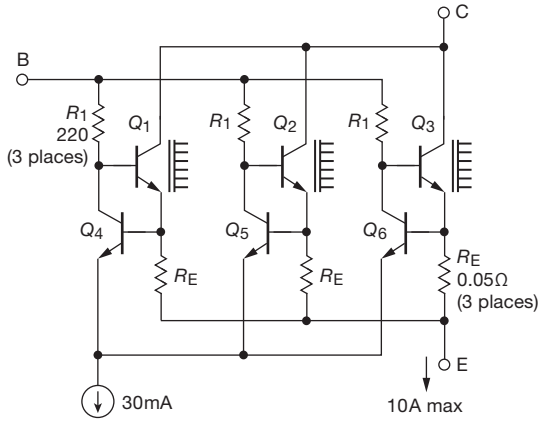


Figure 2.82. Active ballasting of parallel transistors Q_1–Q_3 via feedback from current sensing transistors Q_4–Q_6 lets you configure a parallel power transistors with very low drops across the emitter resistors.

2.4.5 Capacitance and Miller effect

In our discussion so far we have used what amounts to a dc, or low-frequency, model of the transistor. Our simple current amplifier model and the more sophisticated Ebers–Moll transconductance model both deal with voltages, currents, and resistances seen at the various terminals. With these models alone we have managed to go quite far, and in fact these simple models contain nearly everything you will ever need to know to design transistor circuits. However, one important aspect that has serious impact on high-speed and high-frequency circuits has been neglected: the existence of capacitance in the external circuit and in the transistor junctions themselves. Indeed, at high frequencies the effects of capacitance often dominate circuit behavior; at 100 MHz a typical junction capacitance of 5 pF has an impedance of just 320 Ω!

In this brief subsection we introduce the problem, illustrate some of its circuit incarnations, and suggest some methods of circumventing its effects. It would be a mistake to leave this chapter without realizing the nature of this problem. In the course of this brief discussion we will encounter the infamous *Miller effect*, and the use of configurations such as the cascode to overcome it.

[59] Another nice feature of MOSFETs is their lack of second breakdown, thus a wider safe-operating area; see §3.6.4C.

A. Junction and circuit capacitance

Capacitance limits the speed at which the voltages within a circuit can swing ("slew rate"), owing to finite driving impedance or current. When a capacitance is driven by a finite source resistance, you see *RC* exponential charging behavior, whereas a capacitance driven by a current source leads to slew-rate-limited waveforms (ramps). As general guidance, reducing the source impedances and load capacitances and increasing the drive currents within a circuit will speed things up. However, there are some subtleties connected with feedback capacitance and input capacitance. Let's take a brief look.

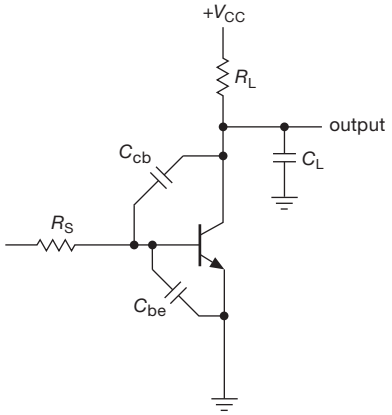


Figure 2.83. Junction and load capacitances in a transistor amplifier.

The circuit in Figure 2.83 illustrates most of the problems of junction capacitance. The output capacitance forms a time constant with the output resistance R_L (R_L includes both the collector and load resistances, and C_L includes both junction and load capacitances), giving a rolloff starting at some frequency $f = 1/2\pi R_L C_L$.

The same is true for the input capacitance, C_{be}, in combination with the source impedance R_S. Of greater significance, at high frequencies the input capacitance robs base current, effectively decreasing the transistor's beta. In fact, transistor datasheets specify a cutoff frequency, f_T, at which the beta has decreased to unity – not much of an amplifier anymore! We discuss this further in Chapter *2x*.

B. Miller effect

The feedback impedance C_{cb} is another matter. The amplifier has some overall voltage gain G_V, so a small voltage wiggle at the input results in a wiggle G_V times larger (and inverted) at the collector. This means that the signal source sees a current through C_{cb} that is $G_V + 1$ times as large as if C_{cb} were connected from base to ground; i.e., for the purpose of input rolloff frequency calculations, the feedback capacitance behaves like a capacitor of value $C_{cb}(G_V + 1)$ from input to ground. This effective increase of C_{cb} is known as the Miller effect. It often dominates the rolloff characteristics of amplifiers, because a typical feedback capacitance of 4 pF can look like several hundred picofarads to ground.

There are several methods available for beating the Miller effect: (a) you can decrease the source impedance driving a grounded-emitter stage by using an emitter follower. Figure 2.84 shows three other possibilities; (b) the differential amplifier circuit with no collector resistor in Q_1 (Figure 2.84A) has no Miller effect; you can think of it as an emitter follower driving a grounded-base amplifier (see below); (c) the famous cascode configuration (Figure 2.84B) elegantly defeats the Miller effect. Here Q_1 is a grounded-emitter amplifier with R_L as its collector resistor: Q_2 is interposed in the collector path to prevent Q_1's collector from swinging (thereby eliminating the Miller effect) while passing the collector current through to the load resistor unchanged. The input labeled V_+ is a fixed-bias voltage, usually set a few volts above Q_1's emitter voltage to pin Q_1's collector and keep it in the active region. This circuit fragment is incomplete, because biasing is not shown; you could either include a bypassed emitter resistor and base divider for biasing Q_1 (as we did earlier in the chapter) or include it within an overall loop with feedback at dc. V_+ might be provided from a divider or zener, with bypassing to keep it stiff at signal frequencies. (d) Finally, the grounded-base amplifier can be used by itself, as shown in Figure 2.84C. It has no Miller effect because the base is driven by zero source impedance (ground), and the amplifier is noninverting from input to output.

Exercise 2.19. Explain in detail why there is no Miller effect in either transistor in the preceding differential amplifier and cascode circuits.

Capacitive effects can be somewhat more complicated than this brief introduction might indicate. In particular: (a) the rolloffs that are due to feedback and output capacitances are not entirely independent; in the terminology of the trade there is *pole splitting*; (b) the transistor's input capacitance still has an effect, even with a stiff input signal source. In particular, current that flows through C_{be} is not amplified by the transistor. This base current "robbing" by the input capacitance causes the transistor's small-signal current gain h_{fe} to drop at high frequencies, eventually reaching unity at a frequency known as f_T. (c) To complicate matters, the junction capacitances depend on voltage: a dominant portion of C_{be} changes proportionally with

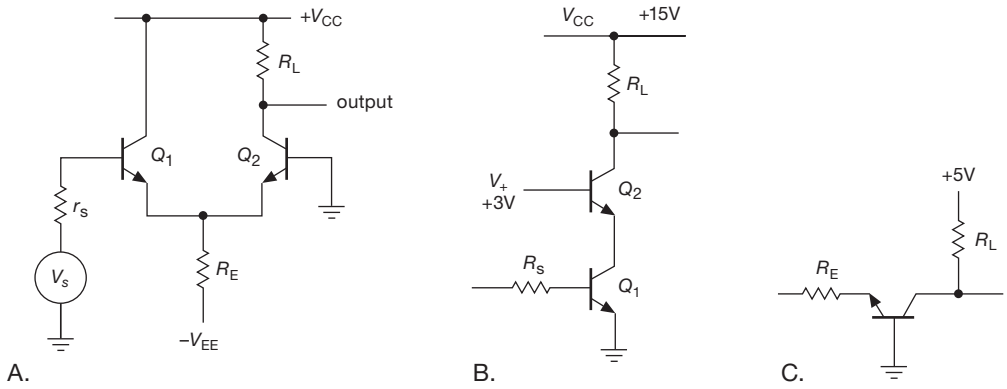


Figure 2.84. Three circuit configurations that avoid the Miller effect. A. Differential amplifier with inverting input grounded. B. Cascode connection. C. Grounded base amplifier.

operating current, so f_T is given instead.[60] (d) When a transistor is operated as a switch, effects associated with charge stored in the base region of a saturated transistor cause an additional loss of speed.

The Miller effect looms large in high-speed and wideband circuits, and we'll be seeing it again and again in subsequent chapters.

2.4.6 Field-effect transistors

In this chapter we have dealt exclusively with BJTs, characterized by the Ebers–Moll equation. BJTs were the original transistors, and they are widely used in analog circuit design. However, it would be a mistake to continue without a few words of explanation about the other kind of transistor, the FET, which we will take up in detail in Chapter 3.

The FET behaves in many ways like an ordinary bipolar transistor. It is a three-terminal amplifying device, available in both polarities, with a terminal (the *gate*) that controls the current flow between the other two terminals (*source* and *drain*). It has a unique property, though: the gate draws no dc current, except for leakage. This means that extremely high input impedances are possible, limited only by capacitance and leakage effects. With FETs you don't have to worry about providing substantial base current, as was necessary with the BJT circuit design of this chapter. Input currents measured in *pico*amperes are commonplace. Yet the FET is a rugged and capable device, with voltage and current ratings comparable to those of bipolar transistors.

Most of the available devices fabricated with BJTs (matched pairs, differential and operational amplifiers, comparators, high-current switches and amplifiers, and RF amplifiers) are also available with FET construction, often with superior performance. Furthermore, digital logic, microprocessors, memory, and all manner of complex and wonderful large-scale digital chips are built almost exclusively with FETs. Finally, the area of micropower design is dominated by FET circuits. It is not exaggeration to say that, demographically, almost all transistors are FETs.[61]

FETs are so important in electronic design that we devote the next chapter to them before treating operational amplifiers and feedback in Chapter 4. We urge the reader to be patient with us as we lay the groundwork in these first three difficult chapters; that patience will be rewarded many times over in the succeeding chapters, as we explore the enjoyable topics of circuit design with operational amplifiers and digital integrated circuits.

2.5 Negative feedback

We've hinted earlier in the chapter that feedback offers a cure to some vexing problems: biasing the grounded-emitter amplifier (§2.3.4 and 2.3.5), biasing the differential amplifier with current-mirror active load (§2.3.8C), and minimizing crossover distortion in push–pull followers (§2.4.1A). It's even better than that – *negative feedback*

[60] See values of f_T versus collector current for 25 transistors, plotted and tabulated in Chapter *2x*'s section titled "BJT Bandwidth and f_T."

[61] Lest this outpouring of enthusiasm leave the wrong impression, we hasten to point out that BJTs are alive and well, largely because they are unbeatable when it comes to characteristics like accuracy and noise (the subjects of Chapters 5 and 8). They excel also in transconductance (i.e., gain). Those muscular power FETs suffer from rather high input capacitance; and, as discrete parts, you cannot get small-signal MOSFETs, only *power* MOSFETs.

is a wonderful technique that can cure all manner of ills: distortion and nonlinearities, frequency dependence of amplifier gain, departure from ideal performance of voltage sources, current sources, or pretty much anything else.

We'll be enjoying the benefits of negative feedback fully in Chapter 4, where we introduce the universal analog component called an *operational amplifier* ("op-amp"), a creature that thrives on negative feedback. But this is a good place to introduce feedback, both because it is widely used in discrete transistor circuits and also because it is present already in our common emitter amplifier, whose improved linearity (compared with that of the grounded-emitter amplifier) is due to negative feedback.

2.5.1 Introduction to feedback

Feedback has become such a well-known concept that the word has entered the general vocabulary. In control systems, feedback consists of comparing the actual output of the system with the desired output and making a correction accordingly. The "system" can be almost anything: for instance, the process of driving a car down the road, in which the output (the position and velocity of the car) is sensed by the driver, who compares it with expectations and makes corrections to the input (steering wheel, throttle, brake). In amplifier circuits the output should be a multiple of the input, so in a feedback amplifier the input is compared with an attenuated version of the output.

As used in amplifiers, negative feedback is implemented simply by coupling the output back in such a way as to cancel some of the input. You might think that this would only have the effect of reducing the amplifier's gain and would be a pretty stupid thing to do. Harold S. Black, who attempted to patent negative feedback in 1928, was greeted with the same response. In his words, "Our patent application was treated in the same manner as one for a perpetual-motion machine."[62] True, it does lower the gain, but in exchange it also improves other characteristics, most notably freedom from distortion and nonlinearity, flatness of response (or conformity to some desired frequency response), and predictability. In fact, as more negative feedback is used, the resultant amplifier characteristics become less dependent on the characteristics of the open-loop (no-feedback) amplifier and finally depend on the properties only of the feedback network itself. Operational amplifiers (the very high-gain differential amplifier building blocks of Chapter 4) are typically used in this high-loop-gain limit, with *open-loop* voltage gain (no feedback) of a million or so.

A feedback network can be frequency dependent, to produce an equalization amplifier (with specific gain-versus-frequency characteristics), or it can be amplitude dependent, producing a nonlinear amplifier (an example is a logarithmic amplifier, built with feedback that exploits the logarithmic V_{BE} versus I_C of a diode or transistor). It can be arranged to produce a current source (near-infinite output impedance) or a voltage source (near-zero output impedance), and it can be connected to generate very high or very low input impedance. Speaking in general terms, the property that is sampled to produce feedback is the property that is improved. Thus, if you feed back a signal proportional to the output current, you will generate a good current source.[63]

Let's look at how feedback works, and how it affects what an amplifier does. We will find simple expressions for the input impedance, output impedance, and gain of an amplifier with negative feedback.

2.5.2 Gain equation

Look at Figure 2.85. To get started we've drawn the familiar common-emitter amplifier with emitter degeneration. Thinking of the transistor in the Ebers–Moll sense, the small-signal voltage from base to emitter (ΔV_{BE}) programs the collector current. But ΔV_{BE} is less than the input voltage V_{in}, because of the drop across R_E. If the output is unloaded, it's easy to get the equation in the figure. In other words, the common-emitter amplifier with emitter degeneration is a grounded-emitter amplifier with negative feedback, as we hinted earlier.

This circuit has some subtleties, which we'd like to sidestep for now by looking instead at the more straightforward configuration shown in Figure 2.85B. Here we've drawn a differential amplifier (with differential gain A), with a fraction of its output signal subtracted from the circuit input v_{in}. That fraction, of course, is given simply by

[62] See the fascinating article in *IEEE Spectrum*, December 1977. His patent for negative feedback (No. 2,102,671, modestly titled "Wave translation system") was granted in 1937, nine years after his initial filing.

[63] Feedback can also be *positive*; that's how you make an oscillator, for instance. As much fun as that may sound, it simply isn't as important as negative feedback. More often it's a nuisance, because a negative-feedback circuit may have large enough phase shifts at some high frequency to produce positive feedback and oscillations. It is surprisingly easy to have this happen, and the prevention of unwanted oscillations is the object of what is called *compensation*, a subject we treat briefly at the end of Chapter 4.

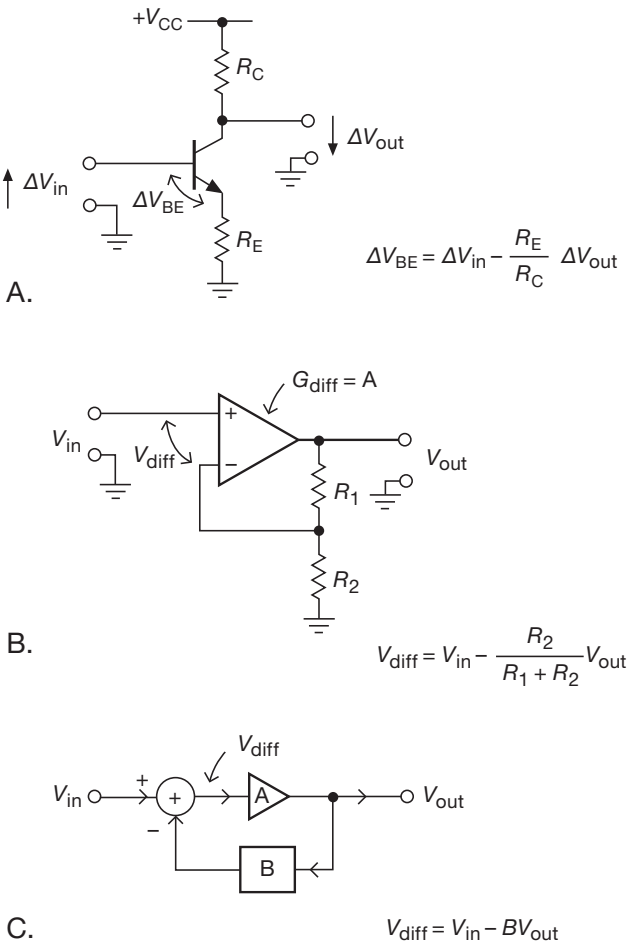


Figure 2.85. Negative feedback subtracts a fraction of the output from the input: A. Common-emitter amplifier. B. Differential amplifier configured as a noninverting voltage amplifier. C. Conventional block diagram.

the voltage divider equation, as shown. This is a very common configuration, widely used with op-amps (Chapter 4), and known simply as a "noninverting amplifier."

When talking about negative feedback, it's conventional to draw a diagram like Figure 2.85C, in which the feedback fraction is simply labeled B. This is useful because it allows more generality than a voltage divider (feedback can include frequency-dependent components like capacitors, and nonlinear components like diodes), and it keeps the equations simple. For a voltage divider, of course, B would simply be equal to $R_2/(R_1+R_2)$.

Let's figure out the gain. The amplifier has open-loop voltage gain A, and the feedback network subtracts a fraction B of the output voltage from the input. (Later we will generalize things so that inputs and outputs can be currents or voltages.) The input to the gain block is then $V_{\rm in} - BV_{\rm out}$. But the output is just the input times A:

$$A(V_{\rm in} - BV_{\rm out}) = V_{\rm out}.$$

In other words,

$$V_{\rm out} = \frac{A}{1+AB} V_{\rm in},$$

and so the closed-loop voltage gain, $V_{\rm out}/V_{\rm in}$, is just

$$G = \frac{A}{1+AB}. \tag{2.16}$$

Some terminology: the standard designations for these quantities are as follows: G = closed-loop gain, A = open-loop gain, AB = loop gain, $1+AB$ = return difference, or desensitivity. The feedback network is sometimes called the beta network (no relation to transistor beta, $h_{\rm fe}$).[64]

2.5.3 Effects of feedback on amplifier circuits

Let's look at the important effects of feedback. The most significant are predictability of gain (and reduction of distortion), changed input impedance, and changed output impedance.

A. Predictability of gain

The voltage gain is $G = A/(1+AB)$. In the limit of infinite[65] open-loop gain A, $G = 1/B$. For finite gain A, feedback acts to reduce the effects of variations of A (with frequency, temperature, amplitude, etc.). For instance, suppose A depends on frequency as in Figure 2.86. This will surely satisfy anyone's definition of a poor amplifier (the gain varies over a factor of 10 with frequency). Now imagine we introduce feedback, with $B = 0.1$ (a simple voltage divider will do). The closed-loop voltage gain now varies from 1000/[1 +(1000×0.1)], or 9.90, to 10,000/[1 + (10,000×0.1)], or 9.99, a variation of just 1% over the same range of frequency! To put it in audio terms, the original amplifier is flat to ±10 dB, whereas the feedback amplifier is flat to ±0.04 dB. We can now recover the original gain of 1000 with nearly this linearity simply by cascading three such stages.

It was for just this reason (namely, the need for extremely flat-response telephone repeater amplifiers) that

[64] We'll see later that amplifiers used with feedback commonly have significant lagging phase shifts from input to output. So the open-loop voltage gain A should properly be represented as a complex number. We'll treat this in §2.5.4; for now we'll adopt the simplification that the amplifier's output voltage is proportional to its input voltage.

[65] Which is not a bad approximation for an op-amp, whose typical open-loop gain is in the neighborhood of $A_{\rm OL} \approx 10^6$.

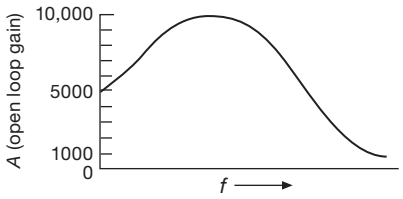


Figure 2.86. Amplifier with open-loop gain A that varies widely with frequency f.

negative feedback in electronics was invented. As the inventor, Harold Black, described it in his first open publication on the invention [*Elec. Eng.*, **53**, 114, (1934)], "by building an amplifier whose gain is made deliberately, say 40 decibels higher than necessary (10,000-fold excess on energy basis) and then feeding the output back to the input in such a way as to throw away the excess gain, it has been found possible to effect extraordinary improvement in constancy of amplification and freedom from nonlinearity." Black's patent is spectacular, with dozens of elegant figures; we reproduce one of them here (Figure 2.87), which makes the point eloquently.

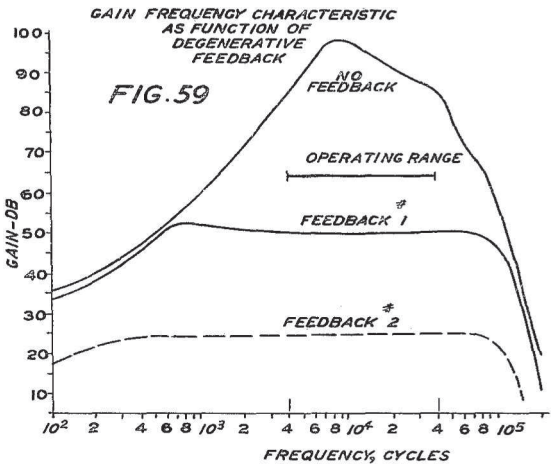


Figure 2.87. Harold Black explains it in his historic 1937 patent, with the unassuming title "Wave translation system."

It is easy to show, by taking the partial derivative of G with respect to A (i.e., $\partial G/\partial A$), that relative variations in the open-loop gain are reduced by the desensitivity:

$$\frac{\Delta G}{G} = \frac{1}{1+AB}\frac{\Delta A}{A}. \qquad (2.17)$$

Thus, for good performance the loop gain AB should be much larger than 1. That's equivalent to saying that the open-loop gain should be much larger than the closed-loop gain.

A very important consequence of this is that nonlinearities, which are simply gain variations that depend on signal level, are reduced in exactly the same way.

B. Input impedance

Feedback can be arranged to subtract a voltage or a current from the input (these are sometimes called *series feedback* and *shunt feedback*, respectively). The noninverting amplifier configuration we've been considering, for instance, subtracts a sample of the output voltage from the differential *voltage* appearing at the input, whereas the feedback scheme in Figure 2.89B subtracts a *current* from the input. The effects on input impedance are opposite in the two cases: voltage feedback multiplies the open-loop input impedance by $1+AB$, whereas current feedback reduces it by the same factor. In the limit of infinite loop gain the input impedance (at the amplifier's input terminal) goes to infinity or zero, respectively. This is easy to understand, since voltage feedback tends to subtract signal from the input, resulting in a smaller change (by the factor AB) across the amplifier's input resistance; it's a form of bootstrapping. Current feedback reduces the input signal by bucking it with an equal current.

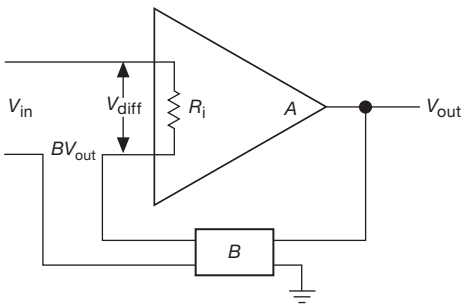


Figure 2.88. Series-feedback input impedance.

Series (voltage) feedback

Let's see explicitly how the effective input impedance is changed by feedback. We illustrate the case of voltage feedback only, since the derivations are similar for the two cases. We begin with a differential amplifier model with (finite) input resistance as shown in Figure 2.88. An input V_{in} is reduced by BV_{out}, putting a voltage $V_{diff} = V_{in} - BV_{out}$ across the inputs of the amplifier. The input current is therefore

$$I_{in} = \frac{V_{in} - BV_{out}}{R_i} = \frac{V_{in}\left(1 - B\dfrac{A}{1+AB}\right)}{R_i} = \frac{V_{in}}{(1+AB)R_i},$$

giving an effective input impedance

$$Z_{\rm in} = V_{\rm in}/I_{\rm in} = (1+AB)R_i.$$

In other words, the input impedance is boosted by a factor of the loop gain plus one. If you were to use the circuit of Figure 2.85B to close the feedback loop around a differential amplifier whose native input impedance is 100 kΩ and whose differential gain is 10^4, choosing the resistor ratio (99:1) for a target gain of 100 (in the limit of infinite amplifier gain), the input impedance seen by the signal source would be approximately 10 MΩ, and the closed-loop gain would be 99.[66]

Shunt (current) feedback

Look at Figure 2.89A. The impedance seen looking into the input of a voltage amplifier with current feedback is reduced by the feedback current, which opposes voltage changes at the input.[67] By considering the current change produced by a voltage change at the input, you find that the input signal sees a parallel combination of (a) the amplifier's native input impedance $R_{\rm i}$ and (b) the feedback resistor $R_{\rm f}$ divided by $1+A$. That is,

$$Z_{\rm in} = R_{\rm i} \,||\, \frac{R_{\rm f}}{1+A}$$

(see if you can prove this). In cases of very high loop gain (e.g, an op-amp), the input impedance is reduced to a fraction of an ohm, which might seem bad. But in fact this configuration is used to convert an input current into an output voltage (a "transresistance amplifier"), for which a low input impedance is a good characteristic. We'll see examples in Chapters 4 and *4x*.

By the addition of an input resistor (Figure 2.89B) the circuit becomes an "inverting amplifier," with input resistance as shown. You can think of this (particularly in the high-loop-gain limit) as a resistor feeding a current-to-voltage amplifier. In that limit $R_{\rm in}$ approximately equals R_1 (and the closed-loop gain approximately equals $-R_2/R_1$).

It is a straightforward exercise to derive an expression for the closed-loop voltage gain of the inverting amplifier with finite loop gain. The answer is

$$G = -A(1-B)/(1+AB)$$

where B is defined as before, $B = R_1/(R_1+R_2)$. In the limit of large open-loop gain A, $G = 1-1/B$ (i.e., $G = -R_2/R_1$).

[66] Of course, knowing that the open-loop gain is approximately 10^4, you might bump the resistor ratio up to 100:1 to compensate. With an op-amp there's no need: with a typical open-loop gain of $\sim 10^6$, the closed-loop gain would be $G_{\rm CL} = 99.99$.

[67] As in the circuit of Figure 2.53 in §2.3.5C.

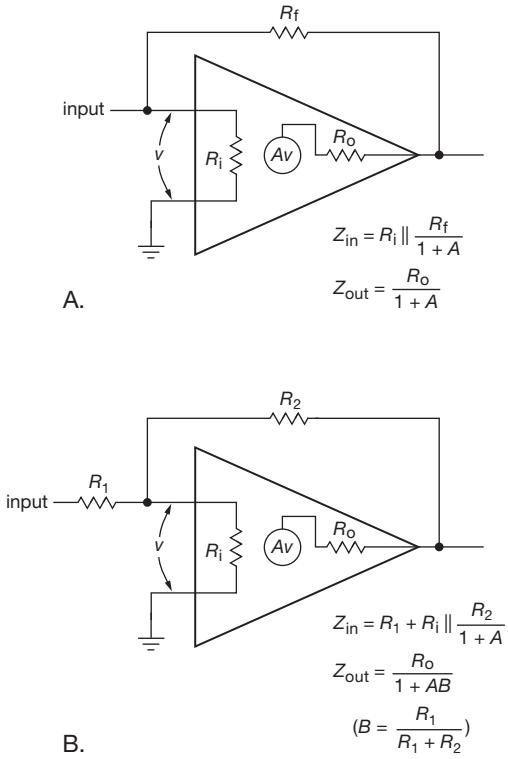


Figure 2.89. Input and output impedances for (A) transresistance amplifier and (B) inverting amplifier.

Exercise 2.20. Derive the foregoing expressions for input impedance and gain of the inverting amplifier.

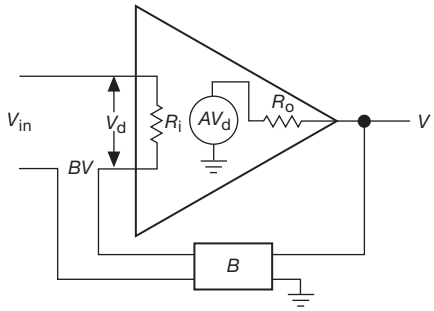


Figure 2.90. Output impedance.

C. Output impedance

Again, feedback can extract a sample of the output voltage or the output current. In the first case the open-loop output impedance will be reduced by the factor $1+AB$, whereas in the second case it will be increased by the same factor. We illustrate this effect for the case of voltage sampling.

We begin with the model shown in Figure 2.90. This time we have shown the output impedance explicitly. The calculation is simplified by a trick: short the input and apply a voltage V to the output; by calculating the output current I, we get the output impedance $R'_o = V/I$. Voltage V at the output puts a voltage $-BV$ across the amplifier's input, producing a voltage $-ABV$ in the amplifier's internal generator. The output current is therefore

$$I = \frac{V-(-ABV)}{R_o} = \frac{V(1+AB)}{R_o}$$

giving an effective output impedance[68] of

$$Z_{out} = V/I = R_o/(1+AB).$$

D. Sensing output *current*

Feedback can be connected instead to sample the output *current*. Then the expression for output impedance becomes

$$Z_{out} = R_o(1+AB).$$

In fact, it is possible to have multiple feedback paths, sampling both voltage and current. In the general case the output impedance is given by Blackman's impedance relation[69]

$$Z_{out} = R_o\frac{1+(AB)_{SC}}{1+(AB)_{OC}},$$

where $(AB)_{SC}$ is the loop gain with the output shorted to ground and $(AB)_{OC}$ is the loop gain with no load attached. Thus feedback can be used to generate a desired output impedance. This equation reduces to the previous results for the usual situation in which feedback is derived from either the output voltage or the output current. See additional discussion in Chapter *2x*.

2.5.4 Two important details

Feedback is a rich subject, which we've simplified shamelessly in this brief introduction. Here are two details that should not be overlooked, however, even at this somewhat superficial level of understanding.

A. Loading by the feedback network

In feedback computations, you usually assume that the beta network doesn't load the amplifier's output. If it does, that must be taken into account in computing the open-loop gain. Likewise, if the connection of the beta network at the amplifier's input affects the open-loop gain (feedback removed, but network still connected), you must use the modified open-loop gain. Finally, the preceding expressions assume that the beta network is unidirectional, i.e., it does not couple any signal from the input to the output.

B. Phase shifts, stability, and "compensation"

The open-loop amplifier gain A is central in the expressions we've found for closed-loop gain and the corresponding input and output impedances. By default one might reasonably assume that A is a real number – that is, that the output is in phase with the input. In real life things are more complex,[70] because of the effects of circuit capacitances (and Miller effect, §2.4.5), and also the limited bandwidth (f_T) of the active components themselves. The result is that the open-loop amplifier will exhibit lagging phase shifts that increase with frequency. This has several consequences for the closed-loop amplifier.

Stability

If the open-loop amplifier's lagging phase shift reaches 180°, then negative feedback becomes *positive* feedback, with the possibility of oscillation. This is not what you want! (The actual criterion for oscillation is that the phase shift be 180° at a frequency at which the loop gain AB equals 1.) This is a serious concern, particularly in amplifiers with plenty of gain (such as op-amps). The problem is only exacerbated if the feedback network contributes additional lagging phase shift (as it often will). The subject of *frequency compensation* in feedback amplifiers deals directly with this essential issue; you can read about it in §4.9.

Gain and phase shift

The expressions we found for closed-loop gain and for the input and output impedances contain the open-loop gain A. For example, the voltage amplifier with series feedback (Figures 2.85B&C, 2.88, and 2.90) has closed-loop gain $G_{CL} = A/(1+AB)$, where $A = G_{OL}$, the amplifier's open-loop gain. Let's imagine that the open-loop gain A is 100, and that we've chosen $B = 0.1$ for a target closed-loop gain of $G_{CL} \approx 10$. Now, if the open-loop amplifier had no phase shifts, then $G_{CL} \approx 9.09$, also without phase shift. If instead the amplifier has a 90° lagging phase shift, then A is pure

[68] If the open-loop gain A is real (i.e. no phase shift), then the output impedance Z_{out} will be real (i.e., resistive: R_{out}). As we'll see in Chapter 4, however, A can be (and often is) complex, representing a lagging phase shift. For op-amps the phase shift is 90° over most of the amplifier's bandwidth. The result is an *inductive* closed-loop output impedance. See for example Figure 4.53 in Chapter 4.

[69] R. B. Blackman, "Effect of feedback on impedance," *Bell. Sys. Tech. J.* **22**, 269 (1943).

[70] That's a pun, get it?

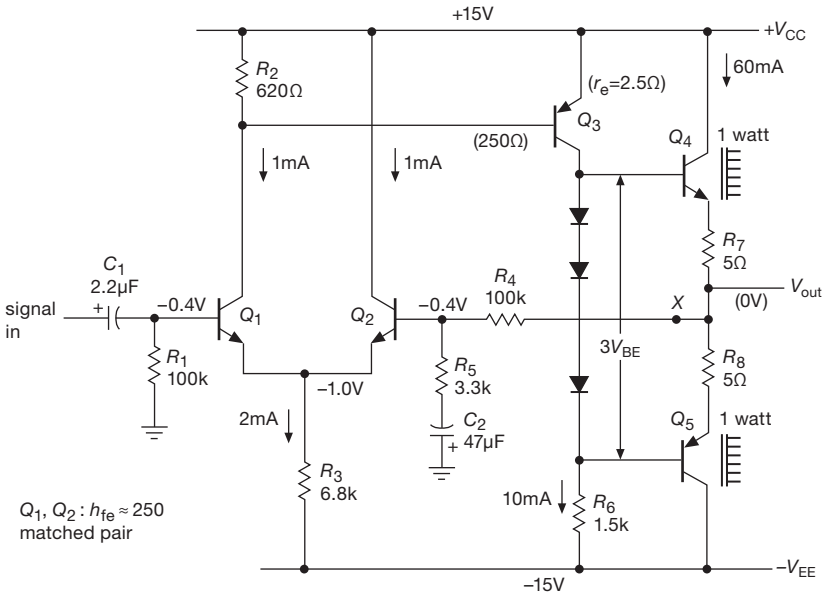


Figure 2.91. Transistor power amplifier with negative feedback.

imaginary ($A=-100j$), and the closed-loop gain becomes $G_{CL} = 9.90-0.99j$. That's a magnitude $|G_{CL}|$=9.95, with a lagging phase shift of approximately 6°. In other words, the effect of a pretty significant (halfway to oscillation!) open-loop phase shift turns out, in fact, to be favorable: the closed-loop gain is only 0.5% less than the target, compared with 9% for the case of the same amplifier without phase shift. The price you pay is some residual phase shift and, of course, an approach to instability.

As artificial as this example may seem, it in fact reflects a reality of op-amps, which usually have an ∼90° lagging phase shift over almost their entire bandwidth (typically from ∼10 Hz to 1 MHz or more). Because of their much higher open-loop gain, the amplifier with feedback exhibits very little phase shift, and an accurate gain set almost entirely by the feedback network. Much more on this in Chapter 4, and in §4.9.

Exercise 2.21. Verify that the above expressions for G_{CL} are correct.

2.5.5 Two examples of transistor amplifiers with feedback

Let's look at two transistor amplifier designs to see how the performance is affected by negative feedback. There's a bit of complexity in this analysis . . . don't be discouraged![71]

[71] Those fearful of discouragement may wish to skip over this section in a first reading.

Figure 2.91 shows a complete transistor amplifier with negative feedback. Let's see how it goes.

A. Circuit description

It may look complicated, but it is extremely straightforward in design and is relatively easy to analyze. Q_1 and Q_2 form a differential pair, with common-emitter amplifier Q_3 amplifying its output. R_6 is Q_3's collector load resistor, and push–pull pair Q_4 and Q_5 form the output emitter follower. The output voltage is sampled by the feedback network consisting of voltage divider R_4 and R_5, with C_2 included to reduce the gain to unity at dc for stable biasing. R_3 sets the quiescent current in the differential pair, and since overall feedback guarantees that the quiescent output voltage is at ground, Q_3's quiescent current is easily seen to be 10 mA (V_{EE} across R_6, approximately). As we discussed earlier (§2.4.1B), the diodes bias the push–pull pair into conduction, leaving one diode drop across the series pair R_7 and R_8, i.e., 60 mA quiescent current. That's class-AB operation, good for minimizing crossover distortion, at the cost of 1 watt standby dissipation in each output transistor.

From the point of view of our earlier circuits, the only unusual feature is Q_1's quiescent collector voltage, one diode drop below V_{CC}. That is where it must sit in order to hold Q_3 in conduction, and the feedback path ensures that it will. (For instance, if Q_1 were to pull its collector closer to ground, Q_3 would conduct heavily, raising the output

voltage, which in turn would force Q_2 to conduct more heavily, reducing Q_1's collector current and hence restoring the status quo.) R_2 was chosen to give a diode drop at Q_1's quiescent current in order to keep the collector currents in the differential pair approximately equal at the quiescent point. In this transistor circuit the input bias current is not negligible ($4\,\mu$A), resulting in a 0.4 V drop across the 100k input resistors. In transistor amplifier circuits like this, in which the input currents are considerably larger than in op-amps, it is particularly important to make sure that the dc resistances seen from the inputs are equal, as shown (a Darlington input stage would probably be better here).

B. Analysis

Let's analyze this circuit in detail, determining the gain, input and output impedances, and distortion. To illustrate the utility of feedback, we will find these parameters for both the open-loop and closed-loop situations (recognizing that biasing would be hopeless in the open-loop case). To get a feeling for the linearizing effect of the feedback, the gain will be calculated at +10 volts and −10 volts output, as well as at the quiescent point (0 V).

Open loop

Input impedance We cut the feedback at point X and ground the right-hand side of R_4. The input signal sees 100k in parallel with the impedance looking into the base. The latter is h_{fe} times twice the intrinsic emitter resistance plus the impedance seen at Q_2's emitter caused by the finite impedance of the feedback network at Q_2's base. For $h_{fe} \approx 250$, $Z_{in} \approx 250 \times [(2 \times 25) + (3.3\text{k}/250)]$; i.e., $Z_{in} \approx 16\text{k}$.

Output impedance Since the impedance looking back into Q_3's collector is high, the output transistors are driven by a 1.5k source (R_6). The output impedance is about 15Ω ($\beta \approx 100$) plus the 5 Ω emitter resistance, or 20 Ω. The intrinsic emitter resistance of 0.4 Ω is negligible.

Gain The differential input stage sees a load of R_2 paralleled by Q_3's base resistance. Since Q_3 is running 10 mA quiescent current, its intrinsic emitter resistance is 2.5 Ω, giving a base impedance of about 250 Ω (again, $\beta \approx 100$). The differential pair thus has a gain of

$$\frac{250 \| 620}{2 \times 25\,\Omega} \quad \text{or} \quad 3.5.$$

The second stage, Q_3, has a voltage gain of 1.5k Ω/2.5 Ω, or 600. The overall voltage gain at the quiescent point is 3.5×600, or 2100. Since Q_3's gain depends on its collector current, there is substantial change of gain with signal swing, i.e., nonlinearity. The gain is tabulated in the following section for three values of output voltage.

Closed loop

Input impedance This circuit uses series feedback, so the input impedance is raised by (1 + loop gain). The feedback network is a voltage divider with $B = 1/30$ at signal frequencies, so the loop gain AB is 70. The input impedance is therefore $70 \times 16\text{k}$, still paralleled by the 100k bias resistor, i.e., about 92k. The bias resistor now dominates the input impedance.

Output impedance Since the output *voltage* is sampled, the output impedance is reduced by (1 + loop gain). The output impedance is therefore 0.3 Ω. Note that this is a small-signal impedance and does not mean that a 1 Ω load could be driven to nearly full swing, for instance. The 5 Ω emitter resistors in the output stage limit the large signal swing. For instance, a 4 Ω load could be driven only to 10 Vpp, approximately.

Gain The gain is $A/(1+AB)$. At the quiescent point that equals 30.84, using the exact value for B. To illustrate the gain stability achieved with negative feedback, the overall voltage gain of the circuit with and without feedback is tabulated at three values of output level at the end of this paragraph. It should be obvious that negative feedback has brought about considerable improvement in the amplifier's characteristics, although in fairness it should be pointed out that the amplifier could have been designed for better open-loop performance, e.g., by using a current source for Q_3's collector load and degenerating its emitter, by using a current source for the differential-pair emitter circuit, etc. Even so, feedback would still make a large improvement.

	Open loop			*Closed loop*		
V_{out}	−10	0	+10	−10	0	+10
$\mathbf{Z}_{in}$	16k	16k	16k	92k	92k	92k
$\mathbf{Z}_{out}$	20Ω	20Ω	20Ω	0.3Ω	0.3Ω	0.3Ω
Gain	1360	2100	2400	30.60	30.84	30.90

C. Series-feedback pair

Figure 2.92 shows another transistor amplifier with feedback. Thinking of Q_1 as an amplifier of its base-emitter voltage drop (thinking in the Ebers–Moll sense), the feedback samples the output voltage and subtracts a fraction of it from the input signal. This circuit is a bit tricky because Q_2's collector resistor doubles as the feedback network. Applying the techniques we used earlier, one can show that G(open loop) ≈ 200, loop gain ≈ 20, Z_{out}(open

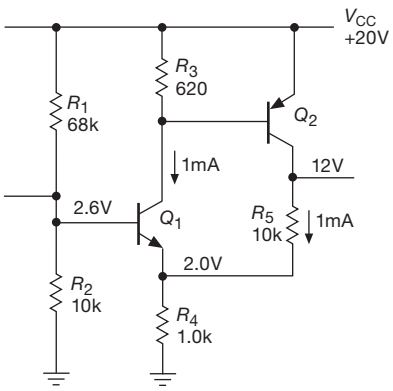


Figure 2.92. Series-feedback pair.

loop) ≈ 10k, Z_{out}(closed loop) $\approx 500\,\Omega$, and G(closed loop) ≈ 9.5.

Exercise 2.22. Go for it!

this circuit to ensure stability (i.e., to prevent oscillation), particularly if the output were capacitively bypassed (as it should be), for reasons we will see later in connection with feedback loop stability (§4.9).

We'll see much more of this subject in Chapter 9.

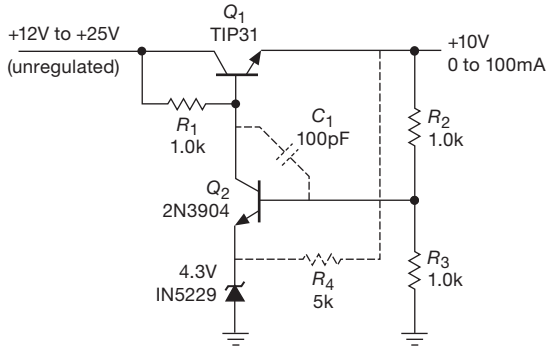


Figure 2.93. Feedback voltage regulator.

2.6 Some typical transistor circuits

To illustrate some of the ideas of this chapter, let's look at a few examples of circuits with transistors. The range of circuits we can cover at this point is necessarily limited, because real-world circuits usually incorporate op-amps (the subject of Chapter 4) and other useful ICs – but we'll see plenty of transistors used alongside ICs in those later chapters.

2.6.1 Regulated power supply

Figure 2.93 shows a very common configuration. R_1 normally holds Q_1 on; when the output reaches 10 volts, Q_2 goes into conduction (base at 5 V), preventing further rise of output voltage by shunting base current from Q_1's base. The supply can be made adjustable by replacing R_2 and R_3 with a potentiometer. In this *voltage regulator* (or "regulated dc supply") circuit, negative feedback acts to stabilize the output voltage: Q_2 "looks at" the output and does something about it if the output isn't at the right voltage.

A few details: (a) Adding a biasing resistor R_4 ensures a relatively constant zener current, so that the zener voltage does not change significantly with load current. It is tempting to provide that bias current from the input, but it is far better to use the regulated output. A warning is in order: whenever you use an output voltage to make something happen within a circuit, make sure that the circuit will start up correctly; here, however, there is no problem (why not?). (b) The capacitor C_1 would probably be needed in

2.6.2 Temperature controller

The schematic diagram in Figure 2.94 shows a temperature controller based on a *thermistor* sensing element, a device that changes resistance with temperature. Differential Darlington Q_1–Q_4 compares the voltage of the adjustable reference divider R_4–R_6 with the divider formed from the thermistor and R_2. (By comparing *ratios* from the same supply, the comparison becomes insensitive to supply variations; this particular configuration is called a Wheatstone bridge.) Current mirror Q_5Q_6 provides an active load to raise the gain, and mirror Q_7Q_8 provides emitter current. Q_9 compares the differential amplifier output with a fixed voltage, saturating Darlington $Q_{10}Q_{11}$ (which supplies power to the heater) if the thermistor is too cold. R_9 is a current-sensing resistor that turns on protection transistor Q_{12} if the output current exceeds about 6 amps; that steals base drive from $Q_{10}Q_{11}$, preventing damage. And R_{12} adds a small amount of positive feedback, to cause the heater to snap on and off abruptly; this is the same trick (a "Schmitt trigger") as in Figure 2.13.

2.6.3 Simple logic with transistors and diodes

Figure 2.95 shows a circuit that performs a task we illustrated in §1.9.1F: sounding a buzzer if either car door is open and the driver is seated. In this circuit the transistors all operate as switches (either OFF or saturated). Diodes D_1 and D_2 form what is called an OR gate, turning off Q_1 if either door is open (switch closed). However, the collector of

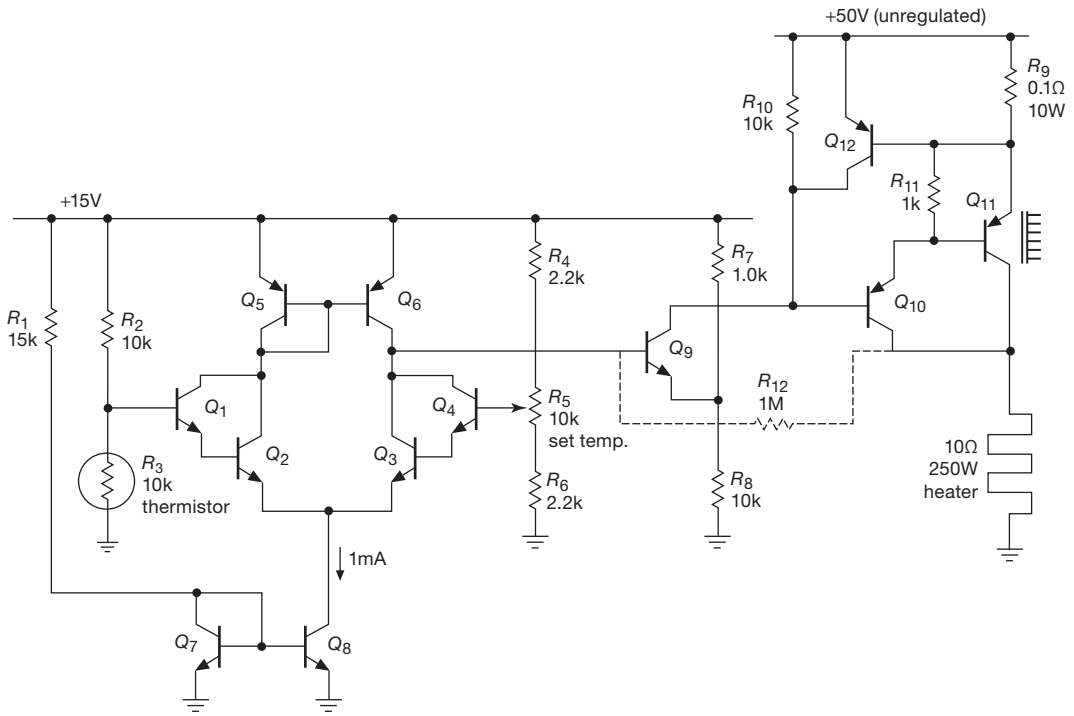


Figure 2.94. Temperature controller for 250 W heater.

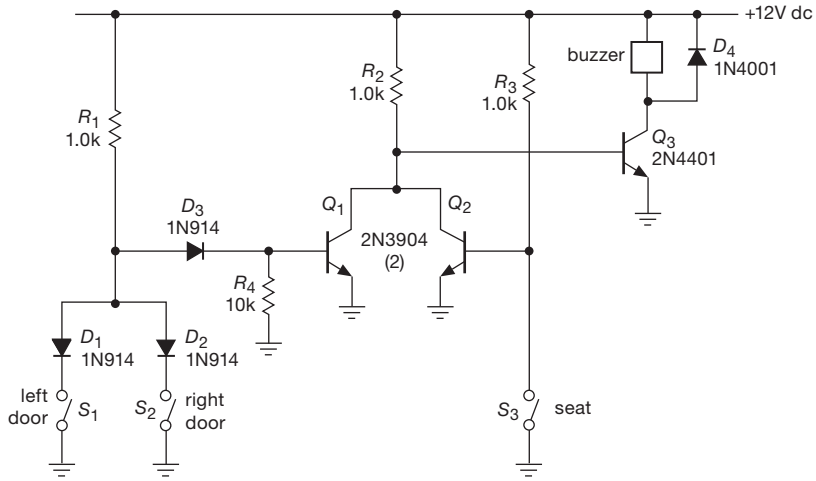


Figure 2.95. Both diodes and transistors are used to make digital logic "gates" in this seat-belt buzzer circuit.

Q_1 stays near ground, preventing the buzzer from sounding unless switch S_3 is also closed (driver seated); in that case R_2 turns on Q_3, putting 12 volts across the buzzer. D_3 provides a diode drop so that Q_1 is OFF with S_1 or S_2 closed, and D_4 protects Q_3 from the buzzer's inductive turn-off transient. In Chapters 10–15 we discuss logic circuitry in detail.

Additional Exercises for Chapter 2

Exercise 2.23. Design a transistor switch circuit that allows you to switch two loads to ground by means of saturated *npn* transistors. Closing switch *A* should cause both loads to be powered, whereas closing switch *B* should power only one load. *Hint*: use diodes.

Exercise 2.24. Consider the current source in Figure 2.96. (a) What is I_{load}? What is the output compliance? Assume V_{BE} is 0.6 V. (b) If β varies from 50 to 100 for collector voltages within the output compliance range, how much will the output current vary? (There are two effects here.) (c) If V_{BE} varies according to $\Delta V_{BE} = -0.0001\ \Delta V_{CE}$ (Early effect), how much will the load current vary over the compliance range? (d) What is the temperature coefficient of output current assuming that β does not vary with temperature? What is the temperature coefficient of output current assuming that β increases from its nominal value of 100 by 0.4%/°C?

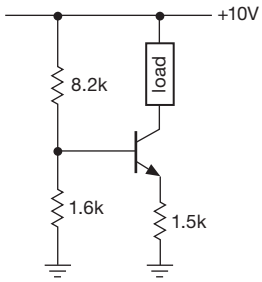


Figure 2.96. Current source exercise.

Exercise 2.25. Design a common-emitter *npn* amplifier with a voltage gain of 15, V_{CC} of +15 V, and I_C of 0.5 mA. Bias the collector at $0.5V_{CC}$, and put the low-frequency 3 dB point at 100 Hz.

Exercise 2.26. Bootstrap the circuit in the preceding problem to raise the input impedance. Choose the rolloff of the bootstrap appropriately.

Exercise 2.27. Design a dc-coupled differential amplifier with a voltage gain of 50 (to a single-ended output) for input signals near ground, supply voltages of ±15 volts, and quiescent currents of 0.1 mA in each transistor. Use a current source in the emitter and an emitter follower output stage.

Exercise 2.28. In this problem you will ultimately design an amplifier whose gain is controlled by an externally applied voltage (in Chapter 3 you will see how to do the same thing with FETs). (a) Begin by designing a long-tailed pair differential amplifier with emitter current source and no emitter resistors (undegenerated). Use ±15 V supplies. Set I_C (each transistor) at 100 μA, and use $R_C = 10$k. Calculate the voltage gain from a single-ended input (other input grounded) to a single-ended output. (b) Now modify the circuit so that an externally applied voltage controls the emitter current source. Give an approximate formula for the gain as a function of controlling voltage. (In a real circuit you might arrange a second set of voltage-controlled current sources to cancel the quiescent-point shift that gain changes produce in this circuit, or a differential-input second stage could be added to your circuit.)

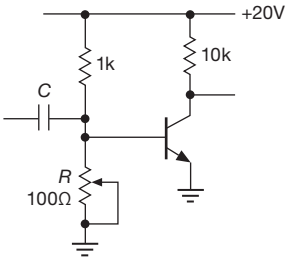


Figure 2.97. Bad biasing.

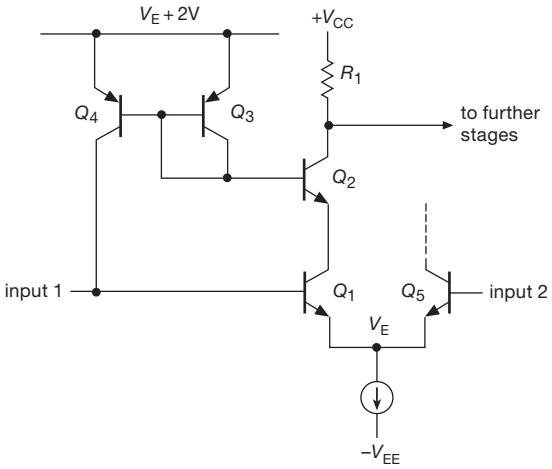


Figure 2.98. Base-current cancellation scheme used in precision operational amplifiers. Bias-current cancellation is discussed in detail in Chapter *4x*.

Exercise 2.29. Disregarding the lessons of this chapter, a disgruntled student builds the amplifier shown in Figure 2.97. He adjusts R until the quiescent point is $0.5V_{CC}$. (a) What is Z_{in} (at high frequencies where $Z_C \approx 0$)? (b) What is the small-signal voltage gain? (c) What rise in ambient temperature (roughly) will cause the transistor to saturate?

Exercise 2.30. Several commercially available precision op-amps (e.g., the venerable OP-07) use the circuit in Figure 2.98 to cancel input bias current (only half of the symmetrical-input differential amplifier is shown in detail; the other half works the same way). Explain how the circuit works. *Note*: Q_1 and Q_2 are a beta-matched pair. *Hint*: it's all done with mirrors.

Review of Chapter 2

An A-to-W review of Chapter 2. This review doesn't follow the exact topic order in the chapter: here we first cover transistor theory, then circle back to discuss some applications. In the chapter circuits have been interspersed with theory to provide motivation and illustrate how to use the theory.

¶A. Pin-Labeling Conventions.
The introduction (§2.1) describes some transistor and circuit-labeling conventions. For example, V_B (with a single subscript) indicates the voltage at the base terminal, and similarly I_B indicates current flowing into the base terminal. V_{BE} (two subscripts) indicates base-to-emitter voltage. Symbols like V_{CC} and V_{EE} (repeated subscripts) indicate the positive and negative supply voltages.

¶B. Transistor Types and Polarities.
Transistors are three-terminal devices capable of amplifying signals. They come in two broad classes, *bipolar junction transistors* (BJTs, the subject of this chapter), and *field-effect transistors* (FETs, the subject of Chapter 3). BJTs have a control terminal called the *base*, and a pair of output terminals, called the *collector* and the *emitter* (the corresponding terminals in a FET are *gate*, *drain*, and *source*). A signal applied to the base controls the current flowing from collector to emitter. There are two BJT polarities available, *npn* and *pnp*; for *npn* devices the collector is more positive than the emitter, and the opposite is true for *pnp*. Figure 2.2 illustrates this and identifies intrinsic diodes that are part of the transistor structure, see ¶D and ¶F below. The figure also illustrates that the collector current and the (much smaller) base current combine to form the emitter current.

Operating modes Transistors can operate as *switches* – turned ON or OFF – or they can be used as *linear* devices, for example as amplifiers, with an output current proportional to an input signal. Put another way, a transistor can be in one of three states: *cutoff* (non-zero V_{CE} but zero I_C), *saturated* (non-zero I_C but near-zero V_{CE}), or in the *linear region* (non-zero V_{CE} and I_C). If you prefer prose (and using "voltage" as shorthand for collector-to-emitter voltage V_{CE}, and "current" as shorthand for collector current I_C), the cutoff state has voltage but no current, the saturated state has current but near-zero voltage, and the linear region has both voltage and current.

¶C. Transistor Man and Current Gain.
In the simplest analysis, §2.1.1, the transistor is simply a current amplifier, with a *current gain* called *beta* (symbol β, or sometimes h_{FE}). A current into the base causes a current β times larger to flow from collector to emitter, $I_C = \beta I_B$, if the external circuit allows it. When currents are flowing, the base-emitter diode is conducting, so the base is $\sim$0.65 V more positive (for *npn*) than the emitter. The transistor doesn't *create* the collector current out of thin air; it simply throttles current from an available supply voltage. This important point is emphasized by our "transistor man" creation (Figure 2.7), a little homunculus whose job is to continuously examine the base current and attempt to adjust the collector's current to be a factor of β (or h_{FE}) times larger. For a typical BJT the beta might be around 150, but beta is only loosely specified, and a particular transistor type may have a 3:1 spread (or more) in specified beta at some collector current (and further 3:1 spreads of β versus I_C and β versus temperature, see for example Figure 2.76).

¶D. Switches and Saturation.
When operated as a switch, §2.2.1, a current must be injected into the base to keep the transistor "ON." This current must be substantially more than $I_B = I_C/\beta$. In practice a value of 1/10th of the maximum expected collector current is common, but you could use less, depending on the manufacturer's recommendations. Under this condition the transistor is in *saturation*, with 25–200 mV across the terminals. At such low collector-to-emitter voltages the base-to-collector diode in Figure 2.2 is conducting, and it robs some of the base-current drive. This creates an equilibrium at the saturation voltage. We'll return in ¶K to look at some circuit examples. See also the discussion of transistor saturation in Chapter *2x*.

¶E. The BJT is a Transconductance Device.
As we point out in §2.1.1, "A circuit that depends on a particular value for beta is a bad circuit." That's because β can vary by factors of 2 to 3 from the manufacturer's nominal datasheet value. A more reliable design approach is to use other highly-predictable BJT parameters that take into account that it is a *transconductance* device. In keeping with the definition of transconductance (an output current proportional to an input voltage), a BJT's collector current, I_C, is controlled by its base-to-emitter voltage, V_{BE}, see §2.3. (We can then rely on $I_B = I_C/\beta$ to estimate the base current, the other way around from the simple approach in ¶C.) The transconductance view of BJTs is helpful in many circumstances (estimating gain, distortion, tempco), and it is essential in understanding and designing circuits such as differential amplifiers and current mirrors. However, in many situations you can circumvent the beta-uncertainty problem with circuit design tricks such as dc feedback or emitter degeneration, without explicitly invoking Ebers–Moll

(¶F). Note also that, just as it would be a bad idea to bias a BJT by applying a base current calculated from I_C/β (from an assumed β), it would be even worse to attempt to bias a BJT by applying a calculated V_{BE} (from an assumed I_s, see ¶F); more on this in ¶Q, below. We might paraphrase this by saying "a circuit that depends on a particular value for I_s, or for operation at a precise ambient temperature, is a bad circuit."

¶F. Ebers–Moll.
Figure 2.41 shows a typical *Gummel plot*, with V_{BE} dictating I_C, and thus an approximate I_B. Equations (2.8) and (2.9) show the exponential (or logarithmic) nature of this relationship. A simple form of the equation, $I_C=I_s\exp(V_{BE}/V_T)$ and its inverse, $V_{BE}=V_T\log_e(I_C/I_s)$, where the constant $V_T=25\,\text{mV}$ at 25°C, reveals that collector current is determined by V_{BE} and a parameter I_s, the latter related to the transistor die size and its current density. I_s is a very small current, typically some 10^{11} times smaller than I_C. The Ebers–Moll formula accurately holds for the entire range of silicon BJT types, for example those listed in Table 8.1. The integrated-circuit (IC) industry relies on Ebers–Moll for the design of their highly-successful BJT linear circuits.

¶G. Collector Current versus Base Voltage: Rules of Thumb.
See §2.3.2. It's useful to remember a few rules of thumb, which we can derive from Ebers–Moll: I_C increases by a factor of ten for a $\approx$60 mV increase in V_{BE}; it doubles for an $\approx$18 mV V_{BE} increase, and it increases 4% for a 1 mV V_{BE} increase.

¶H. Small Signals, Transconductance and r_e.
See §2.3.2B. It's convenient to assume operation at fixed I_C, and look for the effect of small changes ("small signals"). First, thinking about the rules of thumb above, we can calculate (eq'n 2.13,) the transconductance, $g_m=\partial I_C/\partial V_{BE}=I_C/V_T$. This evaluates to $g_m=40\,\text{mS}$ at 1 mA, with g_m proportional to current. To put it another way, we can assign an effective internal resistance r_e in series with the emitter, $r_e=1/g_m=V_T/I_C$, see eq'n 2.12. (The small *r* indicates *small signal*.) A useful fact to memorize: r_e is about 25 Ω at a collector current of 1 mA, and it scales inversely with current.

¶I. Dependence on Temperature.
See §2.3.2C. In ¶F we said $V_T=25\,\text{mV}$ at 25°C, which suggests it's not exactly a constant, but changes with temperature. Because $V_T=kT/q$ (§2.3.1), you might guess that V_{BE} is proportional to absolute temperature, thus a temperature coefficient of about +2mV/°C (because $V_{BE}\approx600\,\text{mV}$ at $T=300\text{K}$). But the scaling parameter I_s has a large opposite tempco, producing an overall tempco of about $-2.1\,\text{mV}/°\text{C}$. Memorize this fact also! Because V_T is proportional to absolute temperature, the tempco of transconductance at fixed collector current is inversely proportional to absolute temperature (recall $g_m=I_C/V_T$), and thus drops by about 0.34%/°C at 25°C.

¶J. Early Effect.
See §2.3.2D. In our simple understanding so far, base voltages (or currents) "program" a BJT's collector current, independent of collector voltage. But in reality I_C increases slightly with increasing V_{CE}. This is called the *Early effect*, see eq'n 2.14 and Figure 2.59, which can be characterized by an *Early voltage* V_A, a parameter independent of operating current; see eq'n 2.15. If the Early voltage is low (a common drawback of *pnp* transistors) the effect can be quite large. For example, a *pnp* 2N5087 with $V_A=55\,\text{V}$ has $\eta=4\times10^{-4}$, and would experience a 4 mV shift of V_{BE} with a 10 V change of V_{CE}; if instead the base voltage were held constant, a 10 V increase of collector voltage would cause a 17% increase of collector current. We hasten to point out there are circuit configurations, such as *degeneration*, or the *cascode*, that alleviate the Early effect. For more detail see the discussion in Chapter *2x*.

Circuit Examples

With this summary of basic BJT theory, we now circle back and review some circuit examples from Chapter 2. One way to review the circuits is to flip through the chapter looking at the pictures (and reading the captions), and refer to the associated text wherever you are uncertain of the underlying principles.

¶K. Transistor Switches.
BJT switches are discussed in §2.2.1, and circuit examples appear in Figures 2.9 (driving an LED), 2.10 (high-side switching, including level shifting), and 2.16 (with an emitter-follower driver). Simply put, you arrange to drive a current into the base to put the transistor into solid saturation for the anticipated collector load current (i.e., $I_B \gg I_C/\beta$), bringing its collector within tens of millivolts of the emitter. More like this appears in Chapter 12 (Logic Interfacing). Looking forward, the use of *MOSFET* switches often provide a superior switching solution (§§3.4.4 and 3.5); their control terminal (the gate) conveniently requires *no* static gate current, though you may have to provide significant transient currents to charge its gate capacitance during rapid switching.

¶L. Transistor Pulsers.
Basic timer and pulse generator circuits are shown in Figures 2.11 (pulse from a step) and 2.12 (pulse from a pulse). These are simple, but not terribly accurate or stable; better to use a dedicated timer or pulse generator IC, see §7.2.

¶M. Schmitt Trigger.
A *Schmitt trigger* is a threshold level-detecting circuit (Figure 2.13) with hysteresis to prevent multiple transitions when noisy input signals go though the threshold(s). Although you can make a Schmitt trigger circuit with discrete transistors, good design practice favors the use of dedicated *comparator* ICs, see §§4.3.2 and 12.3.

¶N. Emitter Follower.
The emitter follower is a linear amplifier with an ideal voltage gain of unity, see §2.2.3. The beta of the transistor increases the follower's input impedance and reduces its output impedance, see §2.2.3B and eq'n 2.2. There's more detail in §2.3.3 and Figure 2.43, where the effect of the intrinsic emitter resistance r_e is taken into account. In simplified form $R_{out}=r_e+R_s/\beta$, where R_s is the signal source resistance seen at the base. The dc output voltage is offset from the dc input by V_{BE}, about 0.6 V to 0.7 V, unless a cancelling circuit is used, see §2.2.3D and Figure 2.29. Emitter followers are also used as voltage regulators, see §2.2.4 and Figures 2.21 and 2.22. A precision alternative is the *op-amp follower*, see §4.2.3 in Chapter 4.

¶O. Current Source (or Current Sink).
In contrast to the familiar *voltage source* (which delivers a constant voltage regardless of load current, think of a battery), a *current source* delivers a constant current regardless of the load's voltage drop, see §2.2.6 and Figure 2.31; there's no everyday "battery equivalent." Transconductance devices like BJTs, with their relatively constant collector output currents, are natural candidates for making current sources. For the simplest current source, the base is biased with a voltage, say V_b, with respect to a reference point (often ground), and the emitter is connected through a resistor to the same reference. For an *npn* transistor with ground reference the output (sinking) current will be $I_C=(V_b-V_{BE})/R_E$, see Figure 2.32. For better stability and predictability the V_{BE} term can be cancelled, see Figure 2.33. The operating voltage range of a current source is called its *compliance range*, set on the low end by collector saturation, and on the high end by the transistor's breakdown voltage or by power-dissipation issues. Current sources are frequently created using current-mirror circuits, see ¶P below. Precise and stable current sources can be made with op-amps (§4.2.5); there are also dedicated current-source integrated circuits (§9.3.14).

¶P. Current Mirrors.
A current mirror (§2.3.7) is a three-terminal current-source circuit that generates an output current proportional to an input "programming" current. In a typical configuration (Figures 2.55 and 2.58) the mirror attaches to a dc rail (or to ground), reflecting the programming current, the latter perhaps set by a resistor. The circuit often omits any emitter resistors, thus achieving compliance to within a fraction of a volt of the rail. Ordinarily you wouldn't attempt to apply exactly the right V_{BE} to generate a prescribed I_C (à la Ebers–Moll); but that's exactly what you're doing here. The trick is that one transistor (Q_1) of the matched pair inverts Ebers–Moll, creating from the programming current I_P exactly the right V_{BE} to re-create the same current in the output transistor Q_2. Cute!

These circuits assume matched transistors, such as you would find inside an IC (recall from ¶G that even a 1 mV difference of V_{BE} produces a 4% change of current). Figure 2.62 graphs base-emitter voltage difference versus collector current ratio, $\Delta V_{BE}=V_T \log_e(I_{C2}/I_{C1})$. You can exploit this effect to generate a "ratio mirror," as discussed in Chapter *2x*.

As nice as it looks, the basic current mirror of Figure 2.55 suffers from Early-effect change of output current when the output voltage changes. The effect is particularly serious with *pnp* transistors: in the example of a 2N5087 in ¶J above, the 4 mV change of V_{BE} (for a 10 V output change) would cause a 17% current error. One solution (Figure 2.60) is to add emitter degeneration resistors, at the expense both of compliance near the reference rail and of dynamic range. A more elegant solution is the Wilson mirror (Figure 2.61), which defeats Early effect by exploiting the ever-useful *cascode* configuration (Figure 2.84B). Cascode transistor Q_3 passes output transistor Q_2's collector current to the load, while Q_2 operates with a fixed V_{CE} of one diode drop (its own V_{BE}). The Wilson mirror's ingenious configuration also cancels base-current errors (an ordinary mirror with BJTs having $\beta=100$ has a current error of 2%). Degeneration resistors can be added, as shown in circuit B, for additional suppression of Early effect, but they would be omitted in a "pure Wilson mirror." Linear ICs are full of Wilson mirrors. See Chapter *2x* for a discussion of *bipolarity* current mirrors.

¶Q. Common-Emitter Amplifiers.
See §§2.2.7 and 2.3.4, and Figures 2.35, 2.48 and 2.50. The simplest form of BJT amplifier has a grounded emitter, a load resistor R_L from the collector to a supply V_+, and

a dc bias plus a small signal voltage applied to the base. The gain is $G_V = -R_L/r_e$. If the base bias is carefully set so that the collector current pulls the collector halfway to ground, then $I_C = V_s/2R_L$, $r_e = V_T/I_C = 2R_L V_T/V_s$, and so the voltage gain (recall $V_T \approx 25\,\text{mV}$) is $G_V = -20V_s$, where V_s is in units of volts. For $V_s = 20\,\text{V}$, for example, the voltage gain is -400.

That's a lot of gain! Unless the signals are small, however, there's a serious problem: the gain is inverse in r_e, thus proportional to I_C. But the latter changes as the output voltage swings up and down, producing first-order changes in gain, with resulting severe distortion (Figure 2.46). This can be alleviated (at the expense of gain) by adding *emitter degeneration* in the form of an emitter resistor R_E. The gain is then $G_V = -R_L/(R_E + r_e)$, with greatly reduced effect of varying r_e; see Figure 2.47, where emitter degeneration was added to reduce the gain by a factor of ten ($R_E = 9r_e$). This is also a form of negative feedback, see §2.3.4B and ¶W below. You can think of this circuit as a classic current source (¶O) driving a resistor as load; the voltage gain is the current source's transconductance multiplied by the load resistance, $G_V = g_m R_L$, where $g_m = -1/r_e$.

We've sidestepped the important issue of setting the base bias voltage to produce the desired quiescent collector current. But we don't know the appropriate voltage V_{BE}, and a small change has a big effect, see ¶G above (e.g., a 60 mV uncertainty in V_{BE}, which is about what you might encounter from different batches of a given transistor, produces a 10× error in I_C!). There are many circuit solutions, see §2.3.5, but the simplest involves adding emitter degeneration at dc, bypassed as necessary to produce higher gain at signal frequencies (Figure 2.50 and 2.51). Another approach is to use a matching transistor to set the bias, analogous to the current mirror (Figure 2.52); this method is inherent in the widely-used *differential amplifier* (Figure 2.65). A third approach is to exploit feedback to set the bias (Figures 2.53 and 2.54), a method that figures centrally in op-amp circuits (Chapter 4).

¶R. Differential Amplifiers.

The differential amplifier (§2.3.8) is a symmetrical configuration of two matched transistors, used to amplify the difference of two input signals. It may include emitter degeneration (Figure 2.64), but need not (Figure 2.65). For best performance the emitter pulldown resistor is replaced by a current source, and (for highest gain) the resistive collector load is replaced by a current mirror (Figure 2.67). Differential amplifiers should reject strongly any common-mode input signal, achieving a good common-mode rejection ratio (CMRR, the ratio G_{diff}/G_{CM}). Differential amplifiers can be used to amplify single-ended input signals (ground the other input), where the inherent cancellation of V_{BE} offsets allows accurate dc performance (§2.3.8B). Ordinarily you use only one output from a differential amplifier; that is, it is used to to convert a balanced input to a single-ended output. But you can use both outputs (a "fully-differential amplifier," §5.17) to drive a balanced load, or to create a pair of signals 180° out of phase (a *phase splitter*). See also the sections on the emitter-input differential amplifier and on BJT amplifier distortion in Chapter *2x*, and §5.13–§5.16 (precision differential and instrumentation amplifiers).

¶S. Comparators.

A differential amplifier with lots of gain G_{diff} is driven into differential saturation with a small differential input (§2.3.8E). For example, just a few millivolts of input difference is adequate to saturate the output if $G_{diff} = 1000$ (easily accomplished with a current-mirror collector load). When operated in this way, the differential amplifier is a voltage *comparator*, a circuit used widely to sense thresholds or compare signal levels; it's the basis of analog-to-digital conversion, and figures importantly in Chapter 12 (see §12.3 and Tables 12.1 and 12.2).

¶T. Push–Pull Amplifiers.

A single transistor conducts in one direction only (e.g., an *npn* transistor can only sink current from its collector, and source current from its emitter). That makes it awkward to drive a heavy load with alternating polarity (e.g., a loudspeaker, servomotor, etc.), although it can be done, wastefully, with a single-ended stage ("class-A") with high quiescent current, see Figure 2.68. The push-pull configuration uses a pair of transistors connected to opposite supply rails (§2.4.1), an arrangement that can supply large output currents of either polarity with little or no quiescent current. Figure 2.69 shows a push-pull follower with complementary polarities, and with zero quiescent current ("class-B"); this produces some crossover distortion, which can be eliminated by biasing the pair into quiescent conduction ("class-AB," Figure 2.71),. The output transistors can be beta-boosting configurations like the Darlington or Sziklai (¶U), see for example Figure 2.78. The push-pull configuration is widely used in logic circuits (see Figure 10.25), gate driver ICs (see Figure 3.97), and in combination with op-amps to deliver greater output currents (see Figure 4.26).

¶U. The Darlington and Sziklai Connections.

These simple combinations of two transistors create a 3-terminal equivalent transistor with $\beta = \beta_1\beta_2$. The Darlington (Figures 2.74 and 2.75) cascades two transistors of the

same polarity and has a base-emitter drop of $2V_{BE}$; the Sziklai (Figure 2.77) pairs opposite polarities, and has a single base-emitter drop (which is only weakly dependent on output current, thanks to R_B). For either configuration a resistor R_B should be connected across the output transistor's base-emitter terminals. For more about this subject see see the discussion in Chapter *2x*.

¶V. Miller Effect.

Like all electronic components, transistors have interterminal capacitances, designated (by terminal pairs) C_{be}, C_{ce}, and C_{cb}.[72] While C_{be} and C_{ce} slow the input and output waveforms by creating lowpass filters with the source and load resistances, the effect of the feedback capacitance C_{cb} is more insidious: it creates an additional input capacitance to ground equal to C_{cb} multiplied by the stage's inverting voltage gain, thus its effective input capacitance becomes $C_{eff}=(G_V+1)C_{cb}$. This is the infamous *Miller effect* (§2.4.5B), whose impact can be devastating in high-speed and wideband amplifiers. Some circuit solutions include the grounded-base amplifier, the differential amplifier, and the cascode configuration (see the discussion of cascode in Chapter *2x*).

[72] These have many aliases (a common set uses initials for "in" and "out" instead of "base" and "collector," thus C_{ie}, C_{oe}, and C_{ob}, respectively), see the section on BJT bandwidth in Chapter *2x*.

¶W. Negative Feedback.

If there were a Nobel prize for grand-concepts-in-circuit-design, it would surely go to Harold Black for his elegant elucidation of *negative feedback*. In its simplest form, it consists of subtracting, from the input signal, a fraction B of an amplifier's output signal V_{out} (Figure 2.85). If the amplifier's open-loop gain is A, then the closed-loop gain becomes (eq'n 2.16) $G_{cl}=A/(1+AB)$. The quantity AB, which generally is large compared with unity, is called the *loop gain*, and it (more precisely the quantity $1+AB$) is the multiplier by which negative feedback improves the amplifier's performance: improved linearity and constancy of gain, and (in this *series feedback* circuit configuration) raised input impedance and lowered output impedance; see §2.5.3.

Feedback is the essence of linear design, and it is woven deeply into the DNA of op-amp circuits (the subject of Chapter 4), and power circuits (Chapter 9). With negative feedback you can make amplifiers with 0.0001% distortion, voltage sources with 0.001 Ω output impedance, and many other wonders too magnificent here to relate. Stay tuned. Better yet, *read on!*